

Spett.le

**STEALTH COMPANY SRL START UP
INNOVATIVA**

Via Elio Vittorini, n. 103
00144 ROMA (RM)

SOLO VIA EMAIL

NS. RIF. PF/cbg/AA11393US-PCT

ROMA, 8 maggio 2025

Domanda di brevetto Internazionale n. PCT/IB2025/051755

Da domanda di provisional protection in USA n. 63/557,868 del 26.02.2024

**per: "AI – DRIVEN SYSTEM AND METHOD FOR CONTENT DIFFERENTIATION
AND PIRACY TRACEABILITY IN STREAMING MEDIA"**

a nome: STEALTH COMPANY SRL START UP INNOVATIVA

Inventore: Antonio Rossi

Gentili Signori,

Facciamo seguito alla ns. lettera del 19.02.2025 e Vi informiamo che abbiamo appena ricevuto il rapporto di ricerca internazionale della Vs. domanda citata in oggetto.

In allegato Vi inviamo copia della lettera e dei brevetti anteriori.

L'esaminatore ha rintracciato 2 brevetti anteriori al Vostro, di cui 1 segnalato con la lettera X e quindi considerato particolarmente rilevante per la Vs. domanda di brevetto:

- Brevetto negli Stati Uniti n. US 2021/352381 A1 (KILSTEIN GOLDA WEYL ET AL)

E 1 segnalato con la lettera A e quindi considerato non particolarmente rilevante per la Vs. domanda:

- Brevetto in Cina n. CN 117 278 762 A (ARBOO TECH CO LTD)

L'esaminatore, nell'opinione scritta dell'esaminatore, afferma che tutte le rivendicazioni sono nuove e che, pur non godendo del requisito dell'attività inventiva, sono tutte applicabili industrialmente.

Vi informiamo che, a questo punto della procedura, è possibile pagare la tassa d'esame preliminare (€ 3.400,00) ed avere così un parere dell'esaminatore sulla brevettabilità della Vs. domanda PCT.

Vi preghiamo di notare che, poiché l'esito dell'esame preliminare costituisce un mero parere e non è vincolante, la scelta consigliata per la Vs. domanda di brevetto è quella di non pagare la tassa di esame e di aspettare l'apertura delle fasi nazionali/regionali, affrontando quindi l'esame nei vari Paesi in cui confermerete la protezione.

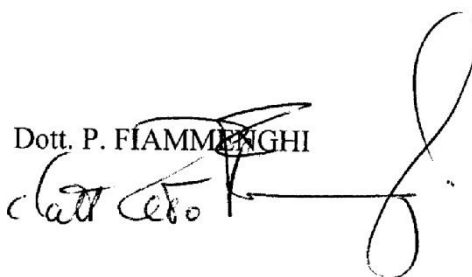
Vi ricordiamo, infatti, che la copertura provvisoria in tutti i 158 Paesi designati al momento del deposito della domanda internazionale è attiva fino a 30 mesi dalla data di deposito della priorità (i.e.: 30 mesi dal 26.02.2024 → termine deposito fasi nazionali / regionali: 26.08.2026).

Sarà ns. premura inviarVi un promemoria con largo anticipo.

Vi informiamo, infine, che poiché la richiesta di esame preliminare è facoltativa, qualora non dovessimo ricevere Vs. istruzioni in merito, non invieremo ulteriori solleciti.

Nel frattempo rimaniamo a Vs. completa disposizione per qualsiasi chiarimento.

Ci è gradita l'occasione per inviarVi i ns. distinti saluti.

Dott. P. FIAMMENGHI


All.to: copia esito ricerca di anteriorità / copia 2 documenti anteriori

PATENT COOPERATION TREATY

From the INTERNATIONAL SEARCHING AUTHORITY

PCT

To:

Praxi Intellectual Property...
Fiammenghi, Eva
c/o PRAXI Intellectual Property S.p.A.
Via Leonida Bissolati, 20
00187 Roma
ITALIE

NOTIFICATION OF TRANSMITTAL OF
THE INTERNATIONAL SEARCH REPORT AND
THE WRITTEN OPINION OF THE INTERNATIONAL
SEARCHING AUTHORITY, OR THE DECLARATION

(PCT Rule 44.1)

	Date of mailing (day/month/year) 13 May 2025 (13-05-2025)
Applicant's or agent's file reference PF-MA-AA11393US-PCT	FOR FURTHER ACTION See paragraphs 1 and 4 below
International application No. PCT/IB2025/051755	International filing date (day/month/year) 19 February 2025 (19-02-2025)
Applicant STEALTH COMPANY SRL START UP INNOVATIVA	

- ☒ The applicant is hereby notified that the international search report and the written opinion of the International Searching Authority have been established and are transmitted herewith.
Filing of amendments and statement under Article 19:
 The applicant is entitled, if he so wishes, to amend the claims of the international application (see Rule 46):
When? The time limit for filing such amendments is normally two months from the date of transmittal of the international search report.
How? Directly to the International Bureau preferably through ePCT, or on paper to:
 The International Bureau of WIPO, 34 chemin des Colombettes, 1211 Geneva 20, Switzerland
For more detailed instructions, see the *PCT Applicant's Guide*, International Phase, paragraphs 9.004 - 9.011.
- ☐ The applicant is hereby notified that no international search report will be established and that the declaration under Article 17(2)(a) to that effect and the written opinion of the International Searching Authority are transmitted herewith.
- ☐ **With regard to any protest** against payment of (an) additional fee(s) under Rule 40.2, the applicant is notified that:
 - ☐ the protest together with the decision thereon has been transmitted to the International Bureau together with any request to forward the texts of both the protest and the decision thereon to the designated Offices.
 - ☐ no decision has been made yet on the protest; the applicant will be notified as soon as a decision is made.
- Reminders**
 The applicant may **submit comments on an informal basis on the written opinion of the International Searching Authority** to the International Bureau. These comments will be made available to the public after international publication. The International Bureau will send a copy of such comments to all designated Offices unless an international preliminary examination report has been or is to be established.
 Shortly after the expiration of **18 months from the priority date, the international application will be published** by the International Bureau. If the applicant wishes to avoid or postpone publication, a notice of withdrawal of the international application, or of the priority claim, must reach the International Bureau before the completion of the technical preparations for international publication (Rules 90bis.1 and 90bis.3).
 Within **19 months** from the priority date, but only in respect of some designated Offices, a demand for international preliminary examination must be filed if the applicant wishes to postpone the entry into the national phase **until 30 months** from the priority date (in some Offices even later); otherwise, the applicant must, **within 20 months** from the priority date, perform the prescribed acts for **entry into the national phase** before those designated Offices. In respect of other designated Offices, the time limit of **30 months** (or later) will apply even if no demand is filed within 19 months. For details about the applicable time limits, Office by Office, see www.wipo.int/pct/en/texts/time_limits.html and the *PCT Applicant's Guide*, National Chapters.
 Within **22 months from the priority date, the applicant may request that a supplementary international search be carried out** by a different International Searching Authority that offers this service (Rule 45bis.1). The procedure for requesting supplementary international search is described in the *PCT Applicant's Guide*, International Phase, paragraphs 8.006-8.032.

Name and mailing address of the International Searching Authority  European Patent Office, P.B. 5818 Patentlaan 2 NL-2280 HV Rijswijk Tel. (+31-70) 340-2040	Authorized officer ISA/EP
--	----------------------------------

PATENT COOPERATION TREATY

PCT

INTERNATIONAL SEARCH REPORT

(PCT Article 18 and Rules 43 and 44)

Applicant's or agent's file reference PF-MA-AA11393US-PCT	FOR FURTHER ACTION see Form PCT/ISA/220 as well as, where applicable, item 5 below.	
International application No. PCT/IB2025/051755	International filing date (day/month/year) 19 February 2025 (19-02-2025)	(Earliest) Priority Date (day/month/year) 26 February 2024 (26-02-2024)
Applicant STEALTH COMPANY SRL START UP INNOVATIVA		

This international search report has been prepared by this International Searching Authority and is transmitted to the applicant according to Article 18. A copy is being transmitted to the International Bureau.

This international search report consists of a total of 3 sheets.

☒ It is also accompanied by a copy of each prior art document cited in this report.

1. **Basis of the report**

a. With regard to the **language**, the international search was carried out on the basis of:

☒ the international application in the language in which it was filed

☐ a translation of the international application into _____, which is the language of a translation furnished for the purposes of international search (Rules 12.3(a) and 23.1(b))

b. ☐ This international search report has been established taking into account the **rectification of an obvious mistake** authorized by or notified to this Authority under Rule 91 (Rule 43.6bis(a)).

c. ☐ With regard to any **nucleotide and/or amino acid sequence** disclosed in the international application, see Box No. I.

2. ☐ **Certain claims were found unsearchable** (See Box No. II)

3. ☐ **Unity of invention is lacking** (see Box No III)

4. With regard to the **title**,

☒ the text is approved as submitted by the applicant

☐ the text has been established by this Authority to read as follows:

5. With regard to the **abstract**,

☒ the text is approved as submitted by the applicant

☐ the text has been established, according to Rule 38.2, by this Authority as it appears in Box No. IV. The applicant may, within one month from the date of mailing of this international search report, submit comments to this Authority

6. With regard to the **drawings**,

a. the figure of the **drawings** to be published with the abstract is Figure No. 1

☒ as suggested by the applicant

☐ as selected by this Authority, because the applicant failed to suggest a figure

☐ as selected by this Authority, because this figure better characterizes the invention

b. ☐ none of the figures is to be published with the abstract

INTERNATIONAL SEARCH REPORT

International application No
PCT/IB2025/051755

A. CLASSIFICATION OF SUBJECT MATTER

INV. H04N21/234 H04N21/2343 H04N21/8358 H04N21/845
ADD.

According to International Patent Classification (IPC) or to both national classification and IPC

B. FIELDS SEARCHED

Minimum documentation searched (classification system followed by classification symbols)

H04N

Documentation searched other than minimum documentation to the extent that such documents are included in the fields searched

Electronic data base consulted during the international search (name of data base and, where practicable, search terms used)

EPO-Internal

C. DOCUMENTS CONSIDERED TO BE RELEVANT

Category*	Citation of document, with indication, where appropriate, of the relevant passages	Relevant to claim No.
X	US 2021/352381 A1 (KILSTEIN GOLDA WEYL [IL] ET AL) 11 November 2021 (2021-11-11) paragraphs [0002], [0025], [0028] - [0029], [0031], [0034], [0037], [0081], [0083]; figure 1 -----	1-18
A	CN 117 278 762 A (ARB00 TECH CO LTD) 22 December 2023 (2023-12-22) paragraphs [0029], [0036], [0047] - [0049] -----	1-18



Further documents are listed in the continuation of Box C.



See patent family annex.

* Special categories of cited documents :

"A" document defining the general state of the art which is not considered to be of particular relevance

"E" earlier application or patent but published on or after the international filing date

"L" document which may throw doubts on priority claim(s) or which is cited to establish the publication date of another citation or other special reason (as specified)

"O" document referring to an oral disclosure, use, exhibition or other means

"P" document published prior to the international filing date but later than the priority date claimed

"T" later document published after the international filing date or priority date and not in conflict with the application but cited to understand the principle or theory underlying the invention

"X" document of particular relevance;; the claimed invention cannot be considered novel or cannot be considered to involve an inventive step when the document is taken alone

"Y" document of particular relevance;; the claimed invention cannot be considered to involve an inventive step when the document is combined with one or more other such documents, such combination being obvious to a person skilled in the art

"&" document member of the same patent family

Date of the actual completion of the international search

30 April 2025

Date of mailing of the international search report

13/05/2025

Name and mailing address of the ISA/

European Patent Office, P.B. 5818 Patentlaan 2

NL - 2280 HV Rijswijk

Tel. (+31-70) 340-2040,

Fax: (+31-70) 340-3016

Authorized officer

Folea, Octavian

INTERNATIONAL SEARCH REPORT

Information on patent family members

International application No

PCT/IB2025/051755

Patent document cited in search report	Publication date	Patent family member(s)	Publication date
US 2021352381 A1	11-11-2021	US 2021352381 A1	11-11-2021
		WO 2021224688 A1	11-11-2021

CN 117278762 A	22-12-2023	NONE	

Information on Search Strategy

The type of information contained in this sheet may change during the pilot for improving the usefulness of this new service.

Application Number

PCT/IB2025/051755

TITLE: AI -DRIVEN SYSTEM AND METHOD FOR CONTENT DIFFERENTIATION AND PIRACY TRACEABILITY IN STREAMING MEDIA

APPLICANT: STEALTH COMPANY SRL START UP INNOVATIVA

IPC CLASSIFICATION: H04N21/234, H04N21/2343, H04N21/8358, H04N21/845

EXAMINER: Folea, Octavian

CONSULTED DATABASES:

CLASSIFICATION SYMBOLS DEFINING EXTENT OF THE SEARCH:

IPC:

CPC: H04N21/23418, H04N21/2187, H04N21/23439, H04N21/8358, H04N21/8456

FI/F-TERMS:

KEYWORDS OR OTHER ELEMENTS FEATURING THE INVENTION:

Watermark and camera switching.

PATENT COOPERATION TREATY

From the
INTERNATIONAL SEARCHING AUTHORITY

PCT

WRITTEN OPINION OF THE INTERNATIONAL SEARCHING AUTHORITY (PCT Rule 43bis.1)

To:

see form PCT/ISA/220

Date of mailing
(day/month/year) see form PCT/ISA/210 (second sheet)

Applicant's or agent's file reference
see form PCT/ISA/220

FOR FURTHER ACTION
See paragraph 2 below

International application No.
PCT/IB2025/051755

International filing date (day/month/year)
19/02/2025

Priority date (day/month/year)
26/02/2024

International Patent Classification (IPC) or both national classification and IPC
INV. H04N21/234 H04N21/2343 H04N21/8358 H04N21/845

Applicant
STEALTH COMPANY SRL START UP INNOVATIVA

1. This opinion contains indications relating to the following items:

- ☒ Box No. I Basis of the opinion
- ☐ Box No. II Priority
- ☐ Box No. III Non-establishment of opinion with regard to novelty, inventive step and industrial applicability
- ☐ Box No. IV Lack of unity of invention
- ☒ Box No. V Reasoned statement under Rule 43bis.1(a)(i) with regard to novelty, inventive step and industrial applicability; citations and explanations supporting such statement
- ☐ Box No. VI Certain documents cited
- ☒ Box No. VII Certain defects in the international application
- ☒ Box No. VIII Certain observations on the international application

2. FURTHER ACTION

If a demand for international preliminary examination is made, this opinion will usually be considered to be a written opinion of the International Preliminary Examining Authority ("IPEA") except that this does not apply where the applicant chooses an Authority other than this one to be the IPEA and the chosen IPEA has notified the International Bureau under Rule 66.1bis(b) that written opinions of this International Searching Authority will not be so considered.

If this opinion is, as provided above, considered to be a written opinion of the IPEA, the applicant is invited to submit to the IPEA a written reply together, where appropriate, with amendments, before the expiration of 3 months from the date of mailing of Form PCT/ISA/220 or before the expiration of 22 months from the priority date, whichever expires later.

For further options, see Form PCT/ISA/220.

Name and mailing address of the ISA:



European Patent Office
80298 MUNICH
GERMANY
Telephone No. + 49 (0)89 2399-0

Date of completion of
this opinion
see form
PCT/ISA/210

Authorised Officer

Folea, Octavian

Telephone No. + 49 (0)89 2399-0



Box No. I Basis of the opinion

1. With regard to the **language**, this opinion has been established on the basis of:
- ☒ the international application in the language in which it was filed.
 - ☐ a translation of the international application into , which is the language of a translation furnished for the purposes of international search (Rules 12.3(a) and 23.1 (b)).
2. ☐ This opinion has been established taking into account the **rectification of an obvious mistake** authorized by or notified to this Authority under Rule 91 (Rule 43bis.1(a))
3. ☐ With regard to any **nucleotide and/or amino acid sequence** disclosed in the international application, this opinion has been established on the basis of a sequence listing:
- a. ☐ forming part of the international application as filed.
 - b. ☐ furnished subsequent to the international filing date for the purposes of international search (Rule 13ter.1(a)),
 - ☐ accompanied by a statement to the effect that the sequence listing does not go beyond the disclosure in the international application as filed.
4. ☐ With regard to any nucleotide and/or amino acid sequence disclosed in the international application, this opinion has been established to the extent that a meaningful opinion could be formed without a WIPO Standard ST.26 compliant sequence listing.
5. Additional comments:

Box No. V Reasoned statement under Rule 43bis.1(a)(i) with regard to novelty, inventive step or industrial applicability; citations and explanations supporting such statement

1. Statement

Novelty (N)	Yes: Claims <u>1-18</u>
	No: Claims

Inventive Steps (IS)	Yes: Claims
	No: Claims <u>1-18</u>

Industrial applicability (IA)	Yes: Claims <u>1-18</u>
	No: Claims

2. Citations and explanations

see separate sheet

Box No. VII Certain defects in the international application

The following defects in the form or contents of the international application have been noted:

see separate sheet

Box No. VIII Certain observations on the international application

The following observations on the clarity of the claims, description, and drawings or on the question whether the claims are fully supported by the description, are made:

see separate sheet

Re Item V

Reasoned statement with regard to novelty, inventive step or industrial applicability; citations and explanations supporting such statement

Reference is made to the following documents:

- D1 US 2021/352381 A1 (KILSTEIN GOLDA WEYL [IL] ET AL) 11 November 2021 (2021-11-11)
- D2 CN 117 278 762 A (ARBOO TECH CO LTD) 22 December 2023 (2023-12-22)

1 Lack of novelty, inventive step

The references that are recited in the following section(s) are found in **D1**.

1.1 Independent claim 1

Anti-piracy system (100) identifying unauthorized distributions of streaming audio-video content; said anti-piracy system (100) comprising:

- *"the transformer 120" in [0025] and figure 1*

(F1.1) -a pipeline receiving live or pre-recorded video ~~captured from one or more cameras (101)~~ ([0029]: "...a single media content storage 101..." and the entire system function in the same paragraph and figure 1) ;

(F1.2) ~~said pipeline creating a reference audio-video content (1);~~

(F1.3) ~~switches between said cameras (101) being called camera cuts;~~

-a mate creation component (110) generating at least one modified version of the video by automatically altering at least one camera-cut timing relative to the reference audio-video content (1), thereby producing one or more unique mates (111) of the reference audio-video content (1);

- [0037]: *"It should be noted that though FIG. 2C illustrates a single server 230 C, components and/or functions of the server 230 C can be distributed to multiple devices. For example, while the variants generator 232 C is*

typically implemented in the video encoding headend, the sequence generator 234 may be implemented in other components of the media chain, e.g., at an encoder, a packager, a multiplexer, an over-the-top (OTT) origin server, an OTT playlist server, a content distribution network (CDN) server (central or edge), or a client device"

a mate (111) being a distinguishable version of the same underlying audio-video content (1);

- **[0083]:** *"In one example, the set of segments can include the video stream, such as applying a fill mode transformation to modify stretch slightly so each client would have a slightly different image dimension to resist collusion attacks. In the example shown in FIG. 6, the transformation is applied segment-by-segment, so that each segment has a different transformation, and for each segment, the user receives one copy of this segment. In order to have a variety of transformations over time, the transformer changes transformations as time goes. As a result, the two copies of the video stream as shown in FIG. 6 may not be different in every point of time (e.g., segment 3 variant 630-B is in both copies), but along the timeline and over time, the two copies of the video stream are different to resist collusion attacks"*

an ensemble (112) being a combination of both the reference audio-video content (1) and its mates (111);

- **[0081]:** *"...generating multiple copies of the media content item by replicating and applying multiple transformations, wherein the multiple transformations applied to the multiple copies of the media content item are statistically different...."*

-a distribution apparatus delivering one or more of said distinguishable versions to at least one recipient;

-a record of associations, made by said distribution apparatus, between said distinguishable versions and respective recipients;

- **[0031]:** *"...with reference to FIG. 6, in headend based transformation, the transformer 120 A determines which transformations to apply and generates multiple copies, e.g., two copies, for each segment and performs a different transformation on each of them. Each client device receives a sequence of transformed segments. For each segment, the client device choses one of the copies, e.g., based on the ID and/or randomly..."*

~~(F1.4) -a detection apparatus applying an artificial intelligence algorithm to analyze a suspected unauthorized distribution by detecting camera-switch timings therein and to identify which of said distinguishable versions matches the suspected unauthorized distribution;~~

said detection apparatus searching said record for a recipient associated to a distinguishable version matching the suspected unauthorized distribution.

- **[0002]:** *"...in case a user leaks certain media content, his/her identity (ID) may be traced using embedded data. The watermark identifier is often encoded in an imperceptible manner while still allowing recovery of the watermark identifier even if the media content is modified."*

The subject-matter of the claim differs from **D1** by the following distinguishing features: F1.1, F1.2, F1.3, F1.4.

The combination of said distinguishing features cannot be considered to be inventive for the following reasons:

- **D1** is silent regarding multiple cameras - however, such system is well-known in the art;
- it is considered that using the multiple cameras do not have a technical synergy with the rest of the claims related to watermarking (modifying the content), besides "camera-switch timings" that corresponds to "scene change" in **[0034]:** *"...When the transformer 120 B applies a set of transformations, the transformations fade in and/or fade out or aligned with scene change for a smooth transition between the transformations...."*;
- since there is no technical synergy between this distinguishing features and the rest of the claim features, and using multiple cameras is well-known in the art, it is considered that these distinguishing features do not contribute to an inventive activity;
- more over, merely referring to "applying an artificial intelligence" without further details cannot be considered to involve an inventive step.

Therefore the subject-matter of the claim is not inventive (Article 33(3) PCT).

1.2 Independent claim 18

Same reasons as for claim **1** apply mutatis mutandis, therefore the subject-matter of the claim is not inventive (Article 33(3) PCT).

1.3 Dependent claims

The subject-matter of the following claims is not inventive (Article 33(3) PCT).

1.3.1 Claims 2-3, 7

- **D1** is silent regarding using AI in this respect. As stated below in the clarity objections, the related features are broad, and is considered that by merely referring the use of AI, without any specific details, does not involve an inventive activity.

1.3.2 Claims 4, 7, 9-10

- **D1** does not disclose specific pipelines and server clusters - however such hardware arrangements are well-known, and therefore they do not involve an inventive activity.

1.3.3 Claim 5

- **D1** relates to watermark - fingerprinting is a well-known alternative, statistical analysis is part of the most watermarking/fingerprinting techniques.

1.3.4 Claim 6

- uni-casting (as multi-casting) is common general knowledge.

1.3.5 Claim 8

- **[0028]**, **[0037]**.

1.3.6 Claims 11-17

- "chunks" are implicit by reference to CDN in **[0028]**: "...the system 100 may also include one or more encoders, packagers, and/or multiplexers, an OTT server, an OTT playlist server, and/or one or more CDNs...".
- manifest files are common general knowledge.

1.3.7 Claim 15

- **[0002]**: *"The process of hiding or embedding information in media content generally comes under the category of watermarking. Digital watermarking (sometimes also known as "forensic watermarking") typically refers to the watermarking process where the embedded data identifies a media content item itself, as well as the content owner, operators, redistribution networks, and/or consumer devices associated with the distribution and consumption of the media content item. For example, in case a user leaks certain media content, his/her*

identity (ID) may be traced using embedded data. The watermark identifier is often encoded in an imperceptible manner while still allowing recovery of the watermark identifier even if the media content is modified."

1.3.8 Claims 16-17

- there is no disclosure of switching camera in **D1**, however it is considered not to involve an inventive activity (as argued for claim **1**).

Re Item VII

Certain defects in the international application (form or content)

- 1 Contrary to the requirements of Rule 5.1(a)(ii) PCT, the relevant background art disclosed in the document D1 is not mentioned in the description, nor is this document identified therein.
- 2 The independent claims are not in the two-part form in accordance with Rule 6.3(b) PCT, which in the present case would be appropriate, with those features known in combination from the prior art (document D1) being placed in the preamble (Rule 6.3(b)(i) PCT) and with the remaining features being included in the characterising part (Rule 6.3(b)(ii) PCT).
- 3 The features of the claims are not provided with reference signs placed in parentheses (Rule 6.2(b) PCT).

Re Item VIII

Certain observations on the international application (clarity)

The application does not meet the requirements of Article 6 PCT, because the following claims are not clear for the reasons expressed below.

4 Lack of clarity

4.1 Claims 1, 16-17

- it is unclear how "adjusting/altering" the "time codes" at "camera cuts" - is it about changing the values of the timestamps ? how in this case it is ensured the integrity of the scene, are these changes small enough in order not to have a visual impact on the rendering process ? the description is not clear in this respect.

4.2 Independent claim 1

- "a mate (111) being a distinguishable version of the same underlying audio-video content..." - it is unclear what technical limitation "distinguishable" entails, since basically a single change (one single bit, for example) renders a copy "distinguishable" for the original one.
- "...applying an artificial intelligence algorithm to analyze a suspected unauthorized distribution by detecting camera-switch timings therein and to identify which of said distinguishable versions matches the suspected unauthorized distribution" seems to be the result to be achieved - more details are provided in claims 2, 3.

4.3 Dependent claims 2-3

- it is unclear whether "said artificial intelligence algorithm" that "...compares the reference audio-video content (1) with the pirate webcasting video..." involves any kind of AI model, or any intrinsic technical matter related to AI.

4.4 Dependent claim 7

- "...machine learning algorithms predicting and preemptively counteracting potential piracy activities by adjusting the time code variations of camera cuts and the configuration of the unique content mates (111), based on historical data patterns and piracy attempt profiles" seems also to be the result to be achieved. The reason is it is not disclosed how "the time code variations" are "adjusted" based on vague terms "historical data patterns" (historic compared to what ? data pattern concerning which kind of data ?) and "piracy attempt profiles" (what is its technical definition ?) - also the description does not provide clarifying details.



US 20210352381A1

(19) **United States**

(12) **Patent Application Publication**
Kilstein et al.

(10) **Pub. No.: US 2021/0352381 A1**

(43) **Pub. Date: Nov. 11, 2021**

(54) **METHODS AND SYSTEMS FOR REDUCING PIRACY OF MEDIA CONTENT**

(52) **U.S. Cl.**

CPC *H04N 21/8456* (2013.01); *H04N 21/8358* (2013.01); *H04N 1/32144* (2013.01); *H04N 21/23418* (2013.01); *H04N 21/812* (2013.01)

(71) Applicant: **Synamedia Limited**, Staines Upon Thames (GB)

(72) Inventors: **Golda Weyl Kilstein**, Jerusalem (IL); **David Livshits**, Jerusalem (IL); **Vered Anikster**, Jerusalem (IL); **Michal Irit Devir**, Jerusalem (IL); **Samie Beheydt**, Geluwe (BE)

(57)

ABSTRACT

Techniques for reducing piracy of media content are described. In some embodiments, a collusion resistant method is performed at a device, where the device receives a first request for a base copy of a media content item. In response, the device determines a first transformation based on a statistical performance criterion and a viewing performance criterion. The device further generates a first copy of the media content item by replicating and applying a first transformation to the base copy, where the first copy of the media content item satisfies the viewing performance criterion, and the first copy of the media content item is statistically different from the base copy or other copies in accordance with the statistical performance criterion. The device then causes transmission of the first copy of the media content item in combination with a first watermark for the base copy of the media content item.

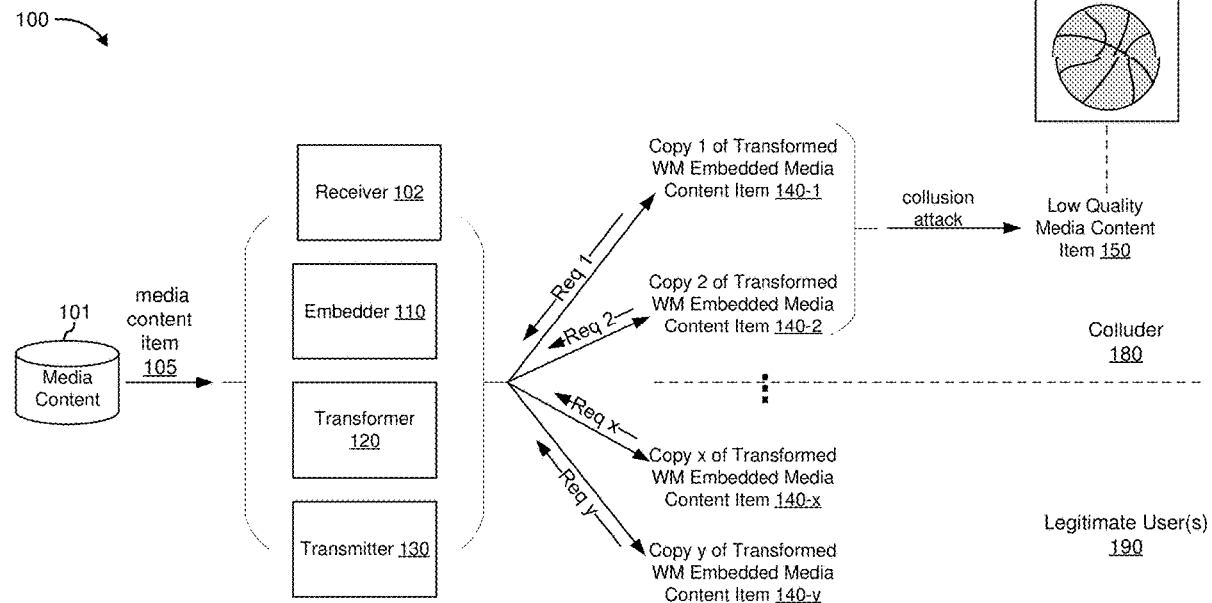
(21) Appl. No.: **16/869,504**

(22) Filed: **May 7, 2020**

Publication Classification

(51) **Int. Cl.**

<i>H04N 21/845</i>	(2006.01)
<i>H04N 21/8358</i>	(2006.01)
<i>H04N 21/81</i>	(2006.01)
<i>H04N 21/234</i>	(2006.01)
<i>H04N 1/32</i>	(2006.01)



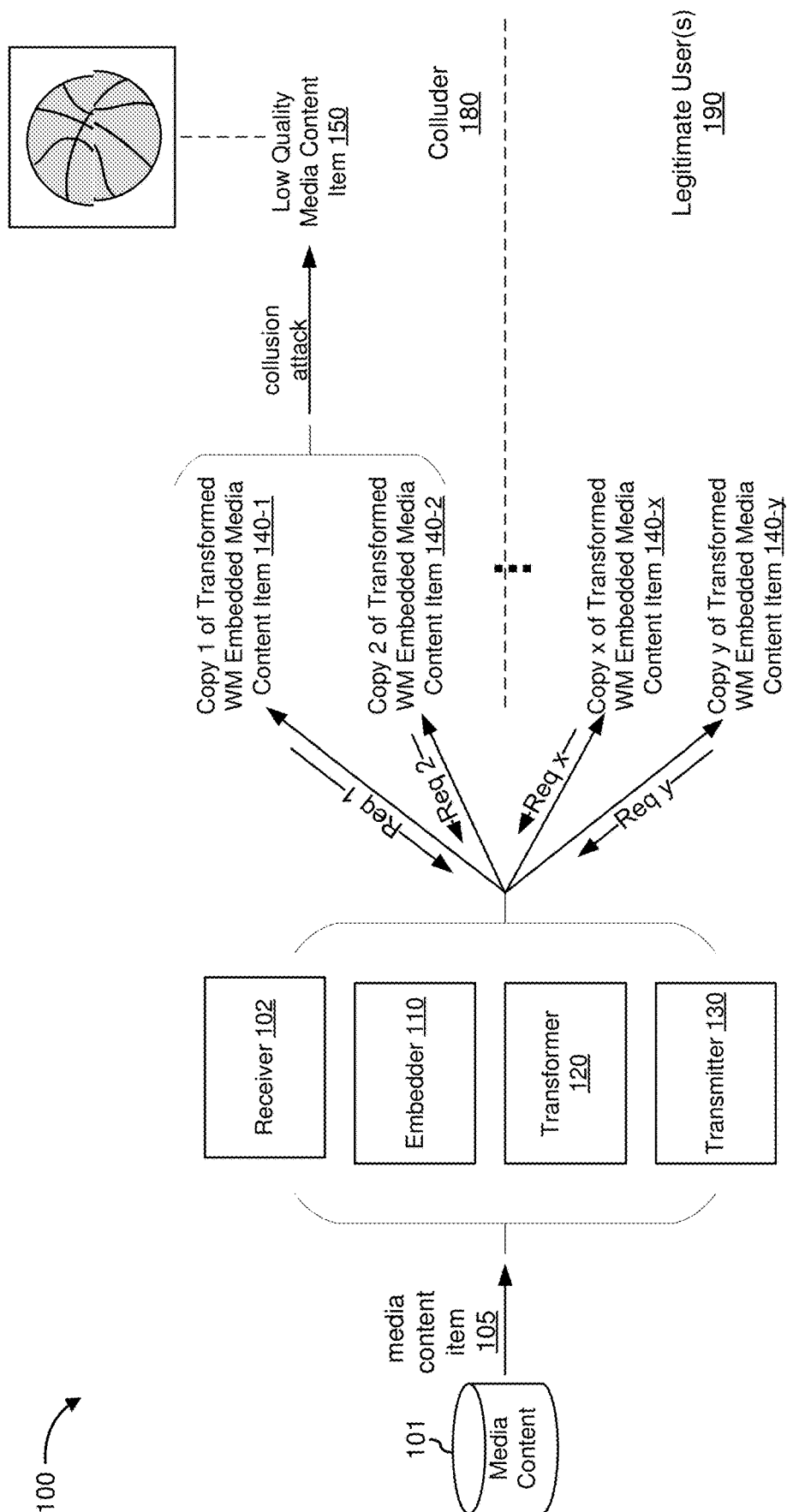
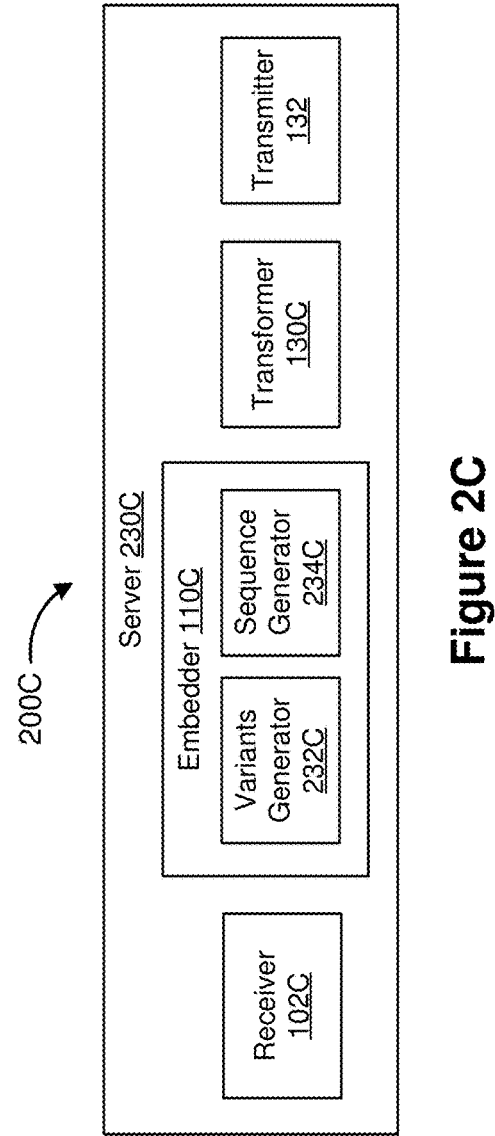
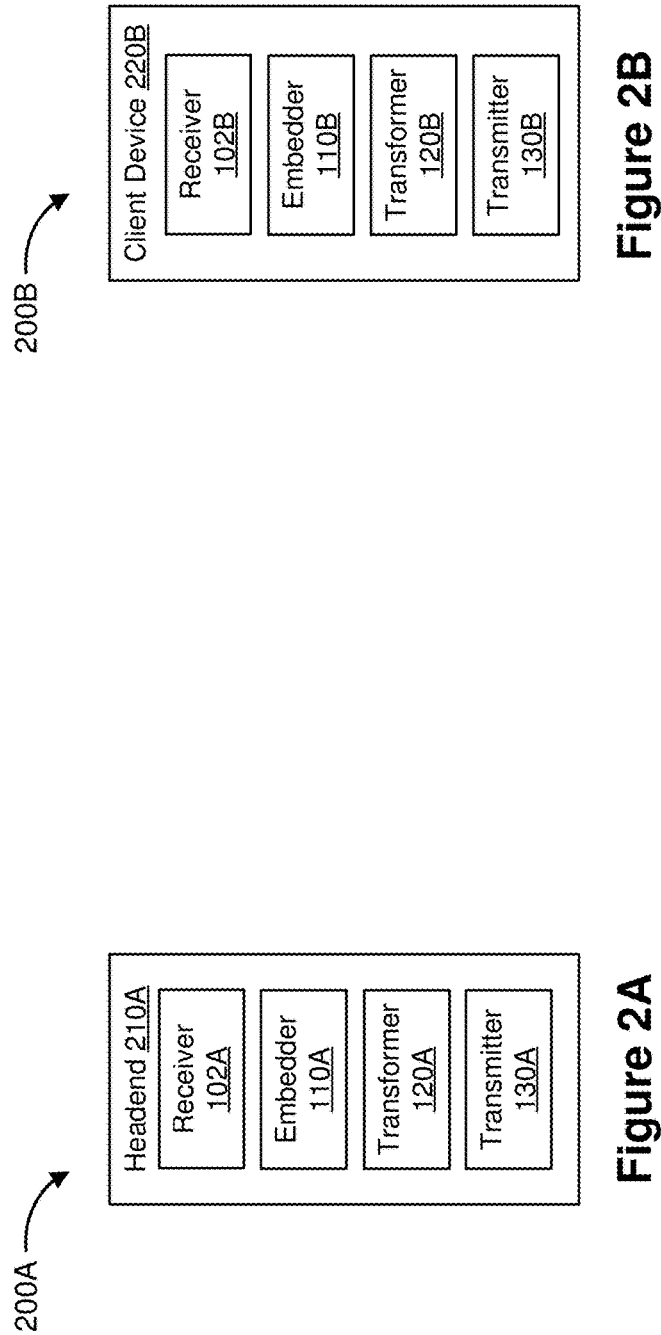


Figure 1



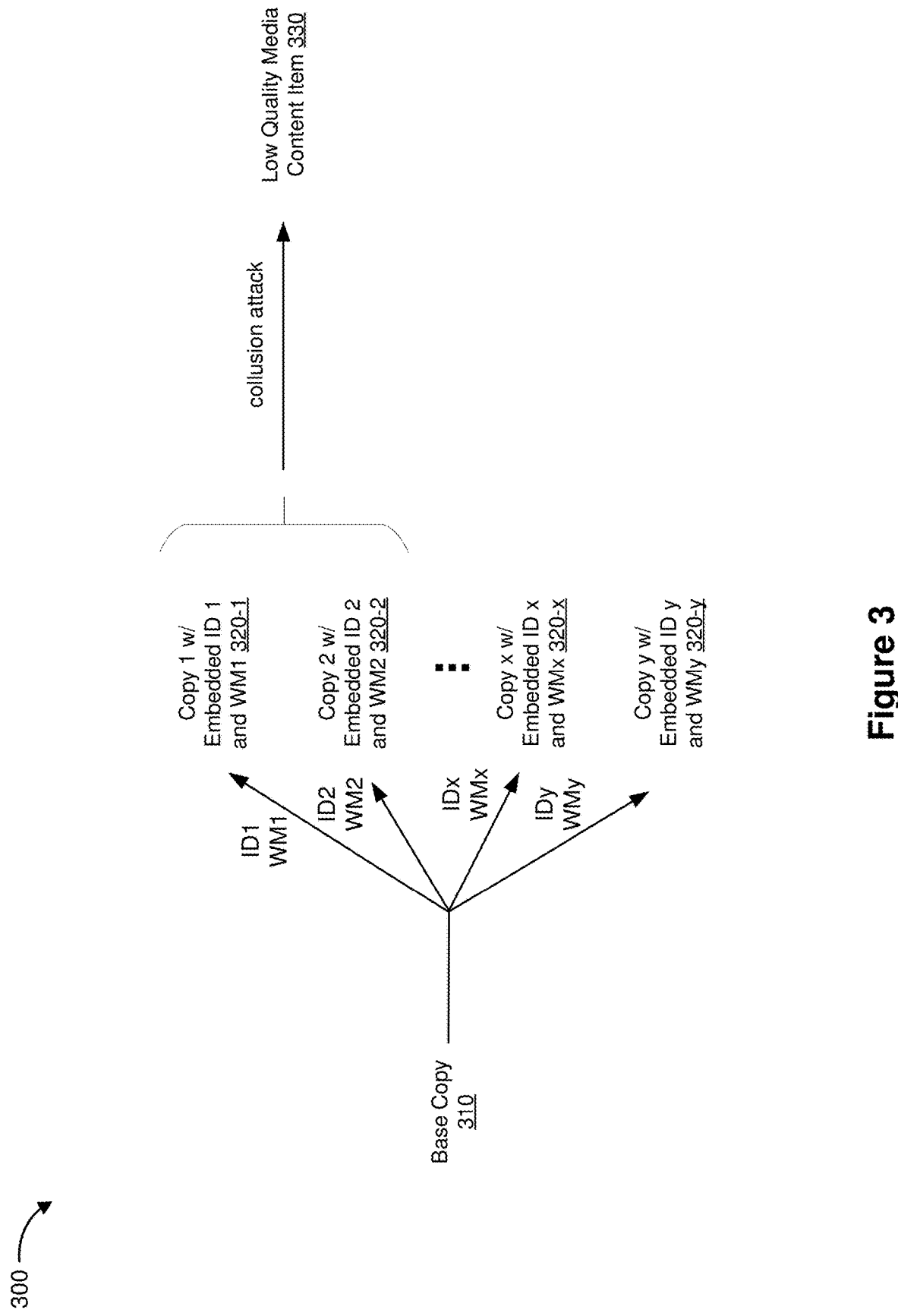


Figure 3

400 →

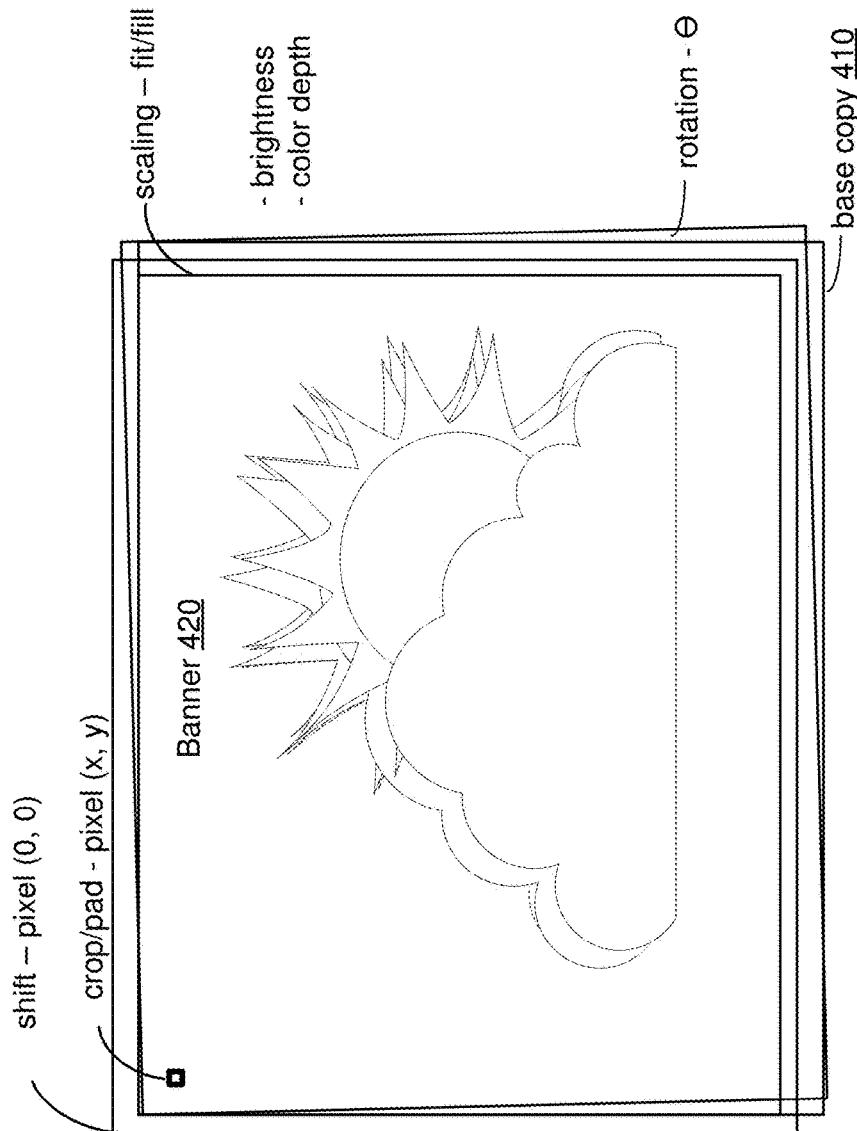


Figure 4

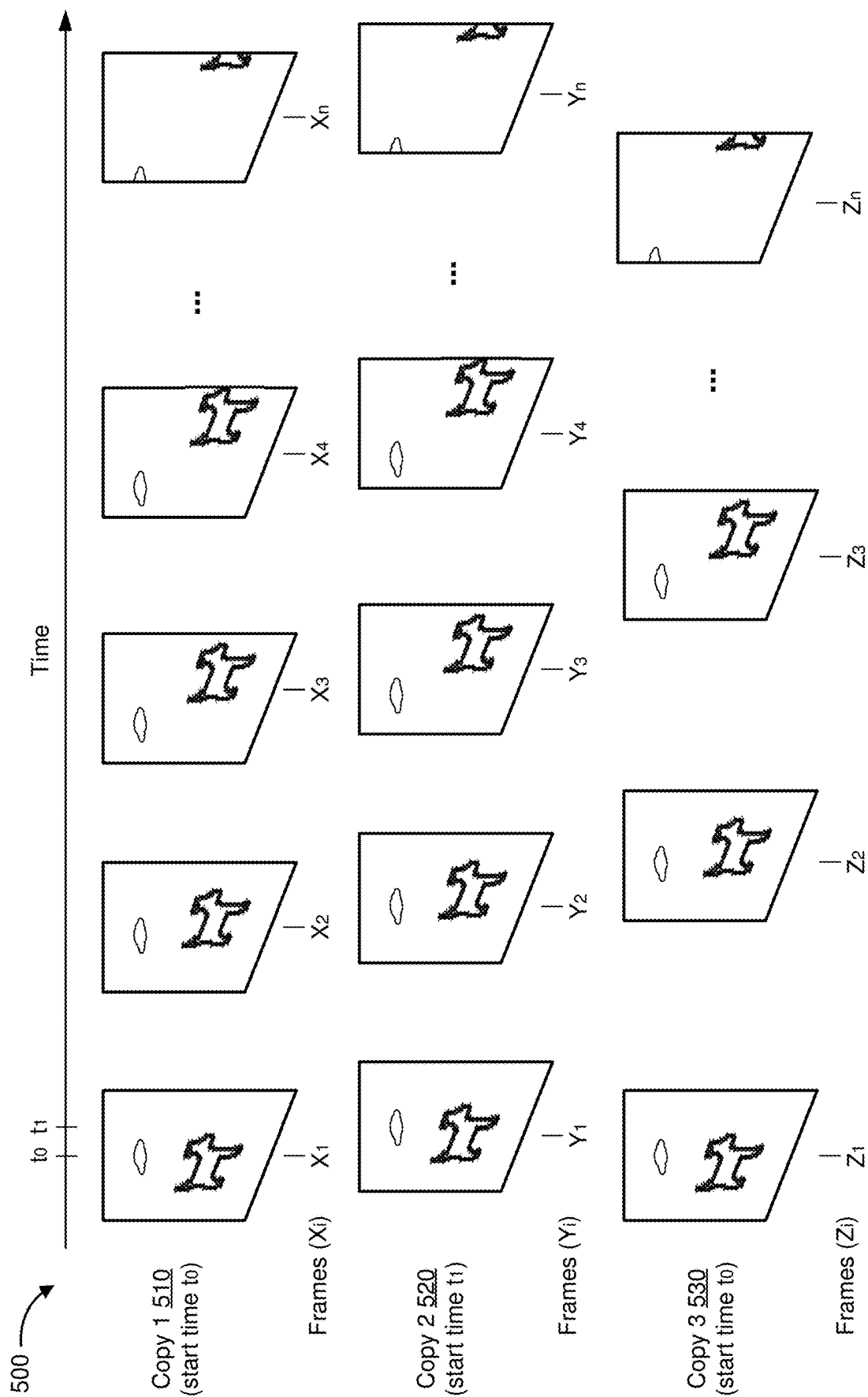


Figure 5

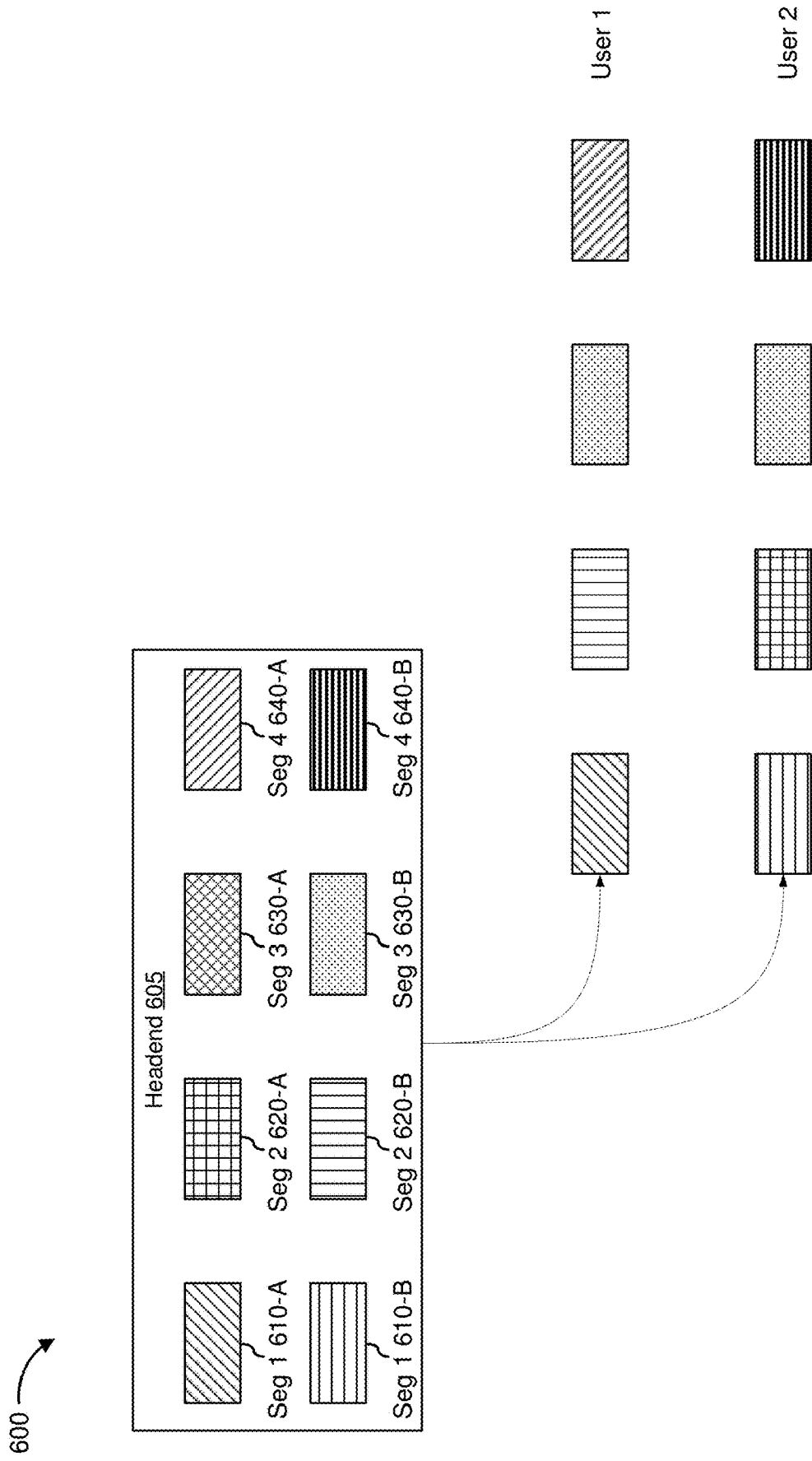


Figure 6

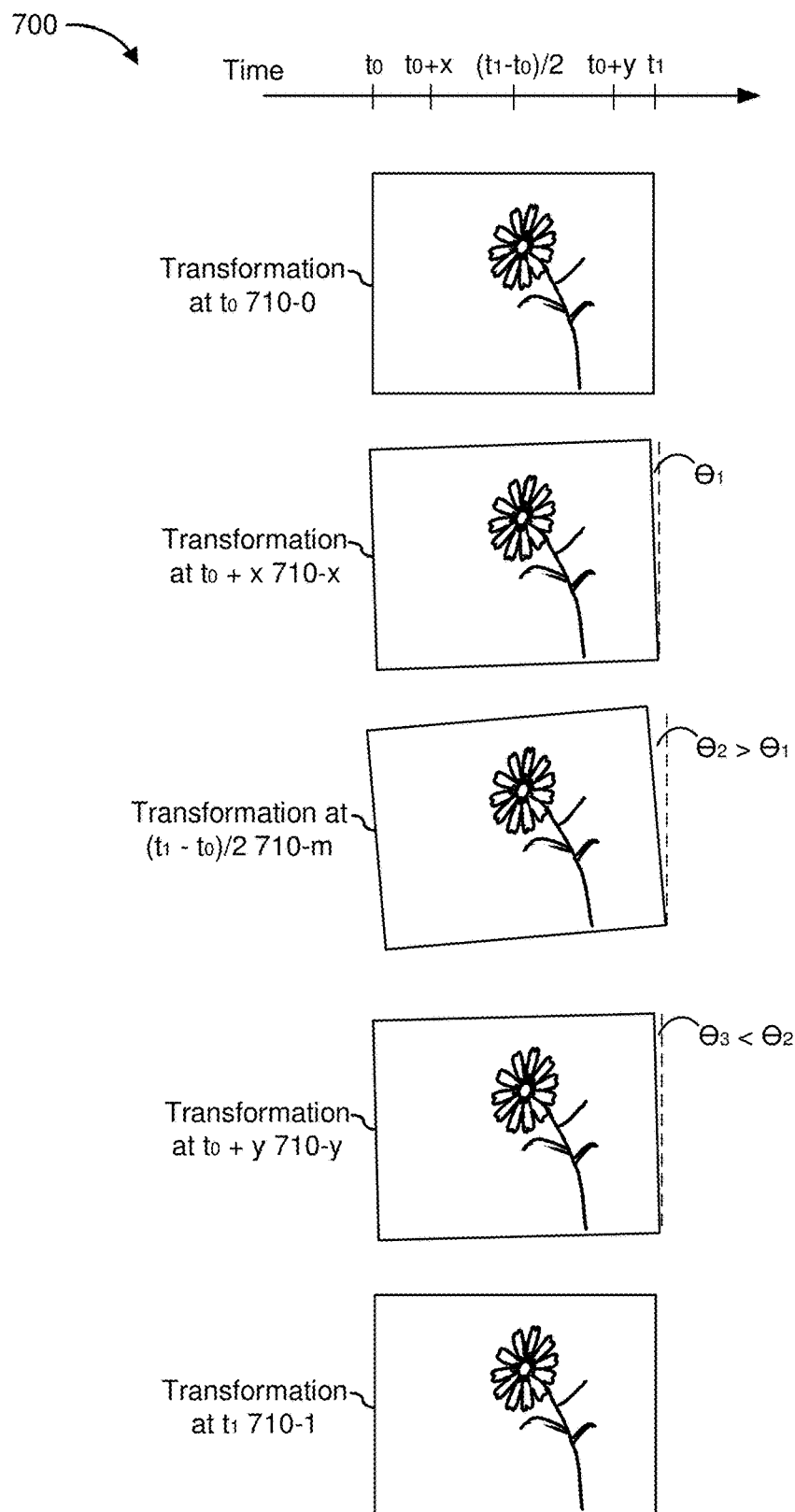


Figure 7

800

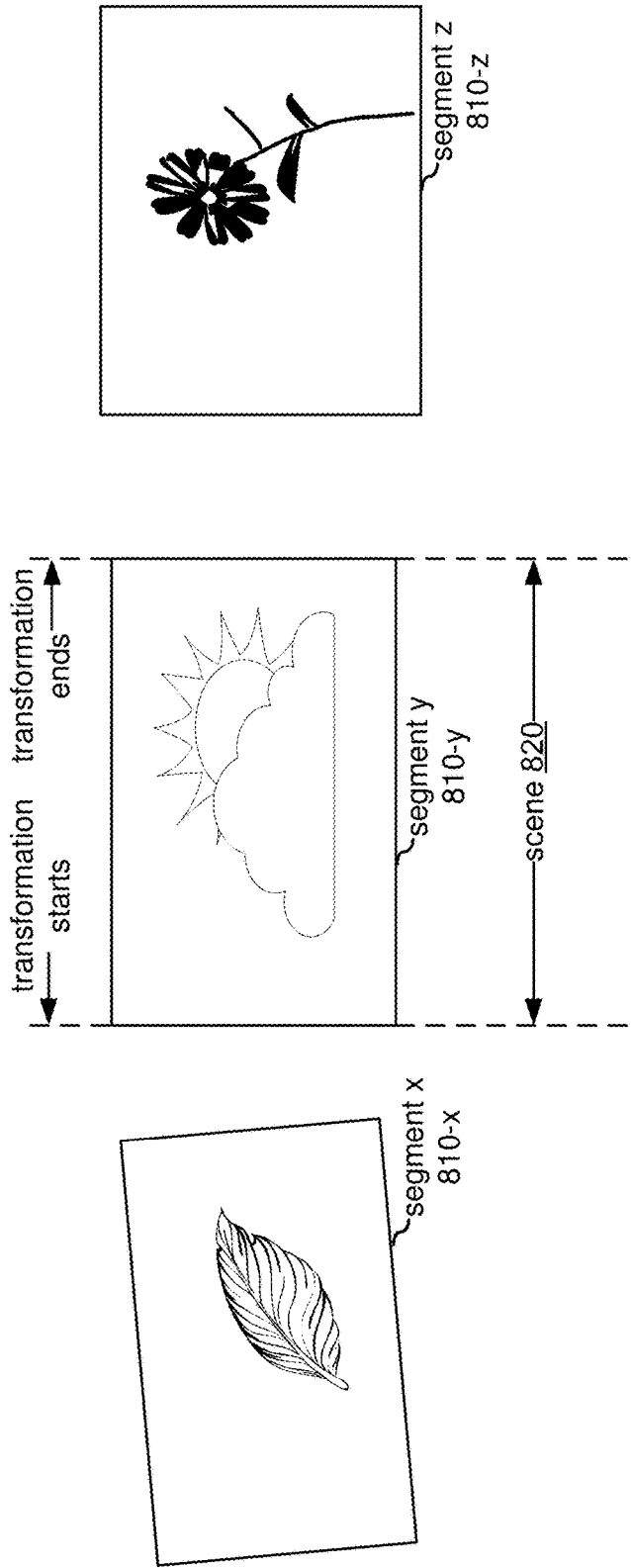
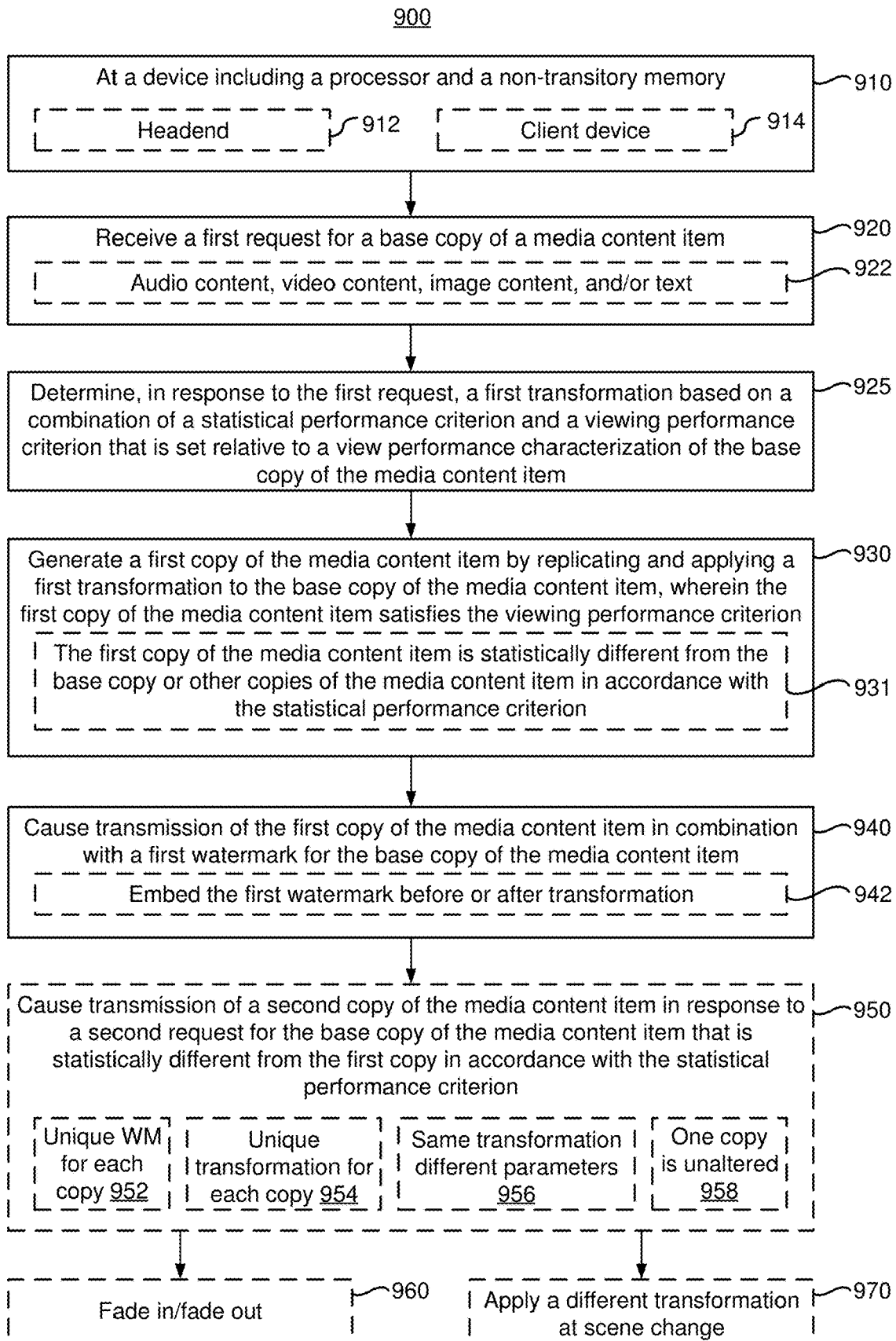


Figure 8



930

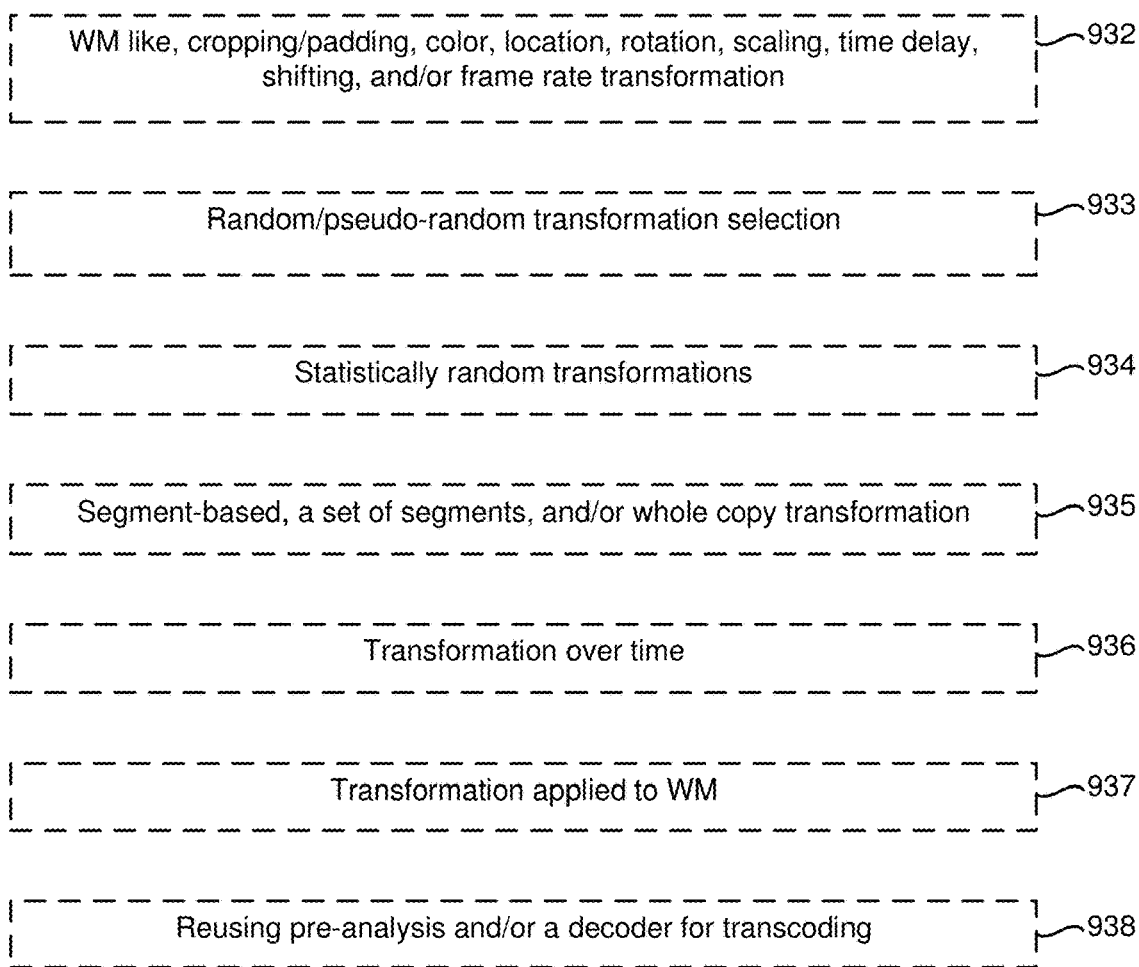


Figure 9B

METHODS AND SYSTEMS FOR REDUCING PIRACY OF MEDIA CONTENT

TECHNICAL FIELD

[0001] The present disclosure relates generally to reducing piracy of media content.

BACKGROUND

[0002] The process of hiding or embedding information in media content generally comes under the category of watermarking. Digital watermarking (sometimes also known as “forensic watermarking”) typically refers to the watermarking process where the embedded data identifies a media content item itself, as well as the content owner, operators, redistribution networks, and/or consumer devices associated with the distribution and consumption of the media content item. For example, in case a user leaks certain media content, his/her identity (ID) may be traced using embedded data. The watermark identifier is often encoded in an imperceptible manner while still allowing recovery of the watermark identifier even if the media content is modified.

[0003] One type of attack on watermarks is collusion attack. When the media content item is a video, collusion attacks typically work by gathering copies of the same video, each embedded with a unique watermark, colluding these copies, and generating a video that is a combination of the original copies. The combination may yield to the watermark corruption, thus making the watermark detection difficult or even impossible.

BRIEF DESCRIPTION OF THE DRAWINGS

[0004] So that the present disclosure can be understood by those of ordinary skill in the art, a more detailed description may be had by reference to aspects of some illustrative embodiments, some of which are shown in the accompanying drawings.

[0005] FIG. 1 is a block diagram of an exemplary collusion resistant system in accordance with some embodiments;

[0006] FIGS. 2A-2C are block diagrams of exemplary configurations of collusion resistant systems in accordance with some embodiments;

[0007] FIG. 3 is a block diagram illustrating one transformation technique in accordance with some embodiments;

[0008] FIG. 4 is a diagram illustrating various transformation techniques in accordance with some embodiments;

[0009] FIG. 5 is a diagram illustrating various other transformation techniques in accordance with some embodiments;

[0010] FIG. 6 is a diagram illustrating an exemplary collusion resistant system where transformations are applied in a headend in accordance with some embodiments;

[0011] FIG. 7 is a diagram illustrating transitioning between transformations with fading in and fading out in accordance with some embodiments;

[0012] FIG. 8 is a diagram illustrating aligning transitions of transformations with scene change in accordance with some embodiments; and

[0013] FIGS. 9A and 9B are flowcharts illustrating a collusion resistant method in accordance with some embodiments.

[0014] In accordance with common practice the various features illustrated in the drawings may not be drawn to scale. Accordingly, the dimensions of the various features

may be arbitrarily expanded or reduced for clarity. In addition, some of the drawings may not depict all of the components of a given system, method or device. Finally, like reference numerals may be used to denote like features throughout the specification and figures.

DESCRIPTION OF EXAMPLE EMBODIMENTS

[0015] Numerous details are described in order to provide a thorough understanding of the example embodiments shown in the drawings. However, the drawings merely show some example aspects of the present disclosure and are therefore not to be considered limiting. Those of ordinary skill in the art will appreciate that other effective aspects and/or variants do not include all of the specific details described herein. Moreover, well-known systems, methods, components, devices and circuits have not been described in exhaustive detail so as not to obscure more pertinent aspects of the example embodiments described herein.

Overview

[0016] Collusion resistant techniques described herein demotivate colluders in order to reduce media content piracy. Previously existing anti-collusion mechanisms focus on making more robust watermarks that can survive various collusion attacks. The collusion resistant techniques described herein degrade the quality of the media content item upon collusion and at the same time preserve the embedded watermarks. Moreover, the collusion resistant techniques described herein maintain the quality of the media content item for legitimate users, e.g., viewers not colluding. In some embodiments, a system in accordance with embodiments described herein generates multiple copies of a media content item and applies different transformations to each copy. As a result, when a colluder attempts to combine multiple copies of the video, the output is a video with downgraded quality that makes the pirated video not attractive. Such a solution is independent of the underlying watermark technique, robust against collusion techniques (including new collusion techniques that may be developed in the future), and yet at the same time does not downgrade the video quality for legitimate users and does not degrade the detectability of the original watermark.

[0017] In accordance with various embodiments, a method is performed at a device including a processor and a non-transitory memory. The method includes receiving a first request for a base copy of a media content item. The method further includes determining, in response to the first request, a first transformation based on a combination of a statistical performance criterion and a viewing performance criterion that is set relative to a view performance characterization of the base copy of the media content item. The method also includes generating a first copy of the media content item by replicating and applying a first transformation to the base copy of the media content item, where the first copy of the media content item satisfies the viewing performance criterion, and the first copy of the media content item is statistically different from the base copy of the media content item in accordance with the statistical performance criterion. The method additionally includes causing transmission of the first copy of the media content item in combination with a first watermark for the base copy of the media content item.

[0018] In accordance with various embodiments, a method is performed at a device including a processor and a non-

transitory memory. The method includes receiving a first request for a base copy of a media content item. The method also includes determining a first transformation based on a combination of a statistical performance criterion and a viewing performance criterion that is set relative to a view performance characterization of the base copy of the media content item. The method additionally includes generating a first copy of a media content item by replicating and applying the first transformation to a base copy of the media content item, where the first copy of the media content item satisfies the viewing performance criterion and the first copy of the media content item is statistically different from the base copy of the media content item and previously generated copies of the media content item generated by the device in accordance with the statistical performance criterion. The method further includes transmitting the first copy of the media content item in combination with a first watermark to one or more display devices.

Example Embodiments

[0019] As explained above, piracy of media content often attacks watermarks. One technique involves collusion attack, in which a few copies of the same media content item, each embedded with a different unique watermark, collude. In case of video content, a colluder often uses devices to synchronize video inputs. As one or more collusion attack devices compare the videos and align them, the difference is assumed to be the watermark. The colluder then generates a new video by modifying or manipulating the parts that are different between the video inputs and applying an alteration. Once the watermark is removed, piracy activities are undetected. Piracy of media content often applies various alteration techniques, e.g., at the pixel level, over parts of a frame, or over a sequence of frames, in order to corrupt the watermarks.

[0020] For example, at the pixel level, a colluder may, for a respective pixel, produce the average value of the respective pixel over the different copies, e.g., having 3 copies each with respective pixel values of 1, 2, and 6, and producing a pixel value of 3. In another example, a colluder may, for a respective pixel, produce the minority or majority value of the respective pixel over the different copies, e.g., having 3 copies each with respective pixel values of 1, 3, and 3, and producing a pixel value of 1 in case of a minority strategy or 3 in case of a majority strategy. Some piracy techniques alter parts of a frame. For example, using a tiling alteration technique, a colluder may split the screen to Y parts, where every part is taken from a different copy. Tiling results in having parts of different watermarks in the video but no complete watermark. Some other piracy techniques alter a sequence of frames. For example, by alternating frames, a colluder may create a new stream using X frames from a first copy and then Y frames from a second copy. For watermark techniques that spread the watermark over a sequence of frames, alternating frames results in either no watermark identifier or a false positive. Other possible collusion techniques, such as machine learning based collusions and/or a combination of collusion techniques implemented at a point in time and/or over time, can be used by a colluder to attack watermarks.

[0021] FIG. 1 illustrates an exemplary collusion resistant system 100 in accordance with some embodiments. In some embodiments, the collusion resistant system 100 obtains a base copy 105 of a media content item from a storage 101

storing media content. In some embodiments, a media content item stored in the storage 101 includes any suitable media data, for example, visual data, audio data, and/or text, etc. In some embodiments, the media content item, which can be live or from a pre-recorded source is stored in clear in the storage 101, thus allowing the base copy 105 of the media content item to be reused in different formats (e.g., different encoding formats for OTT content) and to be packaged dynamically by the packager/encoder. In some embodiments, the media content item can be encoded as MPEG-2, MPEG-4, MP3, AC-3, and/or any other suitable format.

[0022] In some embodiments, the collusion resistant system 100 includes a receiver 102 for receiving requests for the media content item, an embedder 110 for embedding watermarks and a transformer 120 for transforming the media content item 105. In some embodiments, the collusion resistant system 100 further includes a transmitter 130 to send copies of the transformed watermark embedded media content item, e.g., copy 1 of transformed watermark embedded media content item 140-1, copy 2 of transformed watermark embedded media content item 140-2, . . . , copy x of transformed watermark embedded media content item 140-x, and copy y of transformed watermark embedded media content item 140-y (collectively referred to as copies of the transformed watermark embedded media content item 140 or copies 140). Among the copies 140, some copies, e.g., copy x 140-x and copy y 140-y, are sent in response to requests from one or more legitimate users 190, e.g., one or more paid subscribers. In case a colluder 180 attempts collusion attacks by requesting multiple copies from the collusion resistant system 100, the collusion resistant system 100 sends at least two copies, e.g., copy 1 140-1 and copy 2 140-2, to the colluder 180.

[0023] In some embodiments, the embedder 110 embeds watermark(s) in the media content item 105. As explained above, the watermarking process typically involves modifying the media content item 105 to encode a watermark identifier in an imperceptible manner while still allowing recovery of the watermark identifier even if the media content item is modified. The watermark identifier can be used to identify operators, redistribution networks, and/or user devices. There are two major approaches to watermark integrations. One operates in a single step (also referred to as single-step watermarking) and the other has a preprocessing step (also referred to as two-step watermarking or A/B watermarking). Further, there are two major approaches for watermark deployments, i.e., a client-based approach and a server-based approach.

[0024] In single-step watermarking, each of the copies 140 has a uniquely embedded watermark. Single-step watermarking is typically used in client-based deployments. However, it can also be integrated within a video encoder, e.g., for postproduction or screeners, to embed a small number of unique watermark variants. In two-step watermarking, a bitstream is used to facilitate the watermarking. In other words, each element of the original media content item (e.g., each segment in a video stream) is a bit in the bitstream. The bitstream is encoded by using a small number, typically two, variants of the original media content item. In some embodiments, the variants are generated in a preprocessing step. The variants can be seen as pre-watermarked variations of the original media content item. In the second step of such embodiments, the bitstream is created by interchangeably

use portions, e.g., segments, chunks, fragments, or packets, of each media content item variation. As used herein, the terms “segment”, “fragment”, and “chunk” are used interchangeably, unless otherwise explicitly noted. Alternatively, in some other embodiments, the two variants are not generated during the preprocessing step. Instead, the original media content item is preprocessed to identify and instructions for variant generation are generated. In such alternative embodiments, both variant generation and the sequencing are performed in the second step. The preprocessing step is typically implemented in the video encoding headend. The second step may be implemented in other components of the media chain, e.g., at an encoder, a packager, a multiplexer, an over-the-top (OTT) origin server, an OTT playlist server, a content distribution network (CDN) server (central or edge), or a client device.

[0025] In some embodiments, the transformer 120 applies different transformations to the base copy 105 in response to different requests, i.e., copy 1 140-1 is different from copy 2 140-2. When the colluder 180 attempts to generate an output video by combining multiple copies, e.g., copy 1 140-1 and copy 2 140-2, the output is a media content item with downgraded quality 150, e.g., to the level that makes the pirated copy undesirable. On the other hand, for legitimate user(s) 190, the transformations by the transformer 120 do not downgrade the quality. In the exemplary collusion resistant system 100, copy x of transformed watermark embedded media content item 140-x and copy y of transformed watermark embedded media content item 140-y satisfy a viewing performance criterion that is set relative to a viewing performance characterization of the base copy of the media content item 105. As such, the transformations by the transformer 120 have a minor effect on the final media content item video for display to the legitimate user(s) 190.

[0026] As will be shown below with reference to FIGS. 2A-2C, the transformation of the base copy 105 by the transformer 120 is independent of the watermarking techniques utilized by the embedder 110, e.g., whether the watermarking is deployed at a headend or a client device and/or according to single-step watermarking or two-step watermarking. Further, the transformation of the base copy 105 by the transformer 120 can be applied before or after the watermark embedding. When the transformations are applied after embedding the watermarks, since the transformations have a minor effect on the final video, any watermark that meets the minimal requirements for watermark robustness is not expected to be harmed as a result of the transformations. As such, the solution described herein in accordance with various embodiments does not degrade the original watermark detectability.

[0027] As explained above, the assumption of any collusion attack is that multiple copies of the media content are aligned and the only difference is the watermark. The collusion resistant methods and systems disclosed herein cause misalignment among the multiple copies. As a result, any collusion techniques on the transformed media content would produce a poor viewing experience and hence preventing the pirates from generating income on a paid subscriber basis and avoiding being detected. For example, using a tiling alteration technique, the colluder 180 may split the screen to Y parts (e.g., 2 parts as shown in FIG. 1), where every part is taken from a different copy (e.g., the top part from copy 1 140-1 and the bottom part from copy 2 140-2). The combined copy 150 from the two parts produces a

distorted basketball image. A colluder might manually recover from some of the transformations and provide a decent viewing experience to the viewers. However, the manual approach would add significant cost and effort to the piracy operation. Furthermore, as will be described in detail below, in some embodiments, varying transformations over time would make such manual recovery nearly impossible and not worthwhile.

[0028] It should be noted that components are represented in the exemplary collusion resistant system 100 for illustrative purposes. The system 100 can include more, less, and/or different elements than shown in FIG. 1 in various embodiments. For example, the system 100 may also include one or more encoders, packagers, and/or multiplexers, an OTT server, an OTT playlist server, and/or one or more CDNs, etc. Further, elements can be divided, combined, and/or re-configured to perform the collusion resistant functions described herein. Additionally, each of the elements of system 100 includes appropriate hardware, software, and/or firmware to perform the operations attributed to the elements herein. Some examples of appropriate hardware, software, and/or firmware for certain elements will be provided further below. Operation(s) attributed to an element of system 100 herein should not be considered binding and in some embodiments, other element(s) in the system 100 may additionally or alternatively perform such operation(s).

[0029] Further, although a single media content storage 101, a single receiver 102, a single embedder 110, a single transformer 120, and a single transmitter 130 are illustrated in FIG. 1, the system 100 may include one or more media content storages 101, one or more receivers 102, one or more embedders 110, one or more transformers 120, and one or more transmitters 130. For the sake of simplicity, the subject matter will be described hereinafter for the most part with reference to a single media content storage 101, a single receiver 102, a single embedder 110, a single transformer 120, and a single transmitter 130. In some embodiments, one or more receivers 102, one or more embedders 110, one or more transformers 120, and one or more transmitters 130 are distributed on multiple devices. For example, a headend may have one or more receivers 102, one or more embedders 110, one or more transformers 120, and one or more transmitters 130 distributed across one or more servers. In another example, one or more client devices may have one or more receivers 102, one or more embedders 110, one or more transformers 120, and one or more transmitters 130 on one or more client devices.

[0030] FIGS. 2A-2C are block diagrams illustrating various configurations of collusion resistant systems 200A-200C in accordance with some embodiments. In FIG. 2A, a headend 210A includes a receiver 102A (e.g., the receiver 102, FIG. 1) for receiving one or more requests for one or more media content items from one or more client devices, an embedder 110A (e.g., the embedder 110, FIG. 1), a transformer 120A (e.g., the transformer 120, FIG. 1) for transforming media content items, and a transmitter 130A (e.g., the transmitter 130, FIG. 1) for sending copies of transformed watermark embedded media content items to client devices in accordance with some embodiments. In some embodiments, the embedder 110A applies single-step, two-step, or other watermarking techniques to embed watermarks in media content items from the headend 210A. In some embodiments, in headend based watermarking, e.g., the two-step watermarking, the headend 210A generates

multiple copies of each segment, e.g., two copies of each segment. The embedder **110A** then embeds a different watermark in each copy, e.g., 0 is embedded in segment *i* in the first copy, and 1 is embedded in segment *i* in the second copy. When the client device receives a sequence of segments, for each segment, the client device chooses a particular copy (with the corresponding watermark) based on the client device ID. For example, for segment *i*, if a particular bit of the client device ID is 0, the client device receives the first copy of segment *i* with 0 embedded. On the other hand, if the particular bit is 1, the client device receives the second copy with 1 embedded.

[0031] In some embodiments, as will be described in further detail below with reference to FIG. 6, in headend based transformation, the transformer **120A** determines which transformations to apply and generates multiple copies, e.g., two copies, for each segment and performs a different transformation on each of them. Each client device receives a sequence of transformed segments. For each segment, the client device chooses one of the copies, e.g., based on the ID and/or randomly, etc. Further, in some embodiments, the transformer **120A** aligns the start and the end of the transformation with the segment boundaries, e.g., by fading in and/or fading out at the beginning and the end of a segment, or by aligning the segments with a scene change. In some embodiments, at least one of the start or the end of the first transformation is within a respective segment boundary of the segment boundaries. For example, the start of the transformation can appear after the segment beginning, and the transformation may end before the segment ends. Moreover, in some embodiments, when the resolution of each copy is different, the media content is divided into other units, e.g. minutes, instead of into segments. In such embodiments, aligning the transformations includes aligning the transformations with the start and the end of each minute. As such, every copy includes one or more whole transformations (i.e., no partial transformations).

[0032] The transformer **120A** transforms each copy of the media content items before or after the embedder **110A** embeds the watermarks in accordance with various embodiments. In some embodiments, the transformer **120A** is integrated with an encoder so that the transformations are carried out in the headend **210A** before, after, or during the encoding process. As such, when the transmitter **130A** transmits multiple copies of the transformed watermark embedded media content items, these copies are misaligned and/or uncorrelated to resist collusion attacks.

[0033] In FIG. 2B, a client device **220B** includes a receiver **102B** for receiving one or more requests for one or more media content items and/or for receiving one or more media content items, an embedder **110B** (e.g., the embedder **110**, FIG. 1) for applying single-step or other watermarking techniques to embed watermarks in media content items on the client device **220B**, a transformer **120B** (e.g., the transformer **120**, FIG. 1) to transform the media content items, and one or more transmitters **130B** (e.g., the transmitter **130**, FIG. 1) in accordance with some embodiments. In some embodiments, the one or more transmitters **130B** sends one or more requests to a server (e.g., a headend) for a base copy of a media content item. In some embodiments, for client based watermarking, e.g., the single-step watermarking, multiple client devices receive the same media content, e.g., the same video, from the server. The embedder **110B** on each of the client devices embeds a watermark, which represents

its ID. In some embodiments, the embedder **110B** embeds the watermark in full in the payload in each frame. In some other embodiments, the client device embeds the watermark bit-by-bit.

[0034] In case the media content item is a video stream, in some embodiments, as will be described below with reference to FIG. 3, the transformer **120B** applies watermark like transformations, such as applying the single-step watermarking techniques in full on a single frame as a transformation of the frame. As such, in client based transformation, the client device **220B** receives the media content, e.g., a video, and the transformer **120B** performs a transformation or a set of transformations on the video. When the transformer **120B** applies a set of transformations, the transformations fade in and/or fade out or aligned with scene change for a smooth transition between the transformations. When the transmitter **130B** sends the frames to, for example, a rendering unit or module, each frame is transformed with an embedded watermark. As such, every client device **220B** has its own variants, and copies from different client devices are misaligned and/or uncorrelated to resist collusion attacks. In order to have a variety of transformations over time, in some embodiments, the transformer **120** on the client device **220B** also changes a transformation method occasionally, sometimes, and/or periodically. In the exemplary configuration shown in FIG. 2B, because the transformations are applied by the client device **220**, there is no server cost, scalability, or bandwidth constraint that limits the number of transformations for the media content item.

[0035] In some embodiments, in case there are multiple client devices, the transformer **120B** on each client device **220B** determines and chooses (e.g., randomly or pseudo-randomly) the transformations to be performed (or the parameters for the transformations to be used). Because the transformations are applied randomly, statistically one can assume not all client devices connected to the colluder would have the same transformation at least most of the time. In other words, the transformations applied to the multiple copies are statistically different or satisfy a statistical performance criterion. For example, the statistical performance criterion may specify that the probability of the client devices have the same transformation during a threshold amount of time is less than a threshold probability, e.g., $p < 0.01$, etc. As such, the normative case is that the multiple copies used in a collusion attack would be misaligned.

[0036] In FIG. 2C, a server **230C** includes a receiver **102C** (e.g., the receiver **102**, FIG. 1) for receiving multiple requests for a base copy of a media content item, an embedder **110C** (e.g., the embedder **110**, FIG. 1) for applying two-step or other watermarking techniques to embed watermarks in media content items from the server **230C** (e.g., a headend), a transformer **120C** (e.g., the transformer **120**, FIG. 1) to transform the media content items, and a transmitter **130C** (e.g., the transmitter **130**, FIG. 1) for sending copies of transformed watermark embedded media content items to client devices in accordance with some embodiments. In some embodiments, in order to perform the two-step watermarking techniques, the embedder **110C** includes a variants generator **232C** to perform the preprocessing step and a sequence generator **234C** to create the bitstream using variants from the variants generator **232C**. As such, when the transmitter **130C** transmits multiple

copies of the transformed watermark embedded media content items, these copies are misaligned and/or uncorrelated to resist collusion attacks.

[0037] It should be noted that though FIG. 2C illustrates a single server 230C, components and/or functions of the server 230C can be distributed to multiple devices. For example, while the variants generator 232C is typically implemented in the video encoding headend, the sequence generator 234 may be implemented in other components of the media chain, e.g., at an encoder, a packager, a multiplexer, an over-the-top (OTT) origin server, an OTT playlist server, a content distribution network (CDN) server (central or edge), or a client device.

[0038] As explained above, in order to perform a collusion attack, a colluder uses multiple copies (e.g., at least two) of aligned media content items. The collusion resistant techniques described herein modify the multiple copies of the media content items, such that the alignment becomes very difficult or impossible. This would either disable the collusion action or result in a corrupted output after the collusion. The collusion resistant techniques described herein include applying a variety of transformations in each copy of the media content item in accordance with some embodiments. Moreover, in some embodiments, the transformer applies different transformations over time in order to make it harder for the colluder to automatically identify the transformations. As will be described below with reference to FIGS. 3-8, the transformations are designed to have a minor influence on visibility. As such, there is no degradation of the viewing experience for legitimate users. More importantly, the transformations are not expected to affect the watermark quality and detectability regardless of what watermark technique is applied. On the other hand, when combining more than one copy of the media content item, the combination of two (or more) different transformations either fails the collusion attack or yields to a very low-quality and/or corrupted copy of the media content item.

[0039] FIG. 3 is a block diagram 300 illustrating one transformation technique in accordance with some embodiments. In some embodiments, one or more watermarking techniques, e.g., single-step or two-step watermarking, can be adopted for the purpose of transformations. In some embodiments, the transformer (e.g., the transformer 120B, FIG. 2B) obtains a base copy 310 of a media content item and chooses one or more watermarking techniques to embed one or more arbitrary identifiers in addition to the watermark identifier. This is useful especially in the single-step watermarking, where watermarks are applied in full within an element, e.g., a picture, a page, or a frame. As a result, multiple variants are embedded in multiple copies for multiple client devices, i.e., every client device has its own variants. In the example shown in FIG. 3, an arbitrary identifier is embedded in each of the transformed watermark embedded copies in addition to the embedded watermark, e.g., embedding ID₁ and WM₁ in copy 1 320-1, ID₂ and WM₂ in copy 2 320-2, ID_x and WM_x in copy x 320-x, ID_y and WM_y in copy y 320-y, . . . , etc.

[0040] In many cases, a watermark is imperceptible, e.g., a base copy and the watermark embedded copy are perceptually indistinguishable. In some cases, a watermark is perceptible if its presence in the watermark embedded copy is noticeable (e.g., digital on-screen graphics like a network logo, content bug, codes, opaque images). Even in such cases, on media content items, such as videos and images,

the watermarks are often made transparent and/or translucent for the convenience of consumers, since the watermarks may block a portion of the view. As such, adapting watermarking techniques to embed the arbitrary identifiers, the arbitrary identifiers embedded in the copies 320 would have the same perceptibility as the embedded watermarks. Accordingly, the transformation by embedding an arbitrary identifier in each of the copies 320 would not degrade the viewing. In other words, relative to a viewing performance characterization of the base copy 310, each of the transformed watermark embedded copies 320 satisfies a viewing performance criterion. For example, in case the embedded arbitrary identifier and the embedded watermark in copy x 320-x are imperceptible, the base copy 310 and the transformed watermark embedded copy are perceptually indistinguishable. Accordingly, the viewing performance characterization of copy x 320-x satisfies a viewing performance criterion, which is the same as the viewing performance characterization of the base copy 310.

[0041] On the other hand, a colluder may initiate a collusion attack by combining at least two copies of the media content item, e.g., combining copy 1 with embedded ID₁ and WM₁ 320-1 and copy 2 with embedded ID₂ and WM₂ using any collusion attack techniques. However, comparing and/or synchronizing copy 1 320-1 and copy 2 320-2 would fail, because in addition to the watermarks, the arbitrary identifiers are also different among different copies. Thus, colluding two different variants that are misaligned and/or uncorrelated would yield to a low-quality media content item 330.

[0042] FIG. 4 is a diagram 400 illustrating various transformation techniques in accordance with some embodiments. The various transformation techniques can be implemented by a transformer on a server (e.g., the transformer 120A in FIG. 2A or the transformer 120C in FIG. 2C) or a client device. Further, the transformation techniques are independent of watermarking techniques. As such, the collusion resistant techniques can be implemented in conjunction with any watermarking techniques.

[0043] In FIG. 4, a base copy 410 of an image is shown. In some embodiments, transformations of the base copy 410 include cropping and/or padding of pixels. For example, removing a threshold number of pixels from two opposite edges (same number of pixels) and padding with black transform the base copy 410 by introducing variants to the image. Because modifying several pixels within a threshold number is hardly noticeable by human vision, there is no degradation of viewing experience and the transformed image satisfies a viewing performance criterion. In some embodiments, the transformations further include randomly selecting one or more edges and/or the number of pixels to crop and/or pad to vary the degree of transformation by cropping and/or padding.

[0044] In some embodiments, transformations of the base copy 410 including adding small variations in brightness. In some embodiments, changes to the brightness characteristics are below one or more thresholds so that the perceived brightness changes are subtle. In some embodiments, the brightness transformations are gradual, e.g., fading in and fading out. In some embodiments, the brightness transformation changes over time or according to different random profiles. In some embodiments, transformations of the base copy 410 include randomly picking a different color depth within a threshold range or varying between an 8-bit or 10-bit depths.

[0045] In some embodiments, transformations of the base copy 410 include randomizing the location of a banner 420 inserted in the encoder with small variations, e.g., moving the location of the banner 420 within a threshold distance. For example, the transformer may randomly vary the location of a scoring banner in a sports game over time, e.g., occasionally, sometimes, and/or periodically changing the location within a threshold distance from the location of the scoring banner in the base copy 410.

[0046] In some embodiments, transformations of the base copy 410 include rotating the display by small (e.g., -1 , $+1$, or within a threshold degree) and/or random degrees 0. In some embodiments, transformations of the base copy 410 includes scaling the image. For example, in a fill mode, the transformer stretches the image slightly (e.g., within threshold dimensions) so each copy would have a slightly different image. In another example, in a fit mode, the transformer creates small variations in the pillar width (e.g., within a threshold width deviation), thus slightly stretching the display between the pillars. In some embodiments, the transformations of the base copy 410 include shifting, such as moving the display rectangle so pixel (0, 0) would be in slightly different locations for different variants. In some embodiments, the shifting of pixel (0, 0) is within a threshold distance from the location of pixel (0, 0) of the base copy 410.

[0047] Because the transformations in accordance with various embodiments described herein are slight, the changes are generally not perceptible by human eyes. As such, applying any of the transformation techniques or a combination of the various watermarking techniques illustrated in FIG. 4 would not degrade the viewing experience for legitimate users. On the other hand, when a colluder conducts a collusion attack by combining at least two copies of the media content item, because of the transformations, the combined copy would have low quality. As shown in FIG. 4, the image in the combined copy would appear to be blurry. In another example, as shown in FIG. 1, applying the tiling alteration technique, the basket image in the combined copy 150 has distortion and/or is disjointed along the tiling line.

[0048] FIG. 5 is a diagram 500 illustrating various video transformation techniques in accordance with some embodiments. In some embodiments, the transformations include introducing time delay to a video stream. For example, assuming a base copy of a video stream starts at time t_0 and includes a plurality of frames, e.g., $\text{Frames}(X_i) = \{X_1, X_2, X_3, X_4, \dots, X_n\}$. The transformer generates copy 1 510 by replicating the base copy and optionally applying one or more transformations as described above with reference to FIGS. 3 and 4.

[0049] In some embodiments, the transformer generates copy 2 520 by replicating the base copy, e.g., $\text{Frames}(Y_i) = \text{Frames}(X_i)$, and introducing a slight delay in time, e.g., starting at time t_1 that is within a threshold amount of time from t_0 . The slight delay is not perceptible in most cases. However, in case a colluder obtains copy 1 510 and copy 2 520 to attempt collusion attacks, the misalignment between copy 1 510 and copy 2 520 would result in a low-quality or corrupted video.

[0050] In some embodiments, the transformations include varying frame rates. For example, the transformer generates copy 3 530 by replicating the base copy, e.g., $\text{Frames}(Z_i) = \text{Frames}(X_i)$, and varying (e.g., lowering or increasing)

the frame rate within a threshold amount. The slight modification of the frame rate is not perceptible in most cases, e.g., lower the frame rate when the content has little motion. However, in case a colluder obtains copy 1 510 (and/or copy 2 520) and copy 3 530 to attempt collusion attacks, the misalignment between the copies would result in low quality or corrupted video.

[0051] FIG. 6 illustrates an exemplary collusion resistant system 600 where transformations are applied in a headend in accordance with some embodiments. In some embodiments, a headend 605 (e.g., the headend 210A, FIG. 2A, or the server 230C, FIG. 2C) includes a transformer (e.g., the transformer 120A, FIG. 2A, or the transformer 120C) for transforming media content items. In some embodiments, in case the media content item is a stream (e.g., a video stream or an audio stream), the headend 605 generates multiple variants of the stream, where each variant includes a different transformation (or the same transformation but with different parameters) as well as a different watermark (not shown in FIG. 6).

[0052] For example, in FIG. 6, segment 1 610-A and segment 1 610-B are two different variants of segment 1 (as indicated by different patterns of shading), e.g., by applying different transformations as described above with reference to FIGS. 4 and 5. Likewise, segment 2 620-A and segment 2 620-B are two different variants of segment 2; segment 3 630-A and segment 3 630-B are two different variants of segment 3, and segment 4 640-A and segment 4 640-B are two different variants of segment 4, etc. When two users, e.g., user 1 and user 2, request the media content item, the headend 605 streams variants of the segments to the users in response to the requests, e.g., streaming segment 1 610-A, segment 2 620-B, segment 3 630-B, and segment 4 640-A to user 1 and streaming segment 1 610-B, segment 2 620-A, segment 3 630-B, and segment 4 640-B to user 2, etc. As such, the streams received by user 1 and user 2 are uncorrelated to resist collusion attacks.

[0053] In some embodiments, the headend 605 changes the transformations as time goes in order to have a variety of transformations over time. For example, following a segment-based approach (or a set of segments based approach), where the headend 605 applies a different transformation to each segment (or to each set of the segments). For each segment, the user receives one transformed copy of the segment. The two streams generated in response to two requests may include certain portions that are the same. However, along the timeline, the streams are different. As shown in FIG. 6, both user 1 and user 2 receive the same transformed segment 3. However, the stream received by user 1 is different from the stream received by user 2 over time. In some embodiments, each variant of each segment also includes a different watermark. In such embodiments, each stream to a user includes both a unique watermark sequence and one or more unique transformations.

[0054] The transformation techniques shown in FIGS. 4-6 are illustrative and not exhaustive. Other transformation techniques that satisfy the viewing performance criterion may be applied in place of or in combination with the techniques shown in FIGS. 4-6. Further, the collusion resistant systems described herein in accordance with various embodiments can apply one transformation technique or a combination of transformation techniques. Additionally, the transformation techniques and/or the parameters for a

respective transformation technique may vary over time to resist collusion attacks without degrading the viewing experience for legitimate users.

[0055] In some embodiments, the transformations shown in FIGS. 4-6 are performed in the uncompressed domain. Some transformations do not incur an additional cost from re-encoding, such as the time delay transformation as shown in FIG. 5. In FIG. 5, to generate and transmit copy 2 520 with the start time t_1 , the transformer replicates the base copy frames, e.g., $\text{Frames}(Y_i) = \text{Frames}(X_i)$, and delays the transmission of copy 2 520 without performing any re-encoding. To perform other transformations, multiple encoders (e.g., one for each copy) may be deployed in the headend and/or the client device. In some embodiments, to avoid duplication of encoders, the transformer enables transformations on compressed media content items or other cost-effective methods for encoders.

[0056] For example, the transformations as shown in FIG. 4 can share the pre-analysis part of the media content item and the decoder in case of transcoding to reduce the amount of calculation. In particular, for the cropping and/or padding transformation, the transformer can re-use the motion vectors and the mode decisions of the non-transcoded versions of a media content item. In another example, for the banner location transformation, the transformer can re-use the motion vectors and the mode decision for macroblocks that do not refer to regions associated with the banner. In cases where motion vectors are re-calculated (e.g., the brightness transformation and/or the rotation transformation, etc.), the transformer can re-calculate the initial motion vectors based on the motion vectors of the non-transformed version (e.g., the base copy 410, FIG. 4) and re-use the mode decision of the non-transformed version as the initial mode. The transformer can then use the initial motion vector and mode to find a better motion vector and mode in a reduced search space for the transformed copy. Following this approach, the encoder effort is significantly reduced. In cases where the transformer does have a separate encoder in connection with the transformations (e.g., the frame rate transformation as shown in FIG. 5), the separate encoder can still share the pre-analysis part of the video encoder and decoder in case of transcoding for cost savings.

[0057] FIG. 7 is a diagram 700 illustrating one embodiment of transitioning between transformations with fading in and fading out in accordance with some embodiments. As explained above, in some embodiments, the collusion resistant system described herein can change the chosen transformation and can have multiple transformations. To ensure the viewing experience for legitimate users, in some embodiments, the transformations are applied gradually to avoid low-frequency effects. In some embodiments, for headend-based transformations, the transformer (e.g., the transformer 120A, FIG. 2A, or the transformer 120C, FIG. 2C), applies the transformation on chunk boundaries. In some embodiments, the transformer (e.g., the transformer 220B, FIG. 2B) starts at the chunk boundary with some transformation (or without transformation). The transformer then gradually increases the magnitude of the transformation to the middle point of the segment, and gradually decreases the magnitude before moving to another transformation in a gradual manner.

[0058] For example, in FIG. 7, time t_0 and time t_1 mark the chunk boundaries. At the beginning of the chunk boundary, the segment has no transformation at time t_0 710-0. The

transformer gradually increases the magnitude of the transformation. In the case of rotation transformation, as shown in FIG. 7, the rotation degree increases from κ degree to Θ_1 at time t_0+x 710-x. The increase of the rotation degree continues until the middle point of the segment, e.g., the transformation at time $(t_1-t_0)/2$ 710-m peaks with the rotation degree Θ_2 greater than Θ_1 at time t_0+x . Once the middle point of the segment has passed, the transformer gradually reduces the magnitude of the transformation, e.g., the rotation degree decreases from Θ_2 at time $(t_1-t_0)/2$ 710-m to Θ_3 at time t_0+y 710-y. At the end of the chunk boundary, the segment has no transformation at time t_1 710-1.

[0059] FIG. 8 is a diagram 800 illustrating aligning transitions of transformations with scene changes in accordance with some embodiments. In some embodiments, in order to provide a viewing experience to legitimate users that satisfies a viewing performance criterion, the transformer aligns the start and the end of the transformation with scene changes. The idea is based on the observation that human eyes are not sensitive enough to distinguish between two different transformations if an image is significantly changed.

[0060] For headend-based transformations, transformations are often applied on segment-based resolution (or applied on a set of segments). As such, in some embodiments, the segment boundaries are aligned with scene changes to provide a smooth transition. For client-based transformations, in some embodiments, the client device receives a signal when a scene change occurs (possibly an ahead-of-time notification). On the scene change, the transformer (e.g., the transformer 120B, FIG. 2B) chooses a different transformation.

[0061] For example, boundaries of segment y 810-y are aligned with the start and the end of a scene 820. Prior to the scene 820, the transformer, for example, applies the rotation transformation to segment x 810-x. At the start of the scene 820, the transformer chooses a different transformation, e.g., not rotating or scaling, and the chosen transformation ends at the end of the scene 820. Because the image in segment y 810-y has significantly changed relative to segment x 810-x, the transformation changes at the beginning of the scene 820 are indistinguishable. Likewise, at the end of the scene 820, when the transformer chooses a different transformation, e.g., scaling of segment z 810-z, the transformation changes are indistinguishable.

[0062] FIGS. 9A and 9B are flowcharts illustrating a collusion resistant method 900 in accordance with some embodiments. As represented by block 910, in some embodiments, the method 900 is performed at a device including a processor, e.g., for executing instructions for the request handling, the watermark embedding, the applications of transformations, and the transmission handling, and a non-transitory memory for storing the instructions and/or storing media content items. In some embodiments, the device includes at least one of a headend or a client device.

[0063] In other words, as represented by block 912, in some embodiments, the device is a headend (e.g., the headend 210A, FIG. 2A), which includes one or more receivers (e.g., the receiver 102A, FIG. 2A) for receiving requests, one or more embedders (e.g., the embedder 110A, FIG. 2A) for embedding watermarks, one or more transformers (e.g., the transformer 120A, FIG. 2A) for applying transformations to copies of media content items, and one or more transmitters (e.g., the transmitter 120A, FIG. 2A). In

some other embodiments, as represented by block **914**, the device is one or more client devices (e.g., the client device **220B**, FIG. 2B), which includes one or more receivers (e.g., the receiver **102B**), one or more transmitters (e.g., the transmitter **130B**, FIG. 2B) for transmitting requests and sending transformed watermark embedded media content items for display, and one or more transformers (e.g., the transformer **120B**, FIG. 2B) for transforming media content items. In some embodiments, the device includes a plurality of devices to perform the method **900**. Further, one or more components of the embedder and the transformer can be implemented in other components of the media chain. For example, though FIG. 2C illustrates the sequence generator **234C** being located on the server **230C**, the sequence generator **234C** can be implemented, e.g., at the encoding, packaging, or multiplexing, at an OTT origin server, on an OTT playlist server, on a CDN server (central or edge), or in a client device. As such, the collusion resistant method **900** is flexible in headend-based and/or client-based transformation and independent of the watermark techniques, whether the transformation and/or the watermarking are performed at the headend or the client device.

[0064] Briefly, the method **900** includes receiving a first request for a base copy of a media content item; determining, in response to the first request, a first transformation based on a combination of a statistical performance criterion and a viewing performance criterion that is set relative to a view performance characterization of the base copy of the media content item; generating a first copy of the media content item by replicating and applying a first transformation to the base copy of the media content item, wherein the first copy of the media content item satisfies the viewing performance criterion, and the first copy of the media content item is statistically different from the base copy (and/or other copies) of the media content item in accordance with the statistical performance criterion; and causing transmission of the first copy of the media content item in combination with a first watermark for the base copy of the media content item.

[0065] To that end, as represented by block **920**, the method **900** begins with receiving a first request for a base copy of a media content item. In some embodiments, the method **900** further includes receiving a second request for the base copy of a media content item. For example, in FIG. 1, one or more client devices send multiple requests, e.g., request 1, request 2, . . . , request x, and request y, etc., for a base copy of a media content item. In some embodiments, as represented by block **922**, the media content item includes one or more of audio content, video content, image content, and text. As such, the collusion resistant method **900** is flexible and adaptive to tackle collusion attacks on any type of media content.

[0066] Regardless of the type of media content item requested, the method **900** continues, as represented by block **925**, with the transformer determining, in response to the first request, a first transformation based on a combination of a statistical performance criterion and a viewing performance criterion that is set relative to a view performance characterization of the base copy of the media content item. Various methods of generating transformed copies of the media content item are explained below and shown in FIG. 9B. Using any of the various transformation methods described herein in accordance with embodiments, as shown in FIG. 1, each of the transformed watermark

embedded copies **140** satisfy a viewing performance criterion set relative to a viewing performance characterization of the base copy, e.g., the viewer perceived differences between the transformed watermark embedded copies **140** and the non-transformed media content item are negligible. Moreover, by applying the first transformation, the transformations applied by the transformer satisfy a statistical performance criterion, e.g., the probability of the transformations (including the first transformation) are the same during a threshold amount of time is less than a threshold probability.

[0067] As represented by block **930**, the method **900** continues, with the transformer generating a first copy of the media content item by replicating and applying a first transformation to the base copy of the media content item, wherein the first copy of the media content item satisfies the viewing performance criterion. For example, in FIG. 1, the transformer **120** generates multiple copies **140** of the base copy **105** in response to the requests by replicating the base copy (e.g., a non-transformed copy of the media content item) and applying transformations. Various methods of generating transformed copies of the media content item are explained below and shown in FIG. 9B. Using any of the various transformation methods described herein in accordance with embodiments, in FIG. 1, each of the transformed watermark embedded copies **140** satisfy a viewing performance criterion set relative to a viewing performance characterization of the base copy, e.g., the viewer perceived differences between the transformed watermark embedded copies **140** and the non-transformed media content item are negligible. For instance, in FIG. 4, when shifting pixel (0, 0) within a threshold distance, modifying a small number of pixels during cropping and/or padding, slightly brightening or dimming, varying the color depth for a small amount, rotating within a threshold angle range, and/or moving the location of the banner **420** within a threshold distance, etc., the slight transformation is hardly noticeable. Thus, the transformed copy satisfies the viewing performance criterion set relative to the viewing performance characterization of the base copy. As such, the collusion resistant method **900** preserves the quality of the media content item for legitimate users.

[0068] Further, as represented by block **931**, in some embodiments, the first copy of the media content item is statistically different from the base copy and other copies of the media content item in accordance with the statistical performance criterion. For example, although one copy may have no transformation at one time, the copy is statistically different from the base copy of the media content item in accordance with the statistical performance criterion, because other copies may include transformations. In some embodiments, in accordance with the statistical performance criterion, each of the transformed watermark embedded copies **140** is not only statistically different from the base copy of the media content item, but also statistically different from previously generated copies of the media content item generated by the device. For example, a headend may generate multiple copies of the media content item and determine different transformations to apply (including applying no transformations to one or more copies), so that the probability of these multiple copies have the same transformation within a threshold time window is less than a threshold probability. In another example, multiple client devices may each determine and apply transformation(s) to

the copy it receives. Statistically, the copies among these multiple client devices have different transformations at most of the time.

[0069] Still referred to FIG. 9A, the method **900** continues, as represented by block **940**, with the device causing transmission (e.g., by the transmitter) of the first copy of the media content item in combination with a watermark in response to the first request for the base copy of the media content item. For example, in FIG. 1, copy 1 of transformed watermark embedded media content item **140-1** is transmitted in response to request 1. In some embodiments, as represented by block **942**, the first watermark is combined with the first copy of the media content item before or after applying the first transformation to the base copy of the media content item. Because the transformations can be applied before or after watermark embedding, the collusion resistant method **900** is independent of the watermarking technique, e.g., whether embedding the watermark at the headend as shown in FIG. 2A or the client device as shown in FIG. 2B. Further, since the transformations have a minor effect on the media content item, applying transformation after embedding watermarks would not harm any watermark that meets the minimal requirements for watermark robustness. As such, the collusion resistant method **900** does not degrade the detectability of the original watermark.

[0070] In some embodiments, the method **900** includes receiving a second request for the base copy of the media content item, and causing transmission of a second copy of the media content item in response to the second request for the base copy of the media content item, wherein the second copy of the media content item is statistically different from the first copy in accordance with the statistical performance criterion. For example, in FIG. 1, copy 2 of transformed watermark embedded media content item **140-2** is transmitted in response to request 2. In another example, in FIG. 3, multiple client devices may each determine and apply transformation(s) to the copy it receives, e.g., one client device determines ID 1 to be embedded in copy 1 **320-1** and another client device determines ID 2 to be embedded in copy 2 **320-2**, etc. Statistically, the copies **320** among the multiple client devices have different transformations at most of the time and the probability that the copies **320** have the same transformation at least most of the time is less than a threshold probability.

[0071] In some embodiments, as represented by block **952**, the second copy of the media content item includes a second watermark that is different from the watermark, so that each copy includes a unique watermark to facilitate piracy detection. For example, in FIG. 3, copy 1 **320-1** can include watermark 1 that is different from watermark 2 embedded in copy 2 **320-2**. In such embodiments, as represented by block **954**, the second copy of media content item is generated by replicating and applying a second transformation that is different from the first transformation to the base copy of the media content item, where the second copy of the media content item satisfies the viewing performance criterion in accordance with some embodiments. For example, in FIG. 4, in addition to having different watermarks, the transformation to the copy sent to user 1 is different from the transformation to the copy sent to user 2. As such, with each copy having both a unique watermark sequence and unique transformation, the multiple copies are misaligned to be robust against collusion attacks.

[0072] To ensure the uniqueness of each copy, the transformation is not limited to different types of transformations. As represented by block **956**, in some embodiments, the first transformation and the second transformation are of the same type with different transformation parameters. For example, when applying the time delay transformation as shown in FIG. 5, the time delay may vary for different copies. The varying time delay for different copies makes each copy unique and causes misalignment among multiple copies to resist against collusion attacks. In some embodiments, the different transformation parameters include, but are not limited to, different number of pixels cropped and/or padded, different brightness, different color depth, different banner locations, different rotation angles, different scaling aspect in fit or fill mode, different time delay, different coordinates, and/or different frame rate, etc.

[0073] In some embodiments, as represented by block **958**, the second copy is generated by replicating the base copy and embedding a second watermark. In such embodiments, when one copy is transformed and the other is non-transformed or unaltered, the two copies are misaligned to be collusion resistant. In particular, for client-based transformations, the client device may start with no transformation. When starting with no transformation, the viewing performance characterization of the copy is the same as the base copy.

[0074] In some embodiments, as represented by block **960**, to not downgrade the viewing experience for legitimate users, the method **900** provides smooth transition between transformations by applying fading in and fading out techniques. In some embodiments, generating the first copy of the media content item by replicating and applying the first transformation to the base copy of the media content item includes identifying a middle point of a portion in the first copy of the media content item, increasing a magnitude of the first transformation until the middle point of the portion, and decreasing the magnitude of the first transformation past the middle point.

[0075] For example, as explained above in connection with FIG. 7, upon identifying a middle point of a segment duration at time $(t_1 - t_0)/2$. The transformer first gradually increases a magnitude of the rotation transformation (e.g., increasing Θ) until the middle point of the segment duration. The transformer then gradually decreases the magnitude of the rotation transformation (e.g., decreasing Θ) past the middle point. Because of the gradual transformation, the transformed video satisfies the viewing performance criterion set relative to the viewing performance characterization of the base copy.

[0076] In some embodiments, as represented by block **970**, to not downgrade the viewing experience for legitimate users, the method **900** provides smooth transition between transformations by aligning transitions with a scene change. In some embodiments, the method **900** further includes identifying a scene change in the first copy of the media content item, and selecting a different transformation to apply to the first copy of the media content item in accordance with the scene change. For example, as shown in FIG. 8, because the images in different scenes are significantly different, the human eye is not sensitive enough to distinguish the switching from the rotation transformation and then to the scaling transformation along the segment boundaries. As such, the transformed video satisfies the viewing

performance criterion set relative to the viewing performance characterization of the base copy.

[0077] Turning to FIG. 9B, FIG. 9 illustrates various transformation methods for generating a transformed watermark embedded copy represented by block 930 (FIG. 9A). In some embodiments, as represented by block 932, the transformations can include, but are not limited to, a watermark like transformation (FIG. 3), cropping and/or padding (FIG. 4), brightness, color, banner location (FIG. 4), rotation (FIG. 4), scaling (FIG. 4), time delay (FIG. 5), shifting (FIG. 4), and/or frame rate (FIG. 5) transformation.

[0078] For the watermark-like transformation, in some embodiments, applying a first transformation includes embedding an arbitrary identifier in the base copy to generate the first copy of the media content item. For example, as shown in FIG. 3, each of the copies 320 has an embedded arbitrary identifier in addition to a unique watermark. Such watermark-like transformation is useful especially in single-step watermarking but is independent of the watermark technique, e.g., the transformation can be applying in conjunction with the single-step watermark or two-step watermarking.

[0079] For other types of transformations, in some embodiments, the first transformation includes modifying one or more of elements, brightness, color, element location, display angle, scaling aspect, time delay, coordinates, and frame rate associated with transformations of the first copy of the media content item. For example, as shown in FIG. 4, the transformer can apply one transformation selected from cropping and/or padding, brightness, color, banner location, rotation, scaling, time delay, shifting, and/or frame rate transformation. In some embodiments, the transformer can apply more than one type of transformation or apply varying parameters of one type of transformation. Some transformation methods may be more effective against certain types of collusion attacks. Multiple transformations can be applied so that the collusion resistant method 900 is robust, flexible, and adaptive to tackle any collusion attacks.

[0080] In some embodiments, as represented by block 933, the first transformation is selected randomly or pseudo-randomly. For example, when applying transformation in the client device as shown in FIG. 2B, each client chooses, e.g., randomly or pseudo-randomly, the transformation to be performed (or the parameters for the transformations to be used). As such, the collusion resistant system has the ability to dynamically and un-predictably change the transformation strategy against various collusion attacks.

[0081] In some embodiments, as represented by block 934, the method 900 further includes receiving multiple requests for the base copy of the media content item, and generating multiple copies of the media content item by replicating and applying multiple transformations, wherein the multiple transformations applied to the multiple copies of the media content item are statistically different. Because the transformations are applied randomly, statistically, the client devices connected to the colluder would not have the same transformation at least most of the time. As such, the normative case is that the multiple copies used in a collusion attack will be misaligned, and the collusion resistant method 900 is robust against any collusion techniques.

[0082] In some embodiments, as represented by block 935, generating the first copy of the media content item by replicating and applying the first transformation to the base copy of the media content item includes replicating a

subportion of the media content item to generate the first copy and applying the first transformation to the subportion of the media content item. For example, in case the media content item is a video, the subportion can be a segment, a set of segment, or the video stream. Further in such embodiments, as represented by block 936, in some embodiments, applying the first transformation to the base copy includes for each subportion of the media content item (e.g., a segment of a video), applying a different transformation. In other words, the collusion resistant method 900 is flexible to apply transformations to a set of segments that can be one segment, e.g., a segment-based solution, more than one segment, or the video stream.

[0083] In one example, the set of segments can include the video stream, such as applying a fill mode transformation to modify stretch slightly so each client would have a slightly different image dimension to resist collusion attacks. In the example shown in FIG. 6, the transformation is applied segment-by-segment, so that each segment has a different transformation, and for each segment, the user receives one copy of this segment. In order to have a variety of transformations over time, the transformer changes transformations as time goes. As a result, the two copies of the video stream as shown in FIG. 6 may not be different in every point of time (e.g., segment 3 variant 630-B is in both copies), but along the timeline and over time, the two copies of the video stream are different to resist collusion attacks.

[0084] In some embodiments, as represented by block 937, applying the first transformation to the base copy of the media content item includes applying the first transformation to the watermark, wherein the first transformation of the watermark satisfies a watermark robustness criterion. Since the transformations have a minor effect on the media content item, any watermark that meets the minimal requirements for watermark robustness would not be harmed as a result of the transformations. Thus, the collusion resistant method 900 does not degrade the watermark detectability.

[0085] In some embodiments, for cost savings, as represented by block 938, applying the first transformation to the base copy of the media content item includes reusing at least one of pre-analysis (e.g., motion vectors and mode decisions) or a decoder associated with the base copy of the media content item for transcoding the first copy.

[0086] While various aspects of implementations within the scope of the appended claims are described above, it should be apparent that the various features of implementations described above may be embodied in a wide variety of forms and that any specific structure and/or function described above is merely illustrative. Based on the present disclosure one skilled in the art should appreciate that an aspect described herein may be implemented independently of any other aspects and that two or more of these aspects may be combined in various ways. For example, an apparatus may be implemented and/or a method may be practiced using any number of the aspects set forth herein. In addition, such an apparatus may be implemented and/or such a method may be practiced using other structure and/or functionality in addition to or other than one or more of the aspects set forth herein.

[0087] It will also be understood that, although the terms “first,” “second,” etc. may be used herein to describe various elements, these elements should not be limited by these terms. These terms are only used to distinguish one element from another. For example, a first device could be termed a

second device, and, similarly, a second device could be termed a first device, which changing the meaning of the description, so long as all occurrences of the “first device” are renamed consistently and all occurrences of the “second device” are renamed consistently. The first device and the second device are both devices, but they are not the same device.

[0088] The terminology used herein is for the purpose of describing particular embodiments only and is not intended to be limiting of the claims. As used in the description of the embodiments and the appended claims, the singular forms “a”, “an”, and “the” are intended to include the plural forms as well, unless the context clearly indicates otherwise. It will also be understood that the term “and/or” as used herein refers to and encompasses any and all possible combinations of one or more of the associated listed items. It will be further understood that the terms “comprises” and/or “comprising,” when used in this specification, specify the presence of stated features, integers, steps, operations, elements, and/or components, but do not preclude the presence or addition of one or more other features, integers, steps, operations, elements, components, and/or groups thereof.

[0089] As used herein, the term “if” may be construed to mean “when” or “upon” or “in response to determining” or “in accordance with a determination” or “in response to detecting”, that a stated condition precedent is true, depending on the context. Similarly, the phrase “if it is determined [that a stated condition precedent is true]” or “if [a stated condition precedent is true]” or “when [a stated condition precedent is true]” may be construed to mean “upon determining” or “in response to determining” or “in accordance with a determination” or “upon detecting” or “in response to detecting” that the stated condition precedent is true, depending on the context.

1. A method comprising:
 - at a device including a processor and a non-transitory memory;
 - receiving a first request for a base copy of a media content item;
 - determining, in response to the first request, a first transformation based on a combination of a statistical performance criterion and a viewing performance criterion that is set relative to a view performance characterization of the base copy of the media content item;
 - generating a first copy of the media content item by replicating and applying a first transformation to the base copy of the media content item, wherein the first copy of the media content item satisfies the viewing performance criterion, and the first copy of the media content item is statistically different from the base copy of the media content item in accordance with the statistical performance criterion; and
 - causing transmission of the first copy of the media content item in combination with a first watermark for the base copy of the media content item.
2. The method of claim 1, wherein the device includes at least one of a headend or a client device.
3. The method of claim 1, wherein the media content item includes one or more of audio content, video content, image content, and text.
4. The method of claim 1, wherein the first watermark is combined with the first copy of the media content item before or after applying the first transformation to the base copy of the media content item.

5. The method of claim 1, further comprising:
 - receiving a second request for the base copy of the media content item; and
 - causing transmission of a second copy of the media content item in response to the second request for the base copy of the media content item, wherein the second copy of the media content item is statistically different from the first copy in accordance with the statistical performance criterion.
6. The method of claim 5, wherein the second copy of the media content item includes a second watermark different from the first watermark.
7. The method of claim 5, wherein the second copy of media content item is generated by:
 - replicating and applying a second transformation, different from the first transformation to the base copy of the media content item, wherein the second copy of the media content item satisfies the viewing performance criterion.
8. The method of claim 7, wherein the first transformation and the second transformation are of same type with different transformation parameters.
9. The method of claim 1, further comprising aligning a start and an end of the first transformation with segment boundaries.
10. The method of claim 9, wherein at least one of the start or the end of the first transformation is within a respective segment boundary of the segment boundaries.
11. The method of claim 1, wherein generating the first copy of the media content item by replicating and applying the first transformation to the base copy of the media content item includes:
 - identifying a middle point of a portion in the first copy of the media content item;
 - increasing a magnitude of the first transformation until the middle point of the portion; and
 - decreasing the magnitude of the first transformation past the middle point.
12. The method of claim 1, further comprising:
 - identifying a scene change in the first copy of the media content item; and
 - selecting a different transformation to apply to the first copy of the media content item in accordance with the scene change.
13. The method of claim 1, wherein applying the first transformation includes:
 - embedding an arbitrary identifier in the base copy to generate the first copy of the media content item.
14. The method of claim 1, wherein applying the first transformation includes modifying one or more of elements, brightness, color, element location, display angle, scaling aspect, time delay, coordinates, and frame rate associated with the first copy of the media content item.
15. The method of claim 1, further comprising:
 - receiving multiple requests for the base copy of the media content item; and
 - generating multiple copies of the media content item by replicating and applying multiple transformations, wherein the multiple transformations applied to the multiple copies of the media content item are statistically different.

16. The method of claim **1**, wherein generating the first copy of the media content item by replicating and applying the first transformation to the base copy of the media content item includes:

- replicating a subportion of the media content item to generate the first copy; and
- applying the first transformation to the subportion of the media content item.

17. The method of claim **16**, wherein applying the first transformation to the base copy includes:

- for each of the subportion, applying a different transformation.

18. The method of claim **1**, wherein applying the first transformation to the base copy of the media content item includes applying the first transformation to the first watermark, wherein the first transformation of the first watermark satisfies a watermark robustness criterion.

19. The method of claim **1**, wherein applying the first transformation to the base copy of the media content item includes:

- reusing at least one of pre-analysis or a decoder associated the base copy of the media content item for transcoding of the first copy.

20. A system comprising:

- one or more receivers operable to receive a first request from a first client device for a base copy of a media content item;

- one or more transformers operable to:

- determine, in response to the first request, a first transformation based on a combination of a statistical performance criterion and a viewing performance criterion that is set relative to a view performance characterization of the base copy of the media content item;

- generate a first copy of the media content item by replicating and applying a first transformation to the base copy of the media content item, wherein the first copy of the media content item satisfies the viewing performance criterion, and the first copy of the media content item is statistically different from the base copy of the media content item in accordance with the statistical performance criterion; and

- one or more transmitters to cause transmission of the first copy of the media content item in combination with a watermark for the base copy of the media content item.

21. A method comprising:

- at a device including a processor and a non-transitory memory:

- receiving a first request for a base copy of a media content item;

- determining a first transformation based on a combination of a statistical performance criterion and a viewing performance criterion that is set relative to a view performance characterization of the base copy of the media content item;

- generating a first copy of a media content item by replicating and applying the first transformation to the base copy of the media content item, wherein the first copy of the media content item satisfies the viewing performance criterion, and the first copy of the media content item is statistically different from the base copy of the media content item and previously generated copies of

- the media content item generated by the device in accordance with the statistical performance criterion; and

- transmitting the first copy of the media content item in combination with a first watermark to one or more display devices.

22. The method of claim **21**, wherein generating the first copy of the media content item by replicating and applying the first transformation to the base copy of the media content item includes:

- identifying a middle point of a portion in the first copy of the media content item;

- increasing a magnitude of the first transformation until the middle point of the portion; and

- decreasing the magnitude of the first transformation past the middle point.

23. The method of claim **21**, further comprising:

- identifying a scene change in the first copy of the media content item; and

- selecting a different transformation to apply to the first copy of the media content item in accordance with the scene change.

24. The method of claim **21**, wherein applying the first transformation includes:

- embedding an arbitrary identifier in the base copy to generate the first copy of the media content item.

25. The method of claim **21**, wherein applying the first transformation includes modifying one or more of elements, brightness, color, element location, display angle, scaling aspect, time delay, coordinates, and frame rate associated with the first copy of the media content item.

26. The method of claim **21**, wherein generating the first copy of the media content item by replicating and applying the first transformation to the base copy of the media content item includes:

- replicating a subportion of the media content item to generate the first copy; and

- applying the first transformation to the subportion of the media content item.

27. The method of claim **26**, wherein applying the first transformation to the base copy includes:

- for each of the subportion, applying a different transformation.

28. A system comprising:

- one or more receivers operable to receive a first request for a base copy of a media content item;

- one or more transformers operable to:

- determine a first transformation based on a combination of a statistical performance criterion and a viewing performance criterion that is set relative to a view performance characterization of the base copy of the media content item; and

- generate a first copy of a media content item by replicating and applying the first transformation to the base copy of the media content item, wherein the first copy of the media content item satisfies the viewing performance criterion, and the first copy of the media content item is statistically different from the base copy of the media content item and previously generated copies of the media content item generated by the device in accordance with the statistical performance criterion; and

one or more transmitters operable to transmit the first copy of the media content item in combination with a first watermark to one or more display devices.

29. The system of claim **28**, wherein:

the one or more receivers are further operable to receive a second request for the base copy of the media content item; and

the one or more transmitters are further operable to transmit a second copy of the media content item in response to the second request for the base copy of the media content item to the one or more display devices, wherein the second copy of the media content item is statistically different from the first copy in accordance with the statistical performance criterion.

30. The system of claim **29**, wherein the one or more transformers are further operable to generate the second copy of media content item by:

replicating and applying a second transformation, different from the first transformation to the base copy of the media content item, wherein the second copy of the media content item satisfies the viewing performance criterion.

* * * * *



(12) 发明专利申请

(10) 申请公布号 CN 117278762 A

(43) 申请公布日 2023. 12. 22

(21) 申请号 202311220942.6

(22) 申请日 2023.09.20

(71) 申请人 深圳市艾宝科技有限公司

地址 518000 广东省深圳市南山区桃源街
道学苑大道1001号南山智园A5栋9楼

(72) 发明人 李贤哲 陈擎宇 张海

(74) 专利代理机构 北京法筑知识产权代理有限公司 16100

专利代理师 张雨红

(51) Int. Cl.

H04N 19/467 (2014.01)

H04N 19/42 (2014.01)

H04N 19/88 (2014.01)

H04N 21/2347 (2011.01)

H04N 21/8358 (2011.01)

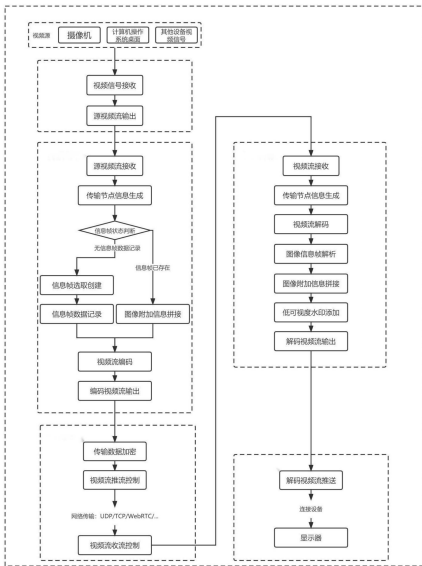
权利要求书3页 说明书17页 附图4页

(54) 发明名称

一种安全可溯源视频编解码系统

(57) 摘要

本申请公开了一种安全可溯源视频编解码系统,涉及特定行业下的视频溯源处理,包括:视频采集模块,采集源视频流并插入水印信息和元数据;视频编码模块,在源视频流中插入编码节点信息并进行压缩编码;传输模块,对编码视频流进行封装和传输;解码模块,对视频流进行解码并插入解码节点信息和水印;显示模块,显示输出的视频流,并从中解析节点信息和水印,生成视频流的传输轨迹;其中,显示模块包含溯源分析单元,追踪图像的传播链路。针对现有技术中存在的安全保密漏洞和视频溯源效率低的问题,本发明通过插入可追溯的信息,并在显示端收集和分析,发生视频非法录制和传播事件后,对视频内容来源进行快速追溯,提升视频处理传输安全保密性。



1. 一种安全可溯源视频编解码系统,包括:

视频采集模块,用于采集源视频流,并在源视频流中插入水印信息和元数据;

视频编码模块,用于接收源视频流,并在源视频流中插入编码节点信息,并对插入编码节点信息后的视频流进行压缩编码;

传输模块,用于对视频编码模块输出的编码视频流进行封装和传输;

解码模块,用于接收传输模块输出的视频流,对接收到的视频流进行解码,在解码后的视频流中插入解码节点信息和水印;

显示模块,用于显示解码模块输出的视频流,并从中解析节点信息和水印,生成视频流的传输轨迹;

其中,显示模块包含溯源分析单元,用于获取解码后的视频流,解析视频流中的节点信息和水印,并追踪图像的传播链路。

2. 根据权利要求1所述的安全可溯源视频编解码系统,其特征在于:

视频采集模块包括:

视频源单元,用于获取视频输出设备产生的源视频信号;

编码标准选择单元,用于根据源视频信号的格式和系统需求,选择相应的视频编码标准,所述视频编码标准包含H.264和H.265;

视频采集单元,用于按照选择的视频编码标准对源视频信号进行采集,并生成符合视频编码标准要求的源视频流;

图像增强单元,用于对源视频流进行图像质量增强处理,图像增强处理包含去噪和校色;

水印信息单元,用于在源视频流的预设位置插入水印信息,所述水印信息包含视频采集时间、地点和设备编号信息;

元数据生成单元,用于按预设间隔提取源视频流中的帧图像作为关键帧,并通过对关键帧的分析,生成描述视频内容的元数据,所述元数据中包含目标检测和场景识别信息;

采集控制单元,用于根据网络状态和用户指令,控制所述视频采集模块的工作模式;

其中,所述视频输出设备包含摄像头和计算机操作系统桌面。

3. 根据权利要求1所述的安全可溯源视频编解码系统,其特征在于:

视频编码模块包括:

信息插入单元,用于接收源视频流,并在源视频流中插入节点信息,生成带有节点信息的视频帧序列,所述节点信息包含编码时间、地点和编码器编号信息;

编码单元,用于接收带有节点信息的视频帧序列,根据网络状态选择H.264或H.265视频编码标准,调整编码器的码率控制参数和变块大小参数,对视频帧序列进行压缩编码,输出压缩编码后的视频帧序列;

加密单元,用于接收压缩后的视频帧序列,利用对称或非对称加密算法对所述视频帧序列进行加密,输出加密后的视频帧序列;

封装单元,用于接收加密后的视频帧序列,并封装成符合视频传输协议标准的输出码流。

4. 根据权利要求1所述的安全可溯源视频编解码系统,其特征在于:

传输模块包括:

加密单元,用于接收视频编码模块输出的编码视频帧序列,利用加密算法对其进行混淆处理,输出混淆后的视频帧序列;

封装单元,连接加密单元,用于将混淆后的视频帧序列封装成传输数据包;

传输模式选择单元,用于根据视频流的实时性需求、网络带宽和稳定性,选择传输模式,输出控制信号,所述传输模式包含UDP、TCP和WebRTC;

传输单元,用于接收传输数据包,根据传输模式选择单元的控制信号,通过网络将传输数据包发送给视频解码模块;

控制单元,用于根据网络状态输出控制信号,调节数据包的生成和传输参数。

5. 根据权利要求1所述的安全可溯源视频编解码系统,其特征在于:

解码模块包括:

解码单元,用于接收视频传输模块输出的视频流,根据自有编解码协议对接收到的视频流进行解码,恢复视频帧序列;

信息读取单元,用于在解码视频帧序列时,读取并解析其中的节点信息;

信息插入单元,用于接收解码后的视频帧序列,并在视频帧序列中插入解码节点信息;

水印嵌入单元,用于在解码恢复的视频帧中选定低可视频帧,嵌入包含各节点信息的水印;

输出单元,用于输出嵌入解码节点信息和水印的视频帧序列。

6. 根据权利要求5所述的安全可溯源视频编解码系统,其特征在于:

水印嵌入单元包括:

水印生成子单元,用于根据节点信息,生成包含校验码的水印;

视频分析子单元,用于分析视频帧的空域信息和时域信息,获取各视频帧的重要性;

嵌入控制子单元,用于根据获取的各视频帧的重要性,选择重要性低于阈值的视频帧作为水印嵌入的目标帧;

水印嵌入子单元,用于在目标帧中嵌入生成的水印;

其中,所述重要性低于阈值的视频帧为图像复杂度低于阈值的视频帧。

7. 根据权利要求5所述的安全可溯源视频编解码系统,其特征在于:

信息插入单元包括:

编码信息补充子单元,用于根据解码时间和设备编号,补充完整的编码解码节点信息;

信息隐写子单元,用于使用隐写技术,将补充后的节点信息引出插入视频帧中。

8. 根据权利要求7所述的安全可溯源视频编解码系统,其特征在于:

隐写技术为数字水印技术。

9. 根据权利要求1所述的安全可溯源视频编解码系统,其特征在于:

显示模块包括:

视频显示单元,用于接收解码后的视频帧序列,在显示设备上渲染和显示;

元数据解析单元,用于检测并解析视频帧序列中的信息帧,获取其中包含的各编码和解码节点的时间、地点和设备编号信息;

水印解析单元,用于检测视频帧序列中的水印帧,解析其中的节点信息水印;

溯源分析单元,用于将元数据解析单元和水印解析单元得到的节点信息进行融合,形成完整的源至目的节点的视频传输轨迹;

存储单元,用于存储视频传输轨迹信息,供溯源查询。

10.根据权利要求9所述的安全可溯源视频编解码系统,其特征在于:

溯源分析单元包括:

信息关联子单元,用于根据时间戳和设备编号信息,将解析得到的各节点信息进行关联;

路由生成子单元,用于根据关联后的节点信息,生成完整的视频流传输路由。

一种安全可溯源视频编解码系统

技术领域

[0001] 本发明涉及视频溯源处理领域,特别涉及一种安全可溯源视频编解码系统。

背景技术

[0002] 近年来,移动互联网、5G网络等技术的进步,促进了视频拍摄设备的普及和网络带宽的增大,导致网络视频内容呈现爆炸式增长。大量视频内容的产生给其来源追溯带来了挑战。当前主流的H.264等视频编码标准主要侧重压缩编码效率,而不具备溯源的能力。其压缩后的视频信息无法直接用于内容溯源。

[0003] 在相关技术中,比如中国专利文献CN115914677A中提供了一种智能视频安全联网装置、服务器,装置包括:视频网关模块、智能分析模块,视频网关模块包括:视频接入单元,用于接收前端图像采集模块采集得到的视频码流数据;视频转码单元,用于对视频码流数据进行解码,得到原始数据流;以及对视频码流数据进行转码处理,得到视频转码数据;智能分析模块用于对原始数据流进行智能分析,得到图像编码和结构化信息;视频转码单元还用于对视频转码数据、图像编码和结构化信息进行签名加密;视频网关模块还包括:视频转发单元用于对签名加密文件进行打包。但是该方案至少存在如下技术问题:智能分析模块仅对原始视频流进行智能分析,获取图像编码和结构化信息。这些信息相对有限,不足以对视频内容进行全面描述,如缺乏场景信息、目标检测信息等。元数据不全面会降低后续溯源的效率;由于元数据和水印信息插入的不连续,会导致生成的溯源信息存在断层,不利于重建完整的源至终端的传播链路。

[0004] 基于此,有必要研究一种安全可溯源视频编解码系统,以提高视频编码传输的安全保密性,在发生视频复发录制和传播事件后,快速完成视频溯源。

发明内容

[0005] 1.要解决的技术问题

[0006] 针对现有技术中存在的视频溯源效率低的问题,本发明提供了一种安全可溯源视频编解码系统,通过在视频处理的每个环节插入可追溯的信息,并在显示端收集和分析这些信息,可有效地对视频内容来源进行追溯,提高视频溯源效率,进而提高安全保密性。

[0007] 2.技术方案

[0008] 本发明的目的通过以下技术方案实现。

[0009] 本说明书实施例提供一种安全可溯源视频编解码系统,包括:视频采集模块,用于采集源视频流,并在源视频流中插入水印信息和元数据;视频编码模块,用于接收源视频流,并在源视频流中插入编码节点信息,并对插入编码节点信息后的视频流进行压缩编码;传输模块,用于对视频编码模块输出的编码视频流进行封装和传输;解码模块,用于接收传输模块输出的视频流,对接收到的视频流进行解码,在解码后的视频流中插入解码节点信息和水印;显示模块,用于显示解码模块输出的视频流,并从中解析节点信息和水印,生成视频流的传输轨迹;其中,显示模块包含溯源分析单元,用于获取解码后的视频流,解析视

频流中的节点信息和水印,并追踪图像的传播链路。

[0010] 进一步地,视频采集模块包括:视频源单元,用于获取视频输出设备产生的源视频信号;编码标准选择单元,用于根据源视频信号的格式和系统需求,选择相应的视频编码标准,视频编码标准包含H.264和H.265;视频采集单元,用于按照选择的视频编码标准对源视频信号进行采集,并生成符合视频编码标准要求的源视频流;图像增强单元,用于对源视频流进行图像质量增强处理,图像增强处理包含去噪和校色;水印信息单元,用于在源视频流的预设位置插入水印信息,水印信息包含视频采集时间、地点和设备编号信息;元数据生成单元,用于按预设间隔提取源视频流中的帧图像作为关键帧,并通过对关键帧的分析,生成描述视频内容的元数据,元数据中包含目标检测和场景识别信息;采集控制单元,用于根据网络状态和用户指令,控制视频采集模块的工作模式;其中,视频输出设备包含摄像头和计算机操作系统桌面。

[0011] 进一步地,视频编码模块包括:信息插入单元,用于接收源视频流,并在源视频流中插入节点信息,生成带有节点信息的视频帧序列,节点信息包含编码时间、地点和编码器编号信息;编码单元,用于接收带有节点信息的视频帧序列,根据网络状态选择H.264或H.265视频编码标准,调整编码器的码率控制参数和变块大小参数,对视频帧序列进行压缩编码,输出压缩编码后的视频帧序列;加密单元,用于接收压缩后的视频帧序列,利用对称或非对称加密算法对视频帧序列进行加密,输出加密后的视频帧序列;封装单元,用于接收加密后的视频帧序列,并封装成符合视频传输协议标准的输出码流。

[0012] 进一步地,传输模块包括:加密单元,用于接收视频编码模块输出的编码视频帧序列,利用加密算法对其进行混淆处理,输出混淆后的视频帧序列;封装单元,连接加密单元,用于将混淆后的视频帧序列封装成传输数据包;传输模式选择单元,用于根据视频流的实时性需求、网络带宽和稳定性,选择传输模式,输出控制信号,传输模式包含UDP、TCP和WebRTC;传输单元,用于接收传输数据包,根据传输模式选择单元的控制信号,通过网络将传输数据包发送给视频解码模块;控制单元,用于根据网络状态输出控制信号,调节数据包的生成和传输参数。

[0013] 进一步的,解码模块包括:解码单元,用于接收视频传输模块输出的视频流,根据自有编解码协议对接收到的视频流进行解码,恢复视频帧序列;信息读取单元,用于在解码视频帧序列时,读取并解析其中的节点信息;信息插入单元,用于接收解码后的视频帧序列,并在视频帧序列中插入解码节点信息;水印嵌入单元,用于在解码恢复的视频帧中选定低可视频帧,嵌入包含各节点信息的水印;输出单元,用于输出嵌入解码节点信息和水印的视频帧序列。

[0014] 进一步地,水印嵌入单元包括:水印生成子单元,用于根据节点信息,生成包含校验码的水印;视频分析子单元,用于分析视频帧的空域信息和时域信息,获取各视频帧的重要性;嵌入控制子单元,用于根据获取的各视频帧的重要性,选择重要性低于阈值的视频帧作为水印嵌入的目标帧;水印嵌入子单元,用于在目标帧中嵌入生成的水印;其中,重要性低于阈值的视频帧为图像复杂度低于阈值的视频帧。

[0015] 进一步地,信息插入单元包括:编码信息补充子单元,用于根据解码时间和设备编号,补充完整的编码解码节点信息;信息隐写子单元,用于使用隐写技术,将补充后的节点信息引出插入视频帧中。

[0016] 进一步的,隐写技术为数字水印技术。

[0017] 进一步的,显示模块包括:视频显示单元,用于接收解码后的视频帧序列,在显示设备上渲染和显示;元数据解析单元,用于检测并解析视频帧序列中的信息帧,获取其中包含的各编码和解码节点的时间、地点和设备编号信息;水印解析单元,用于检测视频帧序列中的水印帧,解析其中的节点信息水印;溯源分析单元,用于将元数据解析单元和水印解析单元得到的节点信息进行融合,形成完整地源至目的节点的视频传输轨迹;存储单元,用于存储视频传输轨迹信息,供溯源查询。

[0018] 进一步地,溯源分析单元包括:信息关联子单元,用于根据时间戳和设备编号信息,将解析得到的各节点信息进行关联;路由生成子单元,用于根据关联后的节点信息,生成完整的视频流传输路由。

[0019] 3.有益效果

[0020] 相比于现有技术,本发明的优点在于:

[0021] (1)通过在视频处理的各个环节(采集、编码、传输、解码、显示)插入可追溯的元数据、水印和节点信息,并在显示端收集和分析这些信息,实现了对视频内容整个处理流程的溯源,达到了细粒度的溯源效果;

[0022] (2)通过数字水印等信息隐写技术,增加了视频中包含的溯源信息量,同时使用加密等手段提高了信息的安全性。多种隐写技术的综合运用增强了溯源信息的容错性;

[0023] (3)通过解析视频处理各节点插入的信息,并进行信息关联和路由生成,能够构建视频内容从源头到终端的完整传播链路,实现了对视频传播全过程的溯源,有助于追踪内容的传播轨迹和确定责任方。

[0024] 综上,本方案实现了针对视频处理流程的精细化溯源设计,能够有效提升视频内容溯源的效率,增强溯源结果的完整性,提高视频溯源效率,进而提高安全保密性。

附图说明

[0025] 本说明书将以示例性实施例的方式进一步描述,这些示例性实施例将通过附图进行详细描述。这些实施例并非限制性的,在这些实施例中,相同的编号表示相同的结构,其中:

[0026] 图1是根据本说明书的一些实施例所示的安全可溯源视频编解码系统的示例性模块图;

[0027] 图2是根据本说明书的一些实施例所示的安全可溯源视频编解码系统的技术路线示意图;

[0028] 图3是根据本说明书一些实施例所示的溯源分析过程的示例性流程图;

[0029] 图4是根据本说明书一些实施例所示的水印嵌入过程的示例性流程图。

具体实施方式

[0030] 名词解释

[0031] 源视频流是指系统输入的未经编码和压缩的原始数字视频序列。它是可溯源视频处理流程的第一个环节;源视频包含连续的视频帧画面和音频数据,以一定的编码格式(如YUV)存储,还包含视频的采集参数信息;源视频流作为整个系统的输入,其质量直接影响后

续视频处理的效果。源视频流越清晰流畅,后续压缩编码越高效;系统通过对源视频流进行编码压缩、加密、封装等处理,生成可传输和分发的编码视频流;源视频中还需要插入可追溯信息如时间戳、地点、设备号等元数据,以及数字水印,为传输视频的溯源提供基础数据;所以源视频流是可溯源系统的信息源头,其采集质量决定整个可追溯分析的效果。综上所述,源视频流是安全可溯源视频编解码系统的原始视频输入,其质量直接影响整个系统的处理和溯源效果。它包含基本视频画面信息,以及可追溯所需的基础数据。

[0032] 水印信息指的是在视频采集和编码环节,使用数字水印技术向源视频帧中植入的信息。这些信息以不可见的形式隐藏在视频信号中;水印信息可以包含视频的采集时间、地点、设备编号、用户账号等可追溯相关数据;在视频传输和解码过程中,水印信息难以被破坏或删除,可经受视频处理和压缩;在对传输视频进行溯源时,可以通过水印信息解析技术提取水印数据,以辅助确认视频的采集来源和传输过程;水印信息与视频元数据不同,元数据以明文形式直接嵌入,而水印信息以隐藏形式嵌入,更难被非法修改;水印技术提高了视频可追溯性和防伪性。水印信息与元数据共同用于溯源分析。综上,在可溯源视频系统中,水印信息利用数字水印技术嵌入视频,作为元数据的有效补充,增强视频溯源的可靠性和抗修改能力。

[0033] 元数据是附加在视频流中的辅助数据,包含对视频内容的描述、标注和补充信息;元数据可以包括视频的采集时间、地点、设备信息、事件信息等,以文本格式或结构化格式嵌入视频流中;元数据在视频采集时产生,在视频处理各阶段可以插入或更新,也可以通过分析视频内容生成;元数据存在于视频编码流中,不影响视频解码播放,但可以被相关解析程序读取;在对视频传输进行溯源时,可以从视频流提取和解析元数据,以获取视频传输的时间地点等信息;元数据直接以明文形式嵌入,易于非法修改,所以需要与数字水印信息相结合,提高可靠性;元数据使视频编码流具有自描述性和可扩展性,丰富了视频信息。综上,元数据为可溯源视频系统提供辅助描述,是视频溯源的重要信息来源之一,与水印信息共同提高溯源分析的效果。

[0034] 编码节点包含对视频编码处理的时间、地点、操作员、设备等信息的数字记录;编码节点在视频编码环节时生成,以元数据或数字水印形式插入编码视频流;一个完整的编码系统可能包含多个编码节点,如编码器识别码、编码标准、编码时间、编码地点等;解码环节也会同时生成解码节点,标识解码时间和地点信息;在对视频传输进行溯源分析时,可以从编码视频流中解析提取编码节点信息;通过把编码节点和解码节点进行关联融合,可以完成从编码到解码的完整视频传输轨迹的溯源;编码节点是视频可溯源的基础信息源,其准确性和可靠性决定了溯源的效果。综上,编码节点记录视频编码处理的关键信息,和解码节点共同构成视频传输链路可追溯的基础数据。它是实现视频溯源的核心信息之一。

[0035] 压缩编码是利用视频压缩算法对源视频进行编码处理,剔除冗余信息,降低视频数据量;常用的视频压缩编码标准有H.264、H.265等,可以显著减少视频数据量;压缩编码可根据网络带宽等条件,调节生成视频的比特率和质量参数;经压缩编码后的视频可获得高效传输与存储,但也引入压缩噪音;在可溯源系统中,压缩编码过程需要保留必要的元数据和水印信息,以供后续溯源;解码环节根据对应的编解码算法,恢复压缩编码后的视频;使用合适的压缩编码标准和参数控制,可获得高压缩率和良好视觉质量。

[0036] 封装是将压缩编码后的视频流和音频流打包成容器格式的过程;常用的视频封装

格式有MP4、MKV、AVI、TS等。这些格式定义了视频、音频及元数据的存储和打包方式；封装格式还定义了码流的打包方式，比如MP4使用MP4文件格式和MPEG—4编码；封装为视频编码流增加了同步信息、时序信息、容错信息等，利于存储和传输；在可溯源系统中，封装过程中需要保留视频中的元数据和水印信息，进行封装容器的打包；解封装过程则解析容器，提取视频流、音频流及元数据，为后续解码提供数据；封装屏蔽了码流传输的复杂性，使存储和传输更加简便。综上，视频封装实现了压缩编码比特流的打包和格式化，是视频处理流程中承上启下的关键环节。

[0037] 传播链路包含视频采集、编码、加密、封装、传输、解码等所有处理环节的时间地点信息；每个环节处理都会产生元数据或水印信息节点，这些节点信息串联起来构成完整的传播链路；传播链路标识了视频的运动轨迹，可追踪视频的来源归属以及传输流向；对视频内容进行溯源时，需要解析视频中的节点信息，并重构完整的传播链路；传播链路提供了视频真实性、可追溯性和来源可考证性的技术保障；传播链路中任何一环节信息被破坏，都会导致溯源分析的失败；完整准确的传播链路是实现视频可溯源的基础，也是判断视频真伪的重要依据。综上，传播链路通过编码各处理环节的时间地点节点，记录视频内容从源头到终端的完整流转路径，是实现视频可溯源的关键信息。

[0038] 视频输出设备位于系统解码模块之后，用于对解码的数字视频信号进行渲染，转换为可见的画面输出；常见的视频输出设备有液晶显示器、投影机、电视机等。它们接受不同形式的视频输入信号；视频输出设备包含视频接口、视频解码器、显示面板/投影部件等，将数字视频转换为可见画面；在可溯源系统中，输出设备展示包含元数据和水印的完整视频内容，同时也可能包括溯源查询信息；输出设备的性能决定了视频画面的质量，如分辨率、色彩还原度、动态范围等；不同的输出设备适合不同的使用场景，如会议用投影机，家庭娱乐用电视机；输出设备是视频处理链路的最后一环，直接面向用户，其性能直接关乎用户观感。综上，视频输出设备是可溯源系统的终端画面输出环节，通过对视频流的解析渲染将数字视频转换为可见画面，直接影响用户体验。

[0039] 编码标准规定了视频压缩编码的具体算法流程、语法格式、实现方法等；常见的视频编码标准有H.264、H.265、AV1等。新标准通常在降低比特率的同时提高编码效率；编码标准确保了视频压缩的语法和过程规范化，使编码视频具有标准兼容性；不同的编码标准针对特定应用场景进行了优化，如高清视频、视频监控、网络视频等；在可溯源系统中，选择合适的编码标准可以平衡视频质量和编码效率；解码端使用与编码端相同的标准，以正确解析和还原压缩视频；随着硬件性能的提升，新一代编码标准被激活使用，如当前的H.265。

[0040] H.264是ITU和ISO联合发布的数字视频压缩编码国际标准。也称为MPEG—4第10部分；相比老标准如MPEG—2，H.264在同等画质下压缩率提高了约2倍；H.264通过更复杂的压缩算法实现更高编码效率，但编码复杂度也增加；H.264广泛用于视频监控、高清数字电视、网络视频、移动设备等领域；在可溯源系统中，H.264可在保证视频质量的前提下显著降低比特率；H.264也可以保留视频中必要的元数据和水印信息，以实现可溯源；解码端只有支持H.264标准的解码器才能正确解析和渲染H.264编码视频。综上，H.264是当前主流的视频编码标准之一，在可溯源系统中可以实现高效率的视频压缩编码，同时也保证视频质量和溯源信息的完整性。

[0041] H.265是ITU及ISO在H.264标准基础上发布的最新视频编码标准；相比H.264，

H.265在同等视觉质量下,压缩率提高了约30%—50%;H.265通过更复杂的压缩算法实现更高编码效率,但编码复杂度也大幅提高;H.265支持4K、8K等高分辨率视频内容的高效压缩;在可溯源系统中,H.265可在保证质量的前提下,显著降低存储和传输成本;H.265编码也可保留视频中的元数据和水印信息,以实现可溯源;解码端需要支持H.265的解码器才能正确解析H.265压缩视频;当前H.265还未完全普及,硬件采用成本较高。

[0042] 去噪是利用数字信号处理技术,消除或降低视频信号中的噪声成分;视频源在采集和传输中容易受到各类噪声的干扰,如电子噪声、编码压缩噪声等;去噪可有效减少视频源存在的随机噪声和锐利噪声,提高画质;常用的去噪算法有中值滤波、小波降噪、自适应去噪等。去噪强度需把握好度;在可溯源系统中,适当去噪可提高视频压缩编码的效率;但去噪也可能造成视频细节损失,需要保证溯源所需特征不被模糊;通过硬件优化及参数调控,可以在降噪的同时保留视频关键信息。

[0043] 校色主要针对视频采集和编码中的颜色失真问题,进行颜色还原和增强;视频源常因光照、设备因素出现颜色偏差、颜色褪色等问题。校色可以校正这些缺陷;常用的校色技术有白平衡、颜色映射、伽马校正等,可以校正视频的对比度、饱和度、色温等;适当的校色可使视频颜色更加逼真生动,提高视觉效果。但也不可过度美化,影响真实性;在可溯源系统中,校色参数必须可控,避免破坏视频原有的色调特征;校色还需要保证人脸及场景中的颜色相对关系不变,以利于溯源比对;通过选取适当的校色算法及强度,既可以提高视频质量,又不影响视频内容的真实性。

[0044] 设备编号可以是数字序列号,也可以含字母、特殊字符的复杂编码;设备编号将视频处理链路中的各设备,如摄像头、编码器、服务器等进行唯一标识;通过设备编号,可以区分多台同型号设备,并针对每台设备进行监控和溯源;设备编号信息可以以元数据形式插入视频流,也可以通过不可见水印嵌入;在对视频传输进行溯源时,设备编号是确认视频源头和处理经过的关键信息;设备编号还可以将视频内容 with 设备使用人员信息进行绑定,作为责任溯源的依据;通过设备编号的提取和解析,有助于快速锁定视频内容的产生和处理环节。综上,唯一的设备编号标识在可溯源系统中起到追踪、定位视频来源的关键作用,是实现溯源的基础信息组成部分。

[0045] 工作模式包括视频编解码的编码模式、解码模式,以及传输模式等;编码模式决定了视频的编码标准、压缩率、编码速度等参数;解码模式决定了视频的解析方式、处理优先级及回放控制等参数;传输模式决定了视频的传输协议、传输优先级、网络QoS等参数的设置;通过选择和切换不同的工作模式,可以实现对视频处理链路的动态优化和控制;在可溯源系统中,不同的工作模式需要保证视频溯源信息的完整性;工作模式的切换过程也需记录日志,以供溯源分析时追踪参数变化;合理的工作模式配置可以在保证视频质量的同时,优化可溯源系统的性能。

[0046] 节点信息包含视频采集、编码、存储、传输、解码等处理节点的时间、地点、设备、操作人员等数据;节点信息以元数据或者数字水印的形式嵌入视频流中;视频处理流程的每个环节都会产生对应的节点信息。所有节点信息链串联起来形成完整的视频处理链路;在对视频内容进行溯源时,需要提取和解析视频流中的节点信息;节点信息可以帮助快速确定视频的产生时间、地点、流转路径和操作历史;节点信息直接决定了可溯源的精细程度。节点信息越详尽,溯源就越准确和可靠;保护节点信息的完整性和安全性是可溯源系统的

重要环节。综上,节点信息是实现视频可溯源的基础数据,其准确性直接影响到溯源的效果。

[0047] 视频帧序列,视频通过每秒传输多张图像帧的方式产生动态影像;每帧图像都包含独立的视频内容信息,连续的图像帧构成视频序列;视频以每秒25至60帧的速率播放帧序列,产生视觉上的运动效果;视频压缩编码通过帧间预测等方式降低帧序列冗余性;在可溯源系统中,对视频帧序列引入时间戳或帧编码信息;通过分析视频帧序列的细节变化,可以进行视频内容的溯源分析;如通过场景和人物在连续帧中的变化进行视频时间地点的考证;保护视频帧序列的完整性对进行可溯源至关重要。综上,视频帧序列包含视频时间信息和内容细节信息,是实现精确视频溯源的基本数据单元。

[0048] 编码器将采集到的视频源内容进行压缩编码,输出压缩比特流;常见的编码器包括软件编码库、硬编码卡、嵌入式编码芯片等;编码器压缩算法需要实现视频压缩标准,如H.264、H.265等;编码器优化视频压缩率,同时控制码流质量和编码速度;在可溯源系统中,编码器还需要在压缩的同时保留关键元数据;编码器的参数设置直接影响视频压缩质量和比特率;编码器的性能决定视频编码的速度,影响整个视频处理链路;选择合适的编码器及参数设置,对可溯源系统至关重要。综上,编码器在可溯源系统中负责视频内容的高效压缩,其性能直接关乎视频质量及编解码速度。

[0049] 对称加密算法使用单一密钥,该密钥用于数据加密和解密方;常见的对称算法有DES、AES、RC4等。其中AES安全性较高;对称加密速度快,算法相对简单,适合大数据量的视频内容加密;加密视频数据可防止未授权访问,保证数据安全性;在可溯源系统中,需要加密视频中包含的元数据信息;但加密也会引入算法噪声,需要控制对视频质量的影响;加密密钥需要安全管理,防止密钥泄露给未授权用户;选择安全的对称加密算法,可以在保障速度的同时,有效保护视频和元数据的安全。

[0050] 非对称加密使用公钥加密,私钥解密,增强了密钥管理的灵活性;常用的非对称算法有RSA、ECC等。安全性高但速度较对称加密慢;非对称加密可用于视频元数据的加密,以及数字签名和证书管理;公钥可公开,私钥由视频内容所有者持有,具有更强的控制能力;在可溯源系统中,可以用非对称加密保护视频处理节点信息的真实性;还可以建立基于数字证书的设备认证机制,确保可溯源系统的安全;非对称加密算法计算复杂,不适合大量视频数据加密,可结合对称加密使用;合理应用非对称加密,可强化可溯源系统的内容保护与权限控制。

[0051] 视频传输协议标准是指在传输模块中,用于视频传输的数据包格式和交互规则。常见的视频传输协议标准包括:RTP,实时传输协议,一种用于互联网上传送音频和视频的传输协议。它可以对多媒体数据进行源标识、序列号、时间标记等处理,实现实时数据传输;RTSP,实时流传输协议,一种应用层协议,用于控制实时数据传输的启动、暂停等操作。常与RTP协议搭配使用;HTTP,超文本传输协议,应用层协议,用于Web视频传输。支持各种加密机制提高安全性;HLS-HTTP实时流传输协议,由Apple提出,基于HTTP协议设计,用于流媒体传输;MPEG-DASH,动态自适应流传输协议,用于高清多媒体内容的传输,支持动态码率切换;WebRTC,网页实时通信技术,支持浏览器对浏览器的视频流传输;SRT,安全可靠传输协议,用于低延迟实时视频流传输。在该安全可溯源视频编解码系统中,可以根据实际需求和网络条件,选择使用适当的视频传输协议标准,以实现高效可靠的视频流传输,从而保证视频

溯源分析的顺利进行。

[0052] 码流是经过视频编码压缩后的数字视频比特流序列,包含编码后的视频帧信息。在该安全可溯源视频编解码系统中,主要出现的码流包括:源视频码流,视频采集模块输出的符合编码标准要求的数字视频比特流;编码码流,视频编码模块经过压缩编码后的数字视频比特流输出;传输码流,传输模块对编码码流进行封装和加密后,用于传输的数字视频比特流;解码码流,解码模块接收并解码传输模块输出的码流,得到的数字视频比特流;输出码流,显示模块输出并显示的包含水印信息的数字视频比特流。通过在不同环节插入元数据、节点信息以及在输出码流中嵌入水印,可以追踪码流的传播链路,实现视频内容的可溯源。码流作为系统的输入和输出,承载了视频溯源所需的重要信息。整体来说,码流是经过压缩数字化处理的视频数据流,在该安全可溯源视频编解码系统中发挥视频存储、传输和显示的关键作用。

[0053] 混淆处理是指传输模块中的加密单元对编码视频帧序列进行的保密处理。其典型目的包括:保护视频内容的安全性,防止未经授权的第三方直接获得和使用原始视频;避免直接传输编码视频帧序列时可能出现的信息泄露;提高后续解码的安全性,只有授权的接收方能正确解码。常见的混淆处理方式有:利用对称或非对称加密算法对视频帧进行全帧或选择帧加密;对视频帧中的视觉内容进行扰乱,破坏原有视觉信息;在视频帧中插入干扰信号或伪随机噪声序列;对视频帧序号或时间信息进行打乱处理;利用网络安全技术,对传输流量进行封装和隐藏。通过混淆处理,可以提高视频传输和解码的安全性,避免信息泄露。是安全可溯源视频编解码系统中必要的安全保障手段之一。只有解码端能正确解析混淆后的视频帧内容。

[0054] 数据包是传输模块对编码视频流进行切分和封装后的传输单位。根据不同的视频传输协议标准,数据包可以包含以下信息:视频数据载荷,包含压缩编码后的视频帧信息;包头,包含数据包信息,如包序号、时间戳、数据长度等;源和目的地址,标识数据包的发送方和接收方地址;校验信息,确保数据包传输的正确性;控制信息,用于调控和控制视频流的传输;加密信息,对视频数据载荷进行加密,保证传输安全。数据包是传输模块根据网络条件、可靠性需求、传输协议标准等进行优化封装后的数据传输单位。通过对数据包内容和传输的控制,可以实现视频流的安全可靠传输,为后续视频帧的解析和溯源分析提供保障。数据包的生成和传输对安全可溯源视频编解码系统网络性能至关重要。

[0055] 网络带宽指视频传输过程中网络连接的最大数据通过量或通信容量。系统的传输模块在选择视频传输模式时,需要考虑网络带宽情况,主要包括:网络连接类型(有线/无线)决定了网络带宽的物理基础;网络带宽直接影响视频编码时的码率控制参数选择;高带宽网络可以传输高质量、低延迟的视频流;低带宽网络需要降低码率,缩小数据包大小,保证基本的视频传输质量;动态监测网络带宽变化,实现码率的自适应调整;不同传输模式(UDP/TCP/WebRTC)对网络带宽有不同需求。充足和动态可控的网络带宽,是保证系统可靠传输视频流、进行顺利视频解析和溯源的基础。搭建合理的网络连接,评估和控制网络带宽,是安全可溯源视频编解码系统的重要环节。

[0056] 网络稳定性指视频传输过程中网络连接的可靠性和抗干扰能力。系统的传输模块在选择传输模式时,需要考虑网络稳定性,主要问题包括:网络连接是否易断和抖动,直接影响视频传输的流畅性;无线网络相比有线网络,容易受到环境干扰,稳定性较差;网络拥

塞会使视频包丢失,需要采取拥塞控制;选择TCP传输可以获得肯定应答和重传机制,提高可靠性;UDP传输虽对实时性要求较低,但网络稳定性直接影响丢包率;可以采用网络质量检测,动态切换传输模式或调整编码参数;部署SDN网络,对数据流进行优化控制,提高稳定性。良好的网络稳定性是系统达到用户体验要求的基础,也是保证视频解析和溯源顺利进行的重要因素。评估和提升网络稳定性非常必要。

[0057] 传输模式指视频传输模块根据视频流特点和网络条件,选择的数据包传输方式。系统中提到的主要传输模式包括:UDP模式,基于用户数据报协议的传输模式,具有实时性强、协议简单等优点,但可靠性较差;TCP模式,基于传输控制协议的传输模式,具有可靠传输、流量控制等特点,但传输延迟可能较高;WebRTC模式,一种浏览器对浏览器的实时多媒体传输技术,基于UDP,可实现低延迟传输。传输模块可以根据视频实时性需求、网络带宽与稳定性等情况,选择合适的传输模式,来获得更好的用户体验。同时,系统也可以实现动态切换传输模式,以适应网络状况的变化。合理的传输模式选择是保证系统传输效率的重要手段。

[0058] UDP(User Datagram Protocol)即用户数据报协议,是一种无连接的传输层协议。该系统中作为一种传输模式时,UDP协议的主要特点包括:提供面向无连接的不可靠服务,传输数据包不进行确认和重传;传输延迟低,更适合用于对实时性要求较高的视频流传输;协议简单且传输效率高,包头开销小,支持一对多的网络组播;数据报可能会丢失或失序,不保证可靠传输;需应用层实现流量控制、重传等机制来提高可靠性;安全性较低,容易遭受拒绝服务类型攻击;传输速率取决于网络带宽和稳定性。系统中可以根据视频流特性和网络条件,适当采用UDP模式传输,获得低延迟和高效率的好处,但需要权衡可靠性问题。

[0059] TCP(Transmission Control Protocol)即传输控制协议,是一种面向连接的、可靠的传输层通信协议。该系统中作为一种传输模式时,TCP协议的主要特点包括:通过三次握手建立连接,点对点传输数据,提供可靠交付服务;支持流量控制,拥塞控制,以避免网络拥堵;采用确认应答机制,支持超时重传,保证数据传输的可靠性;维护传输顺序,处理数据丢包问题;连接导向,安全性较高,难以受到攻击;传输延迟可能较高,不如UDP适合用于对实时性要求极高的场景;头部开销较大,传输效率可能低于UDP。系统中可以根据需求,采用TCP协议的可靠传输特性,确保视频流的完整性,但需要权衡传输延迟问题。

[0060] WebRTC(Web Real-Time Communication)是一种实时通信技术,支持浏览器对浏览器进行实时语音、视频和数据通信。该系统中作为一种传输模式时,WebRTC的主要特点包括:基于UDP协议实现低延迟的实时多媒体传输;支持视频编码如H.264,音频编码如Opus等;通过P2P方式传输数据,不需要服务器中转;在浏览器内部直接实现媒体协商、编解码等功能;通过STUN,TURN等协议穿透防火墙和NAT;应用广泛,可以实现Web页面中的点对点实时视频通话;安全性较高,支持DTLS加密传输。系统中可以使用WebRTC作为传输模式,通过P2P形式传输实时视频流,实现低延迟和高效率,并保证一定的安全性。

[0061] 传输参数是指系统在进行视频流传输时,可以控制和调整的一些设置,以实现更好的传输效果。常见的传输参数包括:传输协议,UDP/TCP/WebRTC等协议的选择;加密算法,对视频流进行加密的算法类型;传输速率,单位时间内传输的数据量;包大小,每个数据包包含的PAYLOAD数据大小;缓冲区大小,接收端用于缓存数据包的队列或缓冲区大小;丢包处理,数据包丢失时的重传或补偿策略;延迟控制,端到端的传输延迟时间控制;错误校验,

对数据进行校验以检查传输错误;流控机制,流量控制算法,避免网络拥塞。调整这些传输参数可以优化视频的编码性能、传输稳定性、时延特性等,提升系统的传输质量,为后续视频解析和溯源保驾护航。

[0062] 编解码协议是视频编解码模块对视频流进行压缩编码和解码时所遵循的标准或规范。该系统中的主要编解码协议可能包括:H.264/AVC,一种广泛使用的视频压缩编码标准,编码效率高,支持多种分辨率;H.265/HEVC,新一代高效视频编码标准,相比H.264有50%以上的压缩效率提升;VP8,一种开源的视频编码格式,由Google提出;VP9-VP8的升级版,采用更多编码工具提高压缩率;AV1,AOMedia联盟推出的开源视频编码格式;MPEG-2,早期数字电视广播采用的视频编码标准;MPEG-4,提供更佳压缩性能的编码标准,支持对象编码。选择合适的编解码协议,可以有效提升视频压缩率,减少存储和传输开销,从而提高系统的性能。但也需要平衡编码复杂度和视频质量之间的关系。

[0063] 低可视频帧是指通过视频编码压缩后,重构质量较差的视频帧。其特点通常包括:分配的码率或比特数较少,导致图像失真严重;帧内容复杂,编码器选择性降低该帧质量,以维持相邻帧的图像质量;存在大面积的模糊区域或马赛克效应;运动变化剧烈的视频区域编码失真明显;帧内预测编码算法的传递误差积累造成质量下降;网络传输丢包、丢帧导致的视频质量损失;解码端由于计算资源限制造成的图像损坏。

[0064] 校验码是用于检验和校验数据传输或存储的正确性的冗余代码。视频流传输和存储过程中加入校验码主要目的是:在传输过程中检测和纠正数据错误,提高可靠性;通过校验发现存储介质上的数据损坏情况;验证数据的完整性和一致性,防止未经授权的篡改。常见的校验码算法有:奇偶校验,简单且效率高,能检测单位错;循环冗余校验(CRC),能检测突发错误,广泛使用;消息认证码(MAC),防止消息伪造;安全哈希算法(SHA),能检测任何修改;校验和,简单求和后取余。

[0065] 空域信息是指视频信号在空间域里所包含的视觉信息。视频编码和压缩过程中会利用和处理空域信息,主要包括:运动预测会判断和利用空域内像素块的移动信息;变换编码会将空域像素转换到频域进行编码;量化会保留空域视觉信息中主要的成分;视频滤波操作直接在空域中对像素进行处理;视频增强也可以在空域对比度、色度等参数进行调整;视频水印等直接嵌入空域像素信息;视频压缩标准会利用空域冗余性进行去相关运算;解码重构帧也是在空域重建视频内容。

[0066] 时域信息是指视频信号随时间变化所包含的信息。视频编码和压缩过程中会利用和处理时域信息,主要包括:计算连续帧之间的运动矢量,判断时域运动变化;利用时域冗余性去除静止背景;调整GOP结构控制关键帧插入时间;调整编码器的帧率和码率控制参数;通过时间层级预测编码获得可扩展性;在传输中根据时域信息实时调整码流;时域滤波可平滑视频时域噪声;对视频片段进行时域裁剪、拼接;通过时域统计分析判断视频内容。

[0067] 图像复杂度(Image complexity)通常是指描述视频图像的空间信息和时间信息的一个指标,可以用来评估视频图像的复杂程度。图像复杂度较低的视频帧作为水印嵌入的目标帧,这样可以避免在高复杂度的图像区域嵌入水印,从而降低水印对视频质量的影响。图像复杂度的计算可以考虑以下几个方面:空域复杂度:反映图像的空间细节和边缘等静态信息,可以用块内标准差、边缘密度等表示;时间复杂度:反映图像序列的动态变化信息,可以用运动矢量大小、残差信息等表示;纹理复杂度:反映图像的方向性信息,可以用梯

度直方图等表示;对象复杂度:反映图像中的主体目标数量、种类等语义信息;噪声复杂度:反映图像的噪声成分。

[0068] 隐写技术可以将编码/解码节点等溯源信息隐藏嵌入到视频帧中,不影响原始视频的显示,但可以通过解析被提取出来。常见的隐写技术包括:最低有效位 (LSB) 嵌入:在视频像素的最低有效位中嵌入信息位;频谱扩展,将信息隐藏在视频信号的高频分量中;量化索引修改,调整视频压缩编码的量化参数,在量化过程中嵌入信息;运动矢量修改,调整视频压缩编码的运动矢量信息来隐写;抖动模糊,通过引入视频抖动来嵌入信息;统计模型嵌入:利用视频压缩的统计特性隐写;深度学习隐写,使用深度神经网络进行隐写。

[0069] 数字水印是一种将信息隐秘嵌入信号的技术,它可以在不影响信号主体的前提下,向视频信号中插入辅助信息,供后续检测和提取。在该可溯源系统中,数字水印技术可以用来将编码、解码等节点信息以隐写的方式嵌入到视频帧中,用于溯源追踪。数字水印的主要特点包括,隐蔽性,水印嵌入后不影响视觉效果;鲁棒性,水印能够抵抗常见的信号处理操作;安全性,水印具有一定的防篡改能力;容量,水印能够嵌入一定数量的有效信息。数字水印的实现方法有,空域方法,直接在像素样本中嵌入;频域方法,在转换域嵌入,如DCT、DWT等;压缩域方法,在视频压缩编码过程中嵌入;几何攻击恢复方法,提高抗几何攻击能力;融合嵌入方法,结合多域信息进行嵌入。

[0070] 渲染,指的是将解码后的视频帧序列转换为可在显示设备上播放的视频信号的过程。具体来说,渲染过程包括以下步骤,颜色空间转换,将视频帧的YUV或RGB颜色空间转换为显示设备使用的颜色空间,如sRGB;扫描转换,将视频的逐行扫描顺序转换为显示设备的逐行或隔行扫描格式;缩放,将视频的分辨率缩放到显示设备的原生分辨率;去隔行,对隔行扫描的视频信号进行去隔行处理;字幕叠加,将字幕图片或文本叠加到视频上;连接器转换,将数字视频信号转换为显示设备使用的模拟信号接口;显示驱动,发出正确的同步信号,驱动显示面板显示视频图像。通过渲染过程,可以在显示设备上正确播放解码恢复的视频序列,为用户提供良好的观看体验。

[0071] 时间戳是指对一个事件发生的时间进行数字化编码的一个标识符。该系统中,各个处理节点在对视频帧进行操作时,会插入编码时间戳或解码时间戳信息,用来标识视频数据的处理时间。这些时间戳信息用于后续溯源分析,以追踪视频的传播链路。

[0072] 传输路由是指数据包在计算机网络中从源地址传输到目的地址所经过的路径。该系统的溯源分析单元可以根据解析得到的各节点信息,生成视频流的数据包所经过的完整传输路由,反映视频流的传播过程。

[0073] 实施例

[0074] 为了更清楚地说明本说明书实施例的技术方案,下面将对实施例描述中所需要使用的附图作简单的介绍。显而易见的,下面描述中的附图仅仅是本说明书的一些示例或实施例,对于本领域的普通技术人员来讲,在不付出创造性劳动的前提下,还可以根据这些附图将本说明书应用于其他类似情景。除非从语言环境中显而易见或另作说明,图中相同标号代表相同结构或操作。

[0075] 应当理解,本说明书中所使用的“系统”“装置”“单元”和/或“模块”是用于区分不同级别的不同组件、元件、部件、部分或装配的一种方法。然而,如果其他词语可实现相同的目的,则可通过其他表达来替换所述词语。

[0076] 如本说明书和权利要求书中所示,除非上下文明确提示例外情形,“一”“一个”“一种”和/或“该”等词并非特指单数,也可包括复数。一般说来,术语“包括”与“包含”仅提示包括已明确标识的步骤和元素,而这些步骤和元素不构成一个排他性的罗列,方法或者设备也可能包含其他的步骤或元素。

[0077] 本说明书中使用了流程图用来说明根据本说明书的实施例的系统所执行的操作。应当理解的是,前面或后面操作不一定按照顺序来精确地执行。相反,可以按照倒序或同时处理各个步骤。同时,也可以将其他操作添加到这些过程中,或从这些过程移除某一步或数步操作。

[0078] 图1是根据本说明书的一些实施例所示的安全可溯源视频编解码系统的示例性模块图,如图1所示,一种安全可溯源视频编解码系统100包括:包括视频采集模块110、视频编码模块120、传输模块130、解码模块140和显示模块。150视频采集模块110用于采集源视频流,并在源视频流中插入水印信息和元数据。视频编码模块120用于接收源视频流,并在源视频流中插入编码节点信息,并对插入编码节点信息后的视频流进行压缩编码。传输模块130用于对视频编码模块120输出的编码视频流进行封装和传输。解码模块140用于接收传输模块130输出的视频流,对接收到的视频流进行解码,在解码后的视频流中插入解码节点信息和水印。显示模块用于显示解码模块140输出的视频流,并从中解析节点信息和水印,生成视频流的传输轨迹。其中,显示模块包含溯源分析单元,用于获取解码后的视频流,解析视频流中的节点信息和水印,并追踪图像的传播链路。通过该安全可溯源视频编解码系统,可以在视频采集、编码、传输、解码各节点插入必要的信息,并在显示端解析这些信息,实现对视频流的传输轨迹进行溯源,确保视频内容的真实性和可追溯性。

[0079] 其中,视频采集模块110在源视频流中插入水印和元数据。水印包含时间、地点、设备编号等信息,元数据包含检测和识别的结果。这为后续溯源提供了关键信息;视频编码模块120在源视频流中继续插入编码时间、地点、设备编号等节点信息,并进行压缩编码。压缩编码降低了视频数据量,有利于传输。插入的节点信息也是溯源的关键;传输模块130对视频流进行封装和传输。封装提高了传输效率,传输将视频发送到解码端;解码模块140对接收到的视频流进行解码,恢复视频帧。同时也在视频流中插入解码时间、地点、设备编号等节点信息;在解码后的视频流中,水印嵌入模块选择合适的帧嵌入包含各节点信息的水印。水印的隐蔽性使信息不易损坏;显示模块通过解析视频流中的节点信息和水印,生成完整的视频传输轨迹。不同形式的信息进行交叉验证,提高溯源准确性。

[0080] 具体地,视频采集模块110在安全可溯源视频编解码系统中处于视频流生成的第一阶段,其主要功能是对来源视频流进行采集、增强和特征标识,为后续的编码、传输和溯源分析生成优化的源视频流。组成单元:视频采集模块110包括视频源单元、编码标准选择单元、视频采集单元、图像增强单元、水印信息单元、元数据生成单元和采集控制单元。视频源单元:用于获取摄像头或计算机桌面产生的原始视频信号。编码标准选择单元:用于根据源视频信号的格式和系统需求,选择H.264、H.265等视频编码标准。视频采集单元:用于按选定的编码标准对源视频信号进行采集,输出符合标准要求的源视频流。图像增强单元:用于对源视频流进行去噪、校色等图像质量增强处理。水印信息单元:用于在源视频流的预设位置插入包含视频采集时间、地点、设备编号等信息的数字水印。元数据生成单元:用于定期提取源视频流中的关键帧,并通过分析生成包含目标检测、场景识别等信息的元数据。采

集控制单元:用于根据网络状态和用户指令,控制视频采集模块110的工作模式。工作流程:获取原始视频信号;根据源信号选择编码标准;按标准采集源视频流;对源流进行图像增强;在源流中插入水印信息;生成描述源流内容的元数据;根据网络和指令控制工作模式视频采集模块110在源视频流生成阶段,通过水印、元数据等信息的插入,实现对源视频流的特征标识,为后续的视频编码、传输和溯源分析提供基础保障。

[0081] 更具体地,水印信息单元在对源视频流插入数字水印时,其预设位置可以有不同设置方式:固定位置:比如视频序列的左上角或右下角。这种方式操作简单,但易被破坏;随机位置:每隔一定帧数在视频帧的随机位置插入水印。这种方式提高了鲁棒性,但检测复杂;自适应位置:根据视频内容自动选择模糊区域或噪声区域插入水印,可减少视觉影响。但需要内容分析。源视频帧进行频域转换,根据块的视觉复杂度,自适应选择复杂度较低的块来嵌入水印,同时引入随机偏移来增加鲁棒性。在解码端,根据密钥、随机种子进行检测,可以实现视觉影响小且鲁棒的水印检测;水印有效载荷信息可以携带采集时间戳、地点信息、设备编号等,并加入校验码以检验水印的有效性。

[0082] 更具体地,元数据生成单元需要定期提取源视频流中的关键帧来分析生成元数据。这里的“定期”可以有几种具体的技术设置方案:固定间隔:比如每隔10秒或每50帧提取一次关键帧。该方法实现简单,但无法适应视频内容的变化;自适应间隔:根据视频内容的变化情况来动态调整关键帧的提取间隔,譬如检测到场景切换时提高提取帧率。该方法可以保证关键帧捕捉到视频内容的变化,但实现较为复杂;事件驱动:当检测到视频流中的特定事件(如物体进入场景、明显运动等)时触发关键帧的提取。这种方法可以提取到重要的内容变化画面。

[0083] 更具体地,关键帧是视频序列中代表整个画面的参考帧,它独立于其他帧,不依赖于其他帧的信息。视频压缩编码通常以关键帧为基础去编码其间的普通帧。关键帧的设置需要考虑:表示图片内容的完整性:关键帧需要全面反映当前画面内容,为后续普通帧编码提供完整参考;编码压缩效率:关键帧直接进行编码,会占用较大比特率,设置太多关键帧会降低压缩效率;随机访问的便利性:关键帧的设置要能允许从任意位置进行解码访问;处理复杂度:关键帧数量直接影响编码计算量;典型的关键帧设置策略有:定时插入:如每过5秒、每50帧等定时插入一个关键帧;场景切换检测:当检测到视频场景发生明显变化时,插入关键帧;移动检测:当图像内容发生大面积变动时,判定为场景变化,执行关键帧插入。

[0084] 更具体地,视频源单元提供原始视频信号输入,是整个模块的输入源;编码标准选择单元根据源信号特点选定编码标准,使后续处理符合标准要求;视频采集单元基于选定的编码标准对源视频信号进行数字化采集和抽样,输出数字视频流;图像增强单元对数字视频流进行信号去噪和颜色校正,提高视频质量;水印信息单元在视频流中嵌入水印,实现视频内容的数字标识。水印技术确保水印不易被去除,增强视频的可溯源性;元数据生成单元通过分析视频内容,提取语义信息生成元数据,用于描述视频场景和目标,配合水印增强视频语义标识;采集控制单元根据网络等外部条件动态调整模块工作参数,保证视频采集的可靠性;各单元协同工作,在保证视频质量的同时,使用水印、元数据等技术手段对视频流进行可溯源标识,为后续处理奠定基础。元数据和水印技术实现视频内容的数字特征提取。综上,视频采集模块110从源视频采集到特征标识,通过各单元的有机配合,实现可靠、可追溯的视频流生成。

[0085] 具体地,视频编码模块120基于视频采集模块110输出的源视频流,对视频流进行编码压缩、加密和封装处理,以生成可传输的编码视频流。组成单元

[0086] 视频编码模块120包含信息插入单元、编码单元、加密单元和封装单元。信息插入单元:用于在源视频流中插入包含编码时间、地点、设备编号等信息的编码节点信息。编码单元:用于根据网络状态选择H.264/H.265标准对视频流进行编码,可调整码率控制和变块大小参数,实现编码压缩。加密单元:用于使用对称/非对称加密算法对编码后的视频流进行加密处理。封装单元:用于根据视频传输协议要求对加密后的视频流进行封装。工作流程:在源视频流中插入编码节点信息;根据网络状态选择编码标准和参数;对视频流进行编码压缩;利用加密算法对编码流进行加密;按传输协议要求完成流封装视频编码模块120在源视频流基础上,通过编码压缩、加密和封装处理,提高了视频流的安全性及传输效率,为后续的视频传输提供了编码优化后的流媒体数据。

[0087] 更具体地,信息插入单元在源视频流中嵌入编码节点信息,这为后续溯源提供节点标识;编码单元根据网络状态选择合适的编码标准和控制参数,使用编码算法对视频数据进行无损或有损压缩,降低视频码率,提高传输效率;加密单元应用加密算法技术对编码后的视频流进行加密,保证视频内容的机密性和安全性。加密算法控制加密强度;封装单元根据传输协议要求,对视频流进行格式封装,生成能够在网络上传输的流式数据包;编码单元压缩视频,降低码率,提高传输效率。加密单元保证内容安全。封装单元实现传输适应。三者配合,优化视频编码以适应网络传输;信息插入单元配合加密和封装单元,在保证效率和安全的同时,也保留了视频的可溯源信息;每个单元根据自身技术原理发挥作用,相互协调,共同完成视频编码、加密、封装的优化处理,使视频流可以高效、安全地在网络上传输。综上,视频编码模块120通过各单元技术的协同应用,实现对源视频流的编码优化,生成可保密性传输的编码视频流。

[0088] 具体地,传输模块130基于编码模块输出的编码视频流,对流进行封装和传输,实现视频流的网络分发。组成单元:传输模块130包含加密单元、封装单元、传输模式选择单元、传输单元和控制单元。加密单元:用于对编码视频流进行额外的加密混淆处理。封装单元:用于将加密后的视频流封装成网络传输的数据包。传输模式选择单元:用于根据视频流实时性需求、网络带宽等条件,选择UDP、TCP或WebRTC等传输模式。传输单元:用于通过网络将封装后的视频数据包发送给解码端,根据选择的传输模式进行相应控制。控制单元:用于根据网络状态输出控制信号,调节数据包生成和传输的参数。工作流程:对编码视频流进行加密混淆;将加密后的流封装成数据包;根据网络条件选择传输模式;通过网络将数据包发送到解码端;根据网络状态控制数据包的参数传输模块130通过编码流的二次加密和封装,选择适合的传输模式,实现了对视频流进行优化传输的控制,是视频能够通过公网安全高效地分发给解码端的保障。

[0089] 更具体地,加密单元应用加密算法和技术对编码视频流进行二次加密,进一步提高数据传输的保密性和抗攻击能力;封装单元根据传输协议,使用打包封装技术生成可在网络上传输的数据包,封装处理屏蔽了底层数据传输过程的复杂性;传输模式选择单元根据实时性需求、网络带宽等条件,选择UDP、TCP或WebRTC等不同的传输协议和算法,实现对数据包传输的控制,选择最优模式;传输单元基于所选协议,通过网络接口发送封装后的数据包,并处理传输控制与反馈;控制单元监控网络状态,动态调整和控制数据包生成与传输

的参数,适应网络变化;加密和封装提高数据安全性。传输模式选择和传输单元实现高效可靠传输。控制单元实现动态调整。各单元技术协同确保视频流可靠分发;通过二次加密、封装技术以及动态控制传输模式与参数,传输模块130大幅提升了视频在公网环境下的安全传输质量与效率。综上所述,传输模块130中各单元通过技术协同发挥作用,实现了视频流的安全可靠传输,是视频传输的核心保障。

[0090] 具体地,解码模块140用于接收经网络传输的视频流,对其进行解码、节点信息插入和水印嵌入,输出带水印的解码视频给显示终端。组成单元,解码模块140包含解码单元、信息读取单元、信息插入单元、水印嵌入单元和输出单元。解码单元,用于根据自有编解码协议对接收的视频流进行解码,恢复视频帧序列。信息读取单元,用于在解码过程中解析视频流中的编码节点信息。信息插入单元,用于在解码帧序列中插入解码节点信息。水印嵌入单元,用于在解码帧中选取低复杂度帧嵌入水印。输出单元,用于输出嵌入节点信息和水印的解码视频帧。水印嵌入单元又包含,水印生成子单元,生成包含校验码的水印;视频分析子单元,分析各帧的空域和时域信息,评估重要性;嵌入控制子单元,根据重要性选择目标帧;水印嵌入子单元,在目标帧嵌入水印。信息插入单元包含,编码信息补充子单元,补充完整的编码解码节点信息;信息隐写子单元,使用数字水印技术将节点信息隐写插入帧中。工作流程,对接收流进行解码恢复帧序列;解析编码节点信息;在解码序列中插入解码节点信息;对解码帧进行重要性分析;在低复杂度帧中嵌入水印;输出嵌入节点和水印的解码视频。解码模块140在对接收视频流进行解码的同时,进一步插入解码节点信息和水印,为视频的溯源解析生成所需的完整节点信息。

[0091] 更具体地,解码单元根据编解码协议,运用视频解码算法对接收流进行解码,恢复视频帧序列,实现压缩编码视频的解析和还原;信息读取单元在解码过程中,使用信息隐写分析技术解析并提取编码时插入的节点信息,为溯源提供基础数据;信息插入单元运用数字水印等隐写技术,在解码帧中写入解码节点信息,补充完整溯源节点链;水印嵌入单元通过视频帧分析和水印生成、嵌入技术,在解码帧中植入隐藏水印,增强溯源可靠性;输出单元组合处理解码帧和溯源信息,输出带水印的解码视频;解码单元实现视频解析。信息读取和插入单元实现溯源信息采集。水印嵌入单元增强溯源可靠性。各单元技术协同配合,在解码视频的同时生成完整溯源信息;通过解码、信息插入和水印嵌入技术的有机配合,解码模块140实现对压缩编码视频的可靠解析和溯源信息的整合。综上,解码模块140中各单元共同应用相关技术,使视频解码和溯源信息提取生成两不误,为后续溯源分析提供基础。

[0092] 更具体地,视频帧的重要性评估需要考虑空域信息和时域信息;空域信息反映静态画面内容,可以通过以下特征进行分析:图像复杂度:反映纹理、边缘等细节信息,可以用梯度直方图表示;目标数量,通过目标检测算法检测画面中的主体数量;场景类别,使用场景识别技术判断当前场景类别;区域焦点,分析图像落在主要目标上的区域分布;时域信息反映动态变化,可以通过以下特征进行分析:运动量,通过运动估计算法计算出当前帧与前一帧的运动矢量大小;残差信息,编码预测获得的帧间差分残差;目标运动,检测到的目标在连续帧间的位置变化量;将上述空域和时域特征进行融合,可以得到针对特定应用的帧重要性评分。

[0093] 更具体地,图像复杂度反映了画面的细节程度,主要考虑空域复杂度和时间复杂度两个方面;空域复杂度可以利用图像的梯度信息进行判断:将视频帧进行梯度提取,得到

梯度幅值图;计算梯度幅值图的标准差或方差,作为空域复杂度指标;设置适当的阈值,当标准差或方差低于阈值时,判断为该帧空域复杂度低;时间复杂度可以利用运动信息和帧差进行判断:计算当前帧与前一帧的运动矢量大小的平均值作为运动量;计算当前帧与运动补偿后的前一帧之间的均方误差作为帧差;如果运动量和帧差都低于阈值,则判定时间复杂度低;将空域复杂度和时间复杂度结合,当两者同时低于阈值时,判定该帧的图像复杂度低于阈值;阈值的设置可以通过训练学习获得,或针对不同应用进行调整。

[0094] 具体地,显示模块用于对解码模块140输出的视频流进行解析、溯源和显示,实现对视频流传输过程的溯源分析。组成单元:显示模块包含视频显示单元、元数据解析单元、水印解析单元、溯源分析单元和存储单元。视频显示单元:用于在显示设备上渲染和显示解码后的视频帧序列。元数据解析单元:用于解析视频帧中的编码解码节点时间、地点、设备编号等信息。水印解析单元:用于检测和解析视频帧中的水印信息。溯源分析单元:用于将解析的节点信息进行融合,生成完整的视频传输轨迹。存储单元:用于存储生成的视频传输轨迹信息,供溯源查询。溯源分析单元包含:信息关联子单元:根据时间戳、设备编号等关联各节点信息;路由生成子单元:根据关联后信息生成完整传输路由。工作流程:在显示设备上渲染显示解码视频;解析视频帧中的编码解码节点信息;检测并解析视频帧中的水印;将解析的节点信息进行关联融合;生成视频的完整传输轨迹;存储视频传输轨迹,供溯源查询显示模块通过解析和关联视频中的节点信息,能够溯源生成视频的完整传输链路,实现对视频流的可追溯分析。

[0095] 更具体地,视频显示单元基于视频编解码和渲染技术,在显示设备上播放解码后的视频帧序列;元数据解析单元运用元数据分析技术,提取并解析视频帧中的编码解码节点信息;水印解析单元使用数字水印解析技术,检测并解析视频帧中的水印信息;溯源分析单元通过信息融合和关联技术,将解析的节点信息进行组合,生成视频的完整传输轨迹;存储单元基于数据库存储技术,存储生成的视频传输轨迹供查询;显示单元实现视频播放。解析单元实现信息提取。分析单元实现溯源生成。存储单元实现数据保存。各单元技术协同配合,实现视频解析与溯源;通过视频解析、信息提取、关联分析等技术的有机应用,显示模块可以溯源生成视频的完整传输链路,实现视频内容的可追溯性分析。综上,显示模块中各单元共同应用相关技术手段,协调配合实现对视频流的解析、溯源和存储功能,形成视频传输可追溯性分析的完整技术方案。

[0096] 综上所述,本方案中的各模块共同完成信息的插入、编码、传输、解码、解析等功能,既降低了数据量,又提供了多种形式的关键节点信息,通过信息的互证实现了高效准确的视频溯源。该协同工作机制充分利用了视频技术的优势,提高了视频溯源效率。

[0097] 图2是根据本说明书的一些实施例所示的安全可溯源视频编解码系统的技术路线示意图,如图2所示,视频采集端对源视频信号进行采集,获得源视频流在源视频流中插入水印和元数据水印包含时间、地点、设备编号等信息元数据包含目标检测、场景识别等内容分析信息;视频编码端对源视频流进行压缩编码,提高网络传输效率在源视频流中插入编码节点信息,包括编码时间、地点、设备编号等;网络视频传输利用UDP、TCP或WebRTC等协议传输视频流可根据网络状况和实时性需求选择合适的传输模式;视频解码端对接收到的视频流进行解码,恢复视频帧序列在视频帧中插入解码节点信息在视频帧内嵌入水印,包含编码解码节点信息;视频显示端显示和解析视频流,生成传输轨迹从水印和元数据中提取

节点信息,进行溯源分析将视频传输轨迹存储,供后续溯源查询这样通过视频处理、传输、解析各环节的协同,可以实现对视频来源和流转过程的可溯源和监控。

[0098] 具体地,从数据流向的角度,在视频采集端:源视频信号→源视频流→插入水印后的源视频流→生成元数据在视频编码端:源视频流→插入编码信息后的源视频流源视频流(插入编码信息后)→编码压缩后的视频流→封装后的视频流在网络视频传输:封装后的视频流→传输数据包→传输模块在视频解码端:传输数据包→解封装后的视频流解封装后的视频流→解码后的视频帧序列→插入解码信息后的视频帧序列插入解码信息后的视频帧序列→嵌入水印后的视频流在视频显示端:嵌入水印后的视频流→视频传输轨迹→溯源分析通过在各处理环节插入信息,并在显示端解析这些信息,实现了从源视频到传输再到显示的完整数据流,有效地连接起了视频的采集、处理、传输、解析各个环节,从而可实现视频内容的可溯源分析。

[0099] 图3是根据本说明书一些实施例所示的溯源分析过程的示例性流程图,如图3所示,溯源分析过程包括:获取解码后的视频流;检测并解析视频流中的节点信息,包括编码节点信息和解码节点信息。节点信息包含时间、地点、设备编号等;检测视频流中的水印帧,解析水印帧中的节点信息;将解析得到的编码节点信息、解码节点信息以及水印中的节点信息进行关联和融合;根据关联后的节点信息,生成视频流的完整传输轨迹;将生成的视频传输轨迹存储,用于后续的溯源查询。

[0100] 图4是根据本说明书一些实施例所示的水印嵌入过程的示例性流程图,如图4所示,水印嵌入过程包括:根据节点信息生成包含校验码的水印;分析视频帧的空域和时域信息,评估各帧的重要性;根据重要性阈值,选择重要性低的帧作为水印嵌入目标帧;在目标帧中嵌入生成的包含节点信息的水印。重要性低的视频帧可以是图像复杂度低于阈值的帧,嵌入过程需要控制水印的强度,保证视频质量,嵌入位置也需要优化选择,保证水印检测率,水印需要加入校验机制,检测水印的完整性,嵌入参数需要根据视频内容动态调整。

[0101] 以上示意性地对本发明创造及其实施方式进行了描述,该描述没有限制性,在不背离本发明的精神或者基本特征的情况下,能够以其他的具体形式实现本发明。附图中所示的也只是本发明创造的实施方式之一,实际的结构并不局限于此,权利要求中的任何附图标记不应限制所涉及的权利要求。所以,如果本领域的普通技术人员受其启示,在不脱离本创造宗旨的情况下,不经创造性地设计出与该技术方案相似的结构方式及实施例,均应属于本专利的保护范围。此外,“包括”一词不排除其他元件或步骤,在元件前的“一个”一词不排除包括“多个”该元件。产品权利要求中陈述的多个元件也可以由一个元件通过软件或者硬件来实现。第一,第二等词语用来表示名称,而并不表示任何特定的顺序。

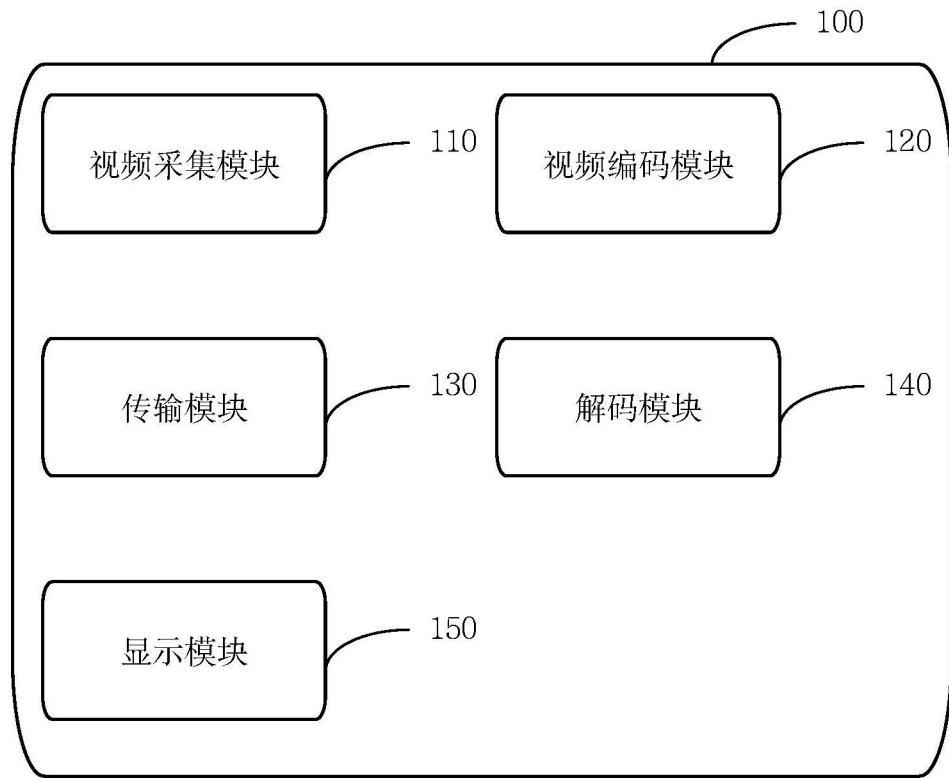


图1

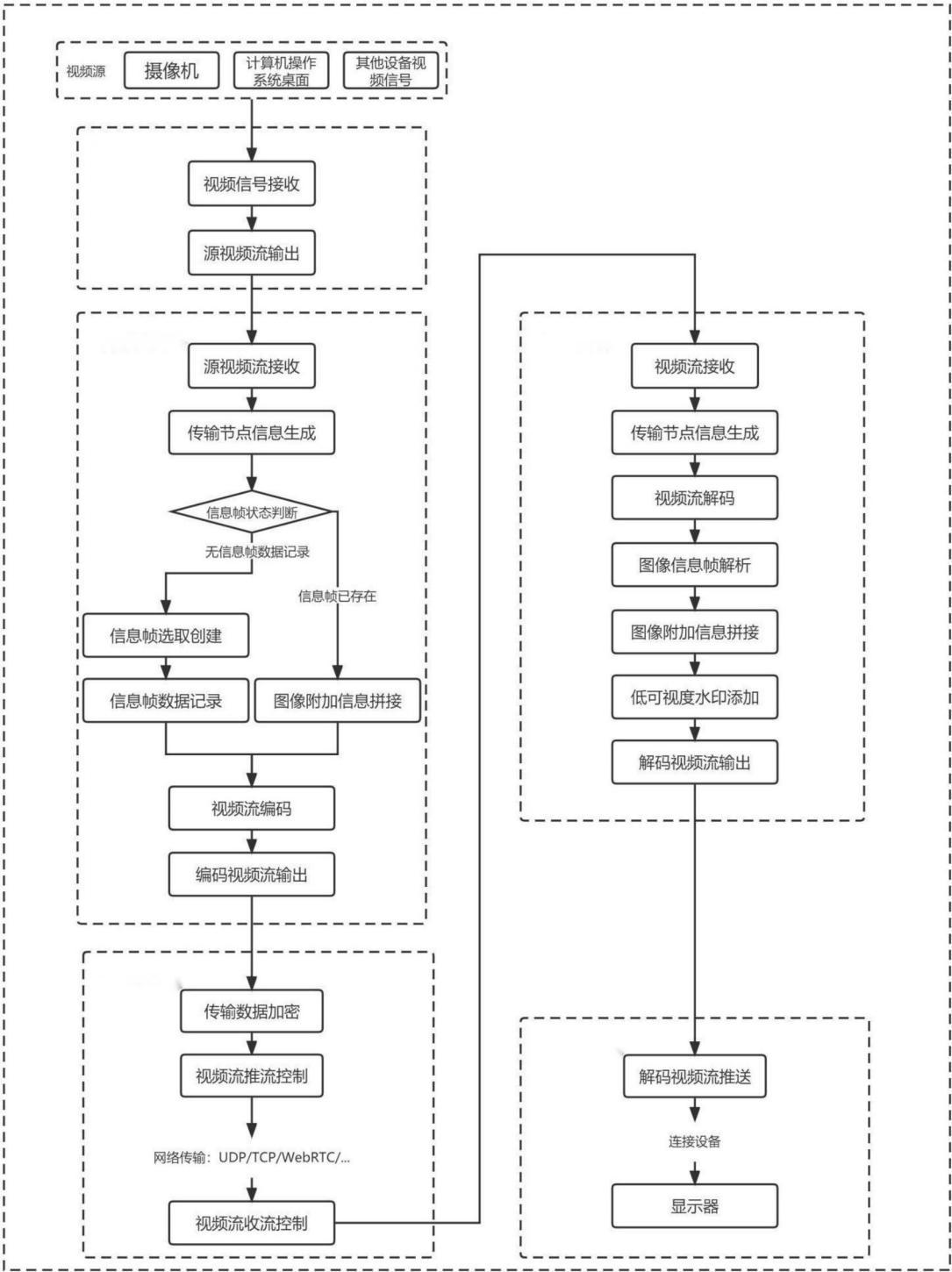


图2

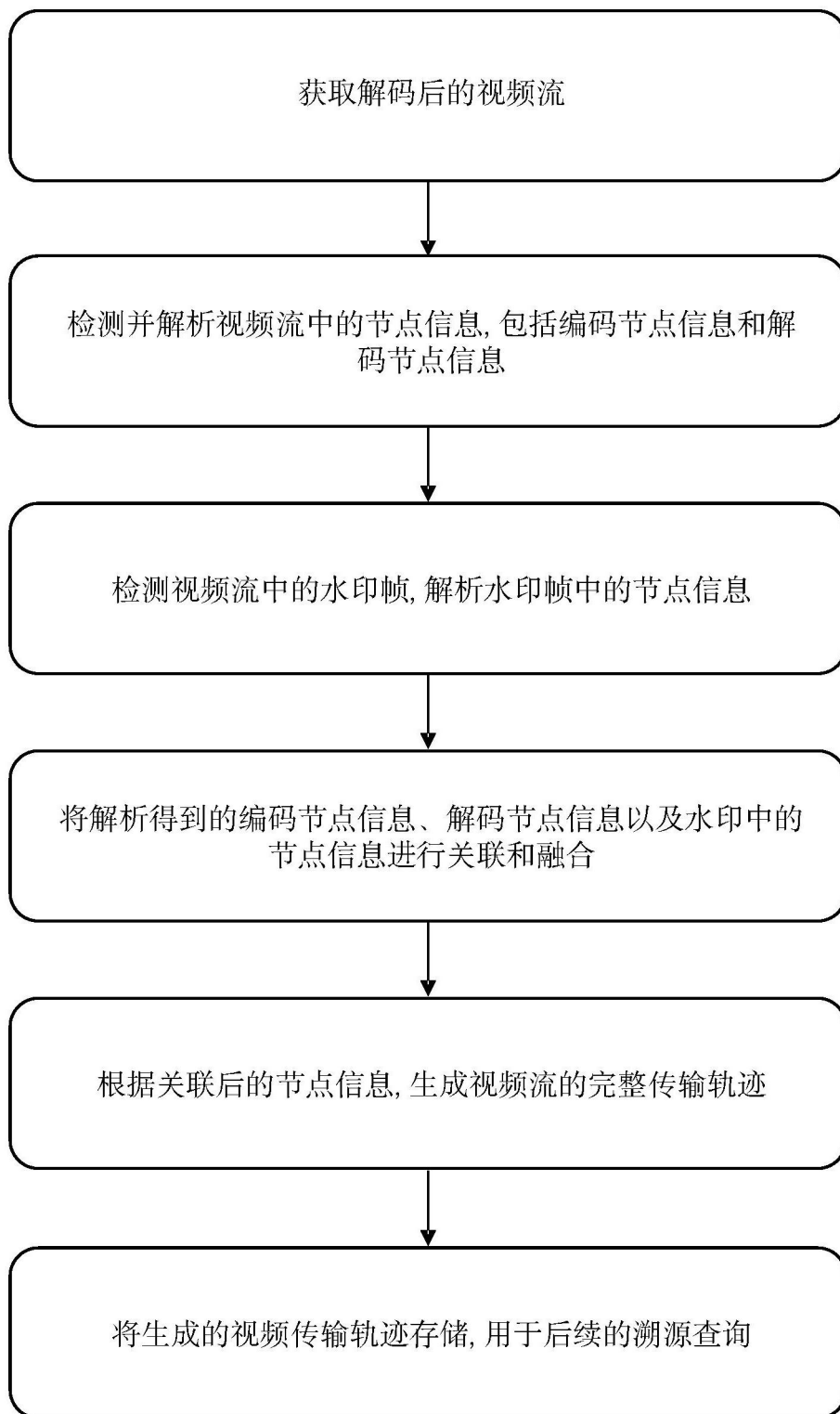


图3

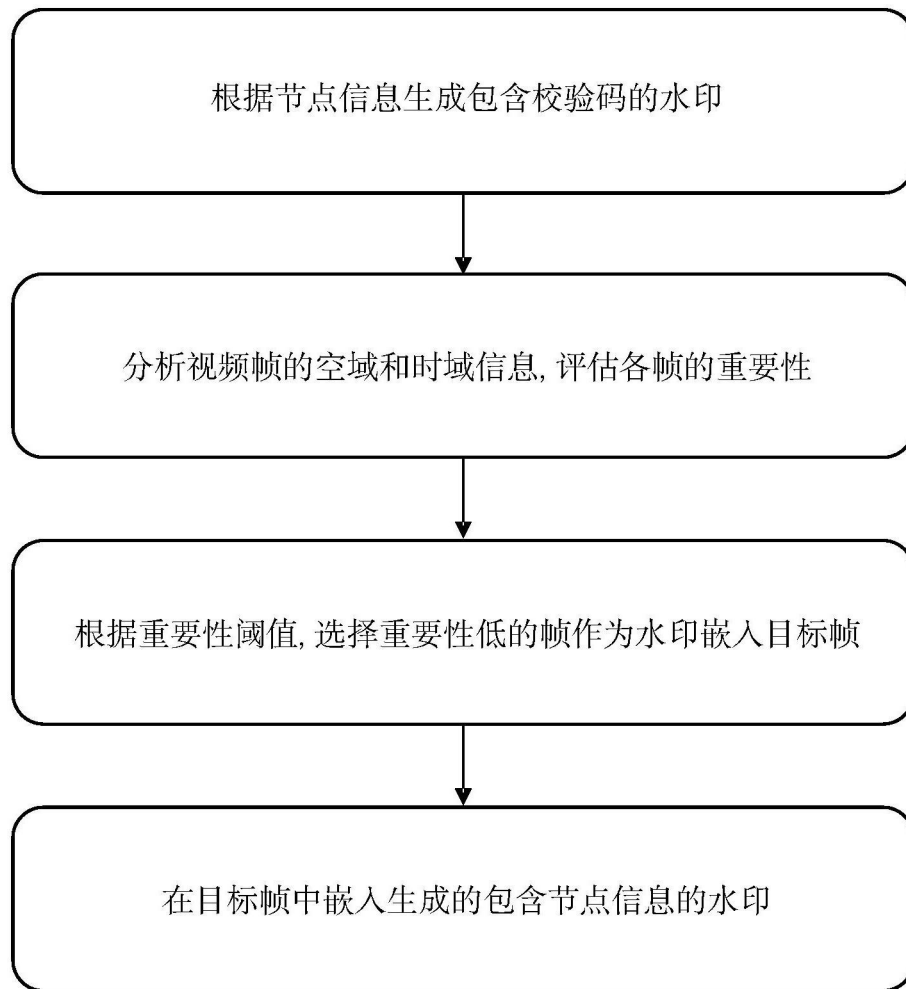


图4